Fasal Flasher

1.0

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Chapter 1

Fasal Flasher

This repository contains Fasal Flasher Source Code. Dive right into the codebase through AppFasal_Run invoked in main.c !

1.1 Table of contents

1.2 Context

Fasal Flasher is an Internal tool meant as a production floor Aid. Its primary purpose is to update the contents flash IC W25Qxx either with the contents of the Fasal Flasher onboard SD-Card or over a serial terminal over XModem protocol. The device has the following salient features

1. Powered by USB either through PC, Mobile phone or any USB source
2. File Input from either Serial console through USB connector or SD-Card
3. Slide switch to select between file source of SD-Card and Serial console (X-modem)
4. The W25Qxx flash IC in external board is powered through the Fasal Flasher and programmed over SPI
5. PushButton to trigger the transfer in the configured Mode
6. Tri-Color LED for visual indication about the process result
7. Logs over the same USB cable with information about the running application
8. File Integrity check via CRC in case of SD-Card transfer mode

9. File Integrity check inbuilt via XModem protocol in case of XModem transfer mode

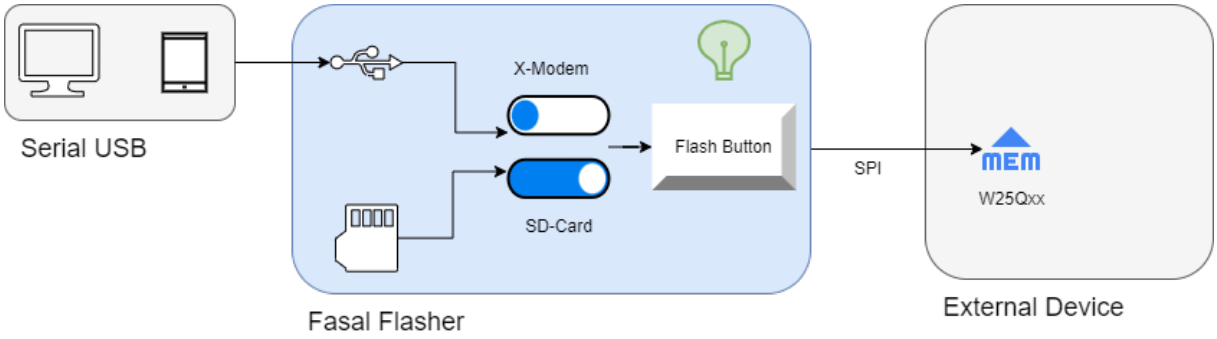


Figure 1.1 Fasal Flasher Usage

1.3 Project-structure

1. The project is divided into the following folders
2. SourceCode/FasalFlasher/Core : Contains CubeMX Generated Files utilizing the STM HAL. These files are mostly unmodified and used as is
3. SourceCode/FasalFlasher/Drivers : STM HAL files
4. SourceCode/FasalFlasher/FATFS : FatFS library
5. SourceCode/FasalFlasher/Middlewares : Contains STM crypto library and FatFS library, used as is without any modifications
6. SourceCode/FasalFlasher/User_Files : All custom code for the Fasal Application is present in this folder
   (a) SourceCode/FasalFlasher/User_Files/AppCommon : Contains modules used across the whole application, these include
      i. SourceCode/FasalFlasher/User_Files/AppCommon/AppIndication : Controls visual indication presented by the board
      ii. SourceCode/FasalFlasher/User_Files/AppCommon/AppUtility : Provides SoftTimer, Circular Buffer and other Misc utility functions
      iii. SourceCode/FasalFlasher/User_Files/AppCommon/CommonDefinitions : Non project specific data structures and Macros
      iv. SourceCode/FasalFlasher/User_Files/AppCommon/ConfigSetting : Module to check Board HW configuration
      v. SourceCode/FasalFlasher/User_Files/AppCommon/PushButton : PushButton module
   (b) SourceCode/FasalFlasher/User_Files/AppConfiguration : Application configuration options are controlled here
   (c) SourceCode/FasalFlasher/User_Files/AppFasal : Application code module
      i. SourceCode/FasalFlasher/User_Files/AppFasal/AppHelper : Helper functions for main application
      ii. SourceCode/FasalFlasher/User_Files/AppFasal/AppResetAndError : Error handling and reset management
   (d) SourceCode/FasalFlasher/User_Files/AppInterface : Modules that help Interface Sensor Adapter with external peripherals
      i. SourceCode/FasalFlasher/User_Files/AppInterface/Console : Debug console for getting controlling as well as reading logs
      ii. SourceCode/FasalFlasher/User_Files/AppInterface/xModem : xModem module built on top of UART based console module

(e) SourceCode/FasalFlasher/User_Files/AppInterrupts : Callbacks and other application specific callbacks coupled closely with board specific peripherals

(f) SourceCode/FasalFlasher/User_Files/AppStorage : OnBoard storage module API

i. SourceCode/FasalFlasher/User_Files/AppStorage/AppFlashFS : Filesystem based on LittleFS file system built atop W25Qxx flash IC

ii. SourceCode/FasalFlasher/User_Files/AppStorage/AppSDFS : Filesystem based on FatFS file system built atop SPI based SD-Card

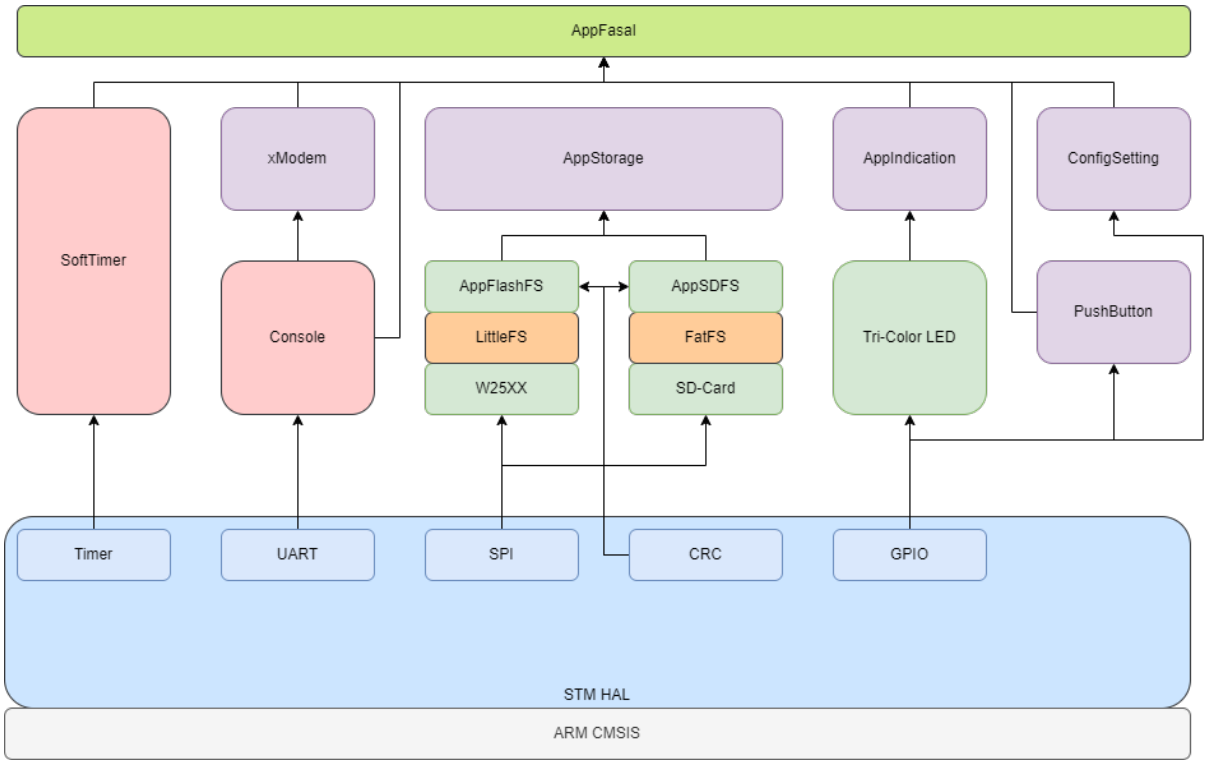


Figure 1.2 Application

```
+---BUILD_DEBUG
+---BUILD_PROD
+---Core
¦   +---Inc
¦   +---Src
¦   +---Startup
+---Drivers
¦   +---CMSIS
¦   ¦   +---Device
¦   ¦   ¦   +---ST
¦   ¦   ¦       +---STM32F1xx
¦   ¦   ¦           +---Include
¦   ¦   ¦           +---Source
¦   ¦   ¦               +---Templates
¦   ¦   +---Include
¦   +---STM32F1xx_HAL_Driver
¦       +---Inc
¦       ¦   +---Legacy
¦       +---Src
+---FATFS
¦   +---App
¦   +---Target
+---Middlewares
¦   +---Third_Party
¦       +---FatFs
¦           +---src
¦               +---option
+---User_Files
    +---AppCommon
    ¦   +---AppIndication
    ¦   ¦   +---TriColorLED
    ¦   +---AppUtility
    ¦   ¦   +---AppProfiler
    ¦   ¦   +---SoftTimer
```

```
¦   +---CommonDefinitions
¦   +---ConfigSetting
¦   +---PushButton
+---AppConfiguration
+---AppFasal
¦   +---AppResetAndError
+---AppInterface
¦   +---Console
¦   +---xModem
+---AppInterrupts
+---AppStorage
    +---AppFlashFS
    ¦   +---LittleFS
    ¦   +---W25Qxx
    +---AppSDFS
```

1.4 Code-flow

1. The code flow and the corresponding modules are documented in this section
2. The code starting point is main. All HAL modules are initialized here along with Application level code housed in AppFasal.h
3. AppFasal_Init Initializes the application, sends startup message and sets up the TriColorLED, console and startup timer
4. The function AppFasal_Run is the application state-machine that drives the application
   (a) Module level initialization is carried out in the first step
   (b) The application then waits for the user to press the flash user button
   (c) External flash is then initialized before progressing to the file transfer phase
   (d) Depending on the hardware slide switch setting, the device can then proceed in one of the following modes
      i. SD-Card Transfer Mode:
         A. SD-Card is initialized and it is ensured that the Golden image is present
         B. Golden image is transferred from the SD card to the external flash
         C. CRC of the same file in SD-Card and in the now transferred flash are computed and checked against each other
      ii. XModem Transfer Mode:
         A. File Must be transferred over XModem 1K option though a serial terminal [Baud: 115200, Data: 8b, Stop Bit: 1b]
   (e) In either mode, On successful reception of Golden Image, the Firmware then enters the termination stage, indicates success and waits on the flash user button stage
   (f) In either mode failure of any of the steps prior to successful transfer results in the application state-machine indicating the failure reason and jumping back flash user button stage

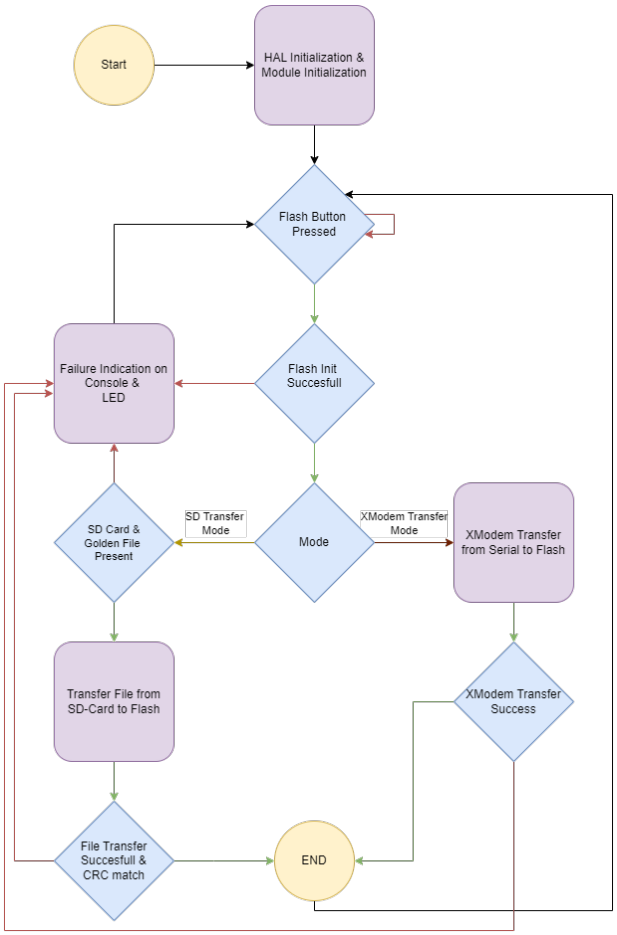


Figure 1.3 APP

1.5 Build-Instruction

1. Built using CubeIDE Version: 1.15.1 Build: 21094_20240412_1041 (UTC)
2. Both BUILD_DEBUG and BUILD_PROD Uses Linker script STM32f103RETX_FLASH.ld with flash offset of 0x0800 0000, application size set to 512KB
3. The following build configurations are built into the codebase
   (a) *BUILD_DEBUG* : Debug build used during development with all debug symbols enabled
      - Optimization level : None
      - Symbols defined : DEBUG | STM32F103xE | USE_HAL_DRIVER

   (b) *BUILD_PROD* : Production build for filed use
      - Optimization level : Ofast
      - Symbols defined : NDEBUG | STM32F103xE | USE_HAL_DRIVER
4. SourceCode/FasalFlasher/User_Files/AppCommon/AppConfiguration for changing compile time build features

1.6 Docs

1. Open Index.html in any browser. Files located here Docs/html
2. Latex Generated PDF file is preset at Docs/Design_Document/FasalNode_FirmwareDesignDocument.pdf
3. The documentation can be generated by running batch script make_designdocument.bat at the project root

1.7 Contact-Me

- vishalbhatta@gmail.com over E-Mail
- LinkedIn

Chapter 2

Data Structure Index

2.1 Data Structures

Here are the data structures with brief descriptions:

Chapter 3

File Index

3.1 File List

Here is a list of all documented files with brief descriptions:

Chapter 4

Data Structure Documentation

4.1 df_dataparam Struct Reference

Data Fields

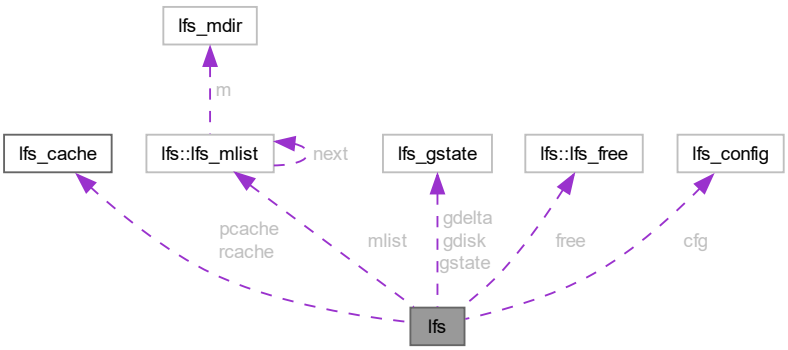
- uint8_t **txbuff** [DF_PAGE_SIZE]
- uint8_t **rxbuff** [DF_PAGE_SIZE]
- uint8_t **txbuff_len**
- uint8_t **rxbuff_len**
- uint8_t **rxbuff_readpos**
- uint8_t **df_mode**

The documentation for this struct was generated from the following file:

- W25Qxx.h

4.2 lfs Struct Reference

Collaboration diagram for lfs:



Data Structures

- struct lfs_free
- struct lfs_mlist

Data Fields

- lfs_cache_t **rcache**
- lfs_cache_t **pcache**
- lfs_block_t **root** [2]
- struct lfs::lfs_mlist * **mlist**
- uint32_t **seed**
- lfs_gstate_t **gstate**
- lfs_gstate_t **gdisk**
- lfs_gstate_t **gdelta**
- struct lfs::lfs_free **free**
- const struct lfs_config * **cfg**
- lfs_size_t **name_max**
- lfs_size_t **file_max**
- lfs_size_t **attr_max**

The documentation for this struct was generated from the following file:

- lfs.h

4.3 lfs_attr Struct Reference

Data Fields

- uint8_t **type**
- void * **buffer**
- lfs_size_t **size**

The documentation for this struct was generated from the following file:

- lfs.h

4.4 lfs_cache Struct Reference

```
#include <lfs.h>
```

Data Fields

- lfs_block_t **block**
- lfs_off_t **off**
- lfs_size_t **size**
- uint8_t * **buffer**

The documentation for this struct was generated from the following file:

- lfs.h

4.5 lfs_commit Struct Reference

Data Fields

- lfs_block_t **block**
- lfs_off_t **off**
- lfs_tag_t **ptag**
- uint32_t **crc**
- lfs_off_t **begin**
- lfs_off_t **end**

The documentation for this struct was generated from the following file:

- lfs.c

4.6 lfs_config Struct Reference

Data Fields

- void ∗ **context**
- int(∗ **read**)(const struct lfs_config ∗c, lfs_block_t block, lfs_off_t off, void ∗buffer, lfs_size_t size)
- int(∗ **prog**)(const struct lfs_config ∗c, lfs_block_t block, lfs_off_t off, const void ∗buffer, lfs_size_t size)
- int(∗ **erase**)(const struct lfs_config ∗c, lfs_block_t block)
- int(∗ **sync**)(const struct lfs_config ∗c)
- lfs_size_t **read_size**
- lfs_size_t **prog_size**
- lfs_size_t **block_size**
- lfs_size_t **block_count**
- int32_t **block_cycles**
- lfs_size_t **cache_size**
- lfs_size_t **lookahead_size**
- void ∗ **read_buffer**
- void ∗ **prog_buffer**
- void ∗ **lookahead_buffer**
- lfs_size_t **name_max**
- lfs_size_t **file_max**
- lfs_size_t **attr_max**
- lfs_size_t **metadata_max**

The documentation for this struct was generated from the following file:

- lfs.h

4.7 lfs_file::lfs_ctz Struct Reference

Data Fields

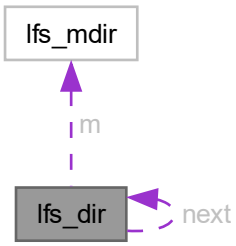
- lfs_block_t **head**
- lfs_size_t **size**

The documentation for this struct was generated from the following file:

- lfs.h

4.8 lfs_dir Struct Reference

Collaboration diagram for lfs_dir:



Data Fields

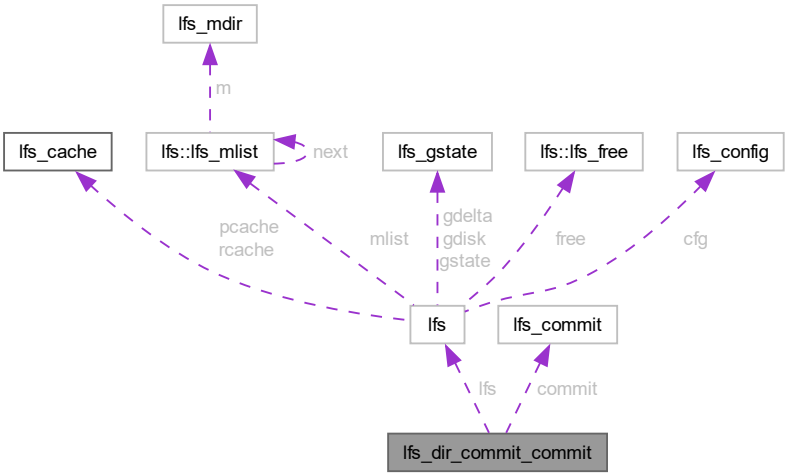
- struct lfs_dir * **next**
- uint16_t **id**
- uint8_t **type**
- lfs_mdir_t **m**
- lfs_off_t **pos**
- lfs_block_t **head** [2]

The documentation for this struct was generated from the following file:

- lfs.h

4.9 lfs_dir_commit_commit Struct Reference

Collaboration diagram for lfs_dir_commit_commit:



Data Fields

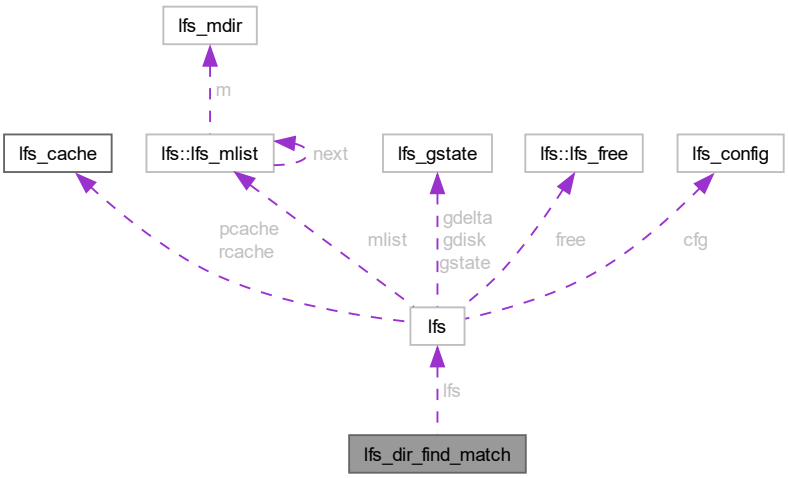
- lfs_t * **lfs**
- struct lfs_commit * **commit**

The documentation for this struct was generated from the following file:

- lfs.c

4.10 lfs_dir_find_match Struct Reference

Collaboration diagram for lfs_dir_find_match:



Data Fields

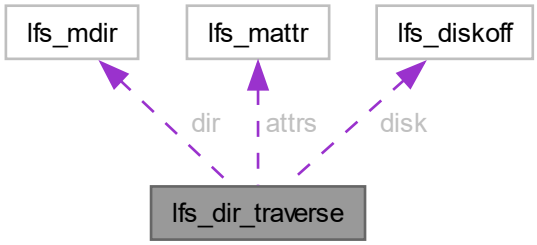
- lfs_t * **lfs**
- const void * **name**
- lfs_size_t **size**

The documentation for this struct was generated from the following file:

- lfs.c

4.11 lfs_dir_traverse Struct Reference

Collaboration diagram for lfs_dir_traverse:



Data Fields

- const lfs_mdir_t * **dir**
- lfs_off_t **off**
- lfs_tag_t **ptag**
- const struct lfs_mattr * **attrs**
- int **attrcount**
- lfs_tag_t **tmask**
- lfs_tag_t **ttag**
- uint16_t **begin**
- uint16_t **end**
- int16_t **diff**
- int(* **cb**)(void *data, lfs_tag_t tag, const void *buffer)
- void * **data**
- lfs_tag_t **tag**
- const void * **buffer**
- struct lfs_diskoff **disk**

The documentation for this struct was generated from the following file:

- lfs.c

4.12 lfs_diskoff Struct Reference

Data Fields

- lfs_block_t **block**
- lfs_off_t **off**

The documentation for this struct was generated from the following file:

- lfs.c

4.13 lfs_fcrc Struct Reference

Data Fields

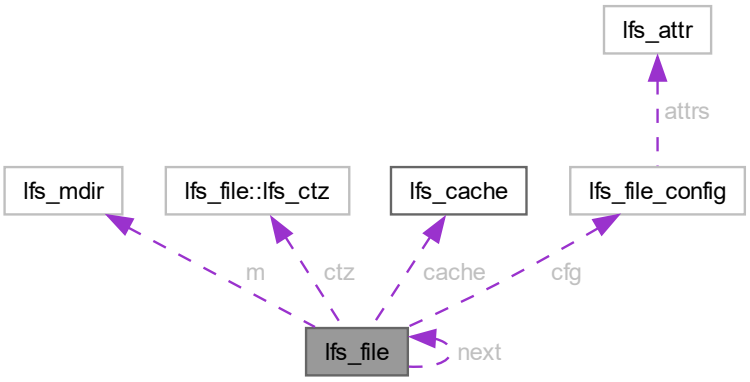
- lfs_size_t **size**
- uint32_t **crc**

The documentation for this struct was generated from the following file:

- lfs.c

4.14 lfs_file Struct Reference

Collaboration diagram for lfs_file:



Data Structures

- struct lfs_ctz

Data Fields

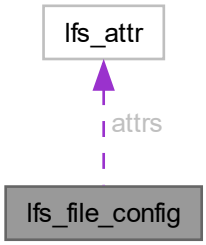
- struct lfs_file * **next**
- uint16_t **id**
- uint8_t **type**
- lfs_mdir_t **m**
- struct lfs_file::lfs_ctz **ctz**
- uint32_t **flags**
- lfs_off_t **pos**
- lfs_block_t **block**
- lfs_off_t **off**
- lfs_cache_t **cache**
- const struct lfs_file_config * **cfg**

The documentation for this struct was generated from the following file:

- lfs.h

4.15 lfs_file_config Struct Reference

Collaboration diagram for lfs_file_config:



Data Fields

- void * **buffer**
- struct lfs_attr * **attrs**
- lfs_size_t **attr_count**

The documentation for this struct was generated from the following file:

- lfs.h

4.16 lfs::lfs_free Struct Reference

Data Fields

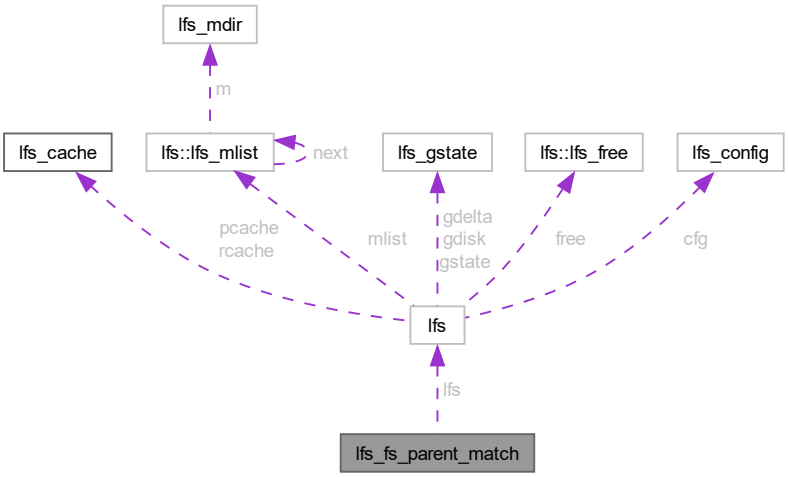
- lfs_block_t **off**
- lfs_block_t **size**
- lfs_block_t **i**
- lfs_block_t **ack**
- uint32_t * **buffer**

The documentation for this struct was generated from the following file:

- lfs.h

4.17 lfs_fs_parent_match Struct Reference

Collaboration diagram for lfs_fs_parent_match:



Data Fields

- lfs_t * **lfs**
- const lfs_block_t **pair** [2]

The documentation for this struct was generated from the following file:

- lfs.c

4.18 lfs_fsinfo Struct Reference

Data Fields

- uint32_t **disk_version**
- lfs_size_t **name_max**
- lfs_size_t **file_max**
- lfs_size_t **attr_max**

The documentation for this struct was generated from the following file:

- lfs.h

4.19 lfs_gstate Struct Reference

Data Fields

- uint32_t **tag**
- lfs_block_t **pair** [2]

The documentation for this struct was generated from the following file:

- lfs.h

4.20 lfs_info Struct Reference

Data Fields

- uint8_t **type**
- lfs_size_t **size**
- char **name** [LFS_NAME_MAX+1]

The documentation for this struct was generated from the following file:

- lfs.h

4.21 lfs_mattr Struct Reference

Data Fields

- lfs_tag_t **tag**
- const void * **buffer**

The documentation for this struct was generated from the following file:

- lfs.c

4.22 lfs_mdir Struct Reference

Data Fields

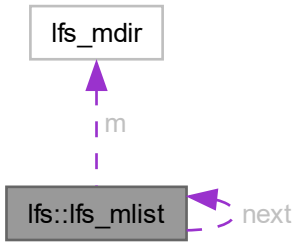
- lfs_block_t **pair** [2]
- uint32_t **rev**
- lfs_off_t **off**
- uint32_t **etag**
- uint16_t **count**
- bool **erased**
- bool **split**
- lfs_block_t **tail** [2]

The documentation for this struct was generated from the following file:

- lfs.h

4.23 lfs::lfs_mlist Struct Reference

Collaboration diagram for lfs::lfs_mlist:



Data Fields

- struct lfs_mlist * **next**
- uint16_t **id**
- uint8_t **type**
- lfs_mdir_t **m**

The documentation for this struct was generated from the following file:

- lfs.h

4.24 lfs_superblock Struct Reference

Data Fields

- uint32_t **version**
- lfs_size_t **block_size**
- lfs_size_t **block_count**
- lfs_size_t **name_max**
- lfs_size_t **file_max**
- lfs_size_t **attr_max**

The documentation for this struct was generated from the following file:

- lfs.h

4.25 sConsoleCommand_t Struct Reference

```
#include <Console.h>
```

Data Fields

- const char * **CommandName**
- char **CommandStr** [BUFFER_SIZE_8]
- volatile bool **IsCommandRaised**
- volatile bool **IsCommandServiced**
- pfConsoleCommandActor_t **pfConsoleCommandActor**

4.25.1 Detailed Description

The documentation for this struct was generated from the following files:

- AppCommon/Console/Console.h
- AppInterface/Console/Console.h

4.26 sElevatedPromptData_t Struct Reference

```
#include <Console.h>
```

Data Fields

- char **buf** [CONSOLE_PROMPT_BUFFER_SIZE]

4.26.1 Detailed Description

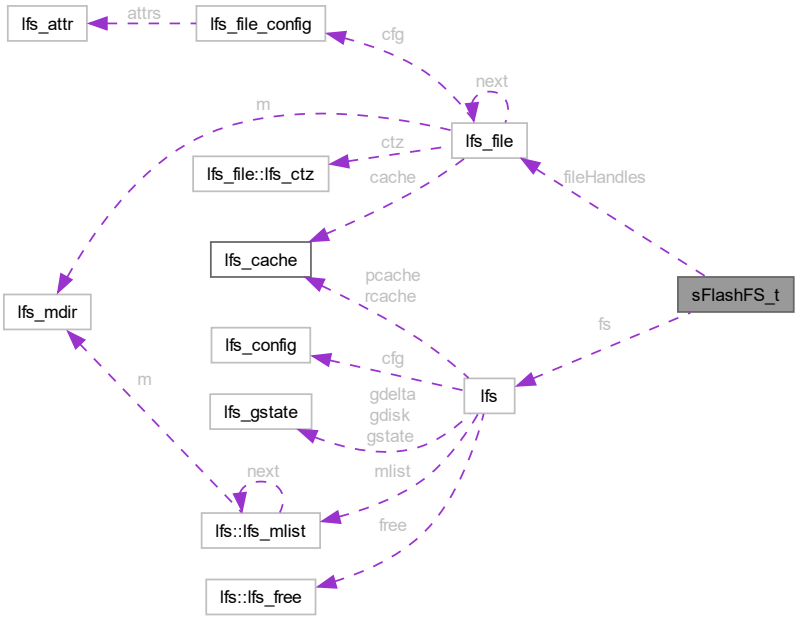
The documentation for this struct was generated from the following files:

- AppCommon/Console/Console.h
- AppInterface/Console/Console.h

4.27 sFlashFS_t Struct Reference

```
#include <AppFlash_API.h>
```

Collaboration diagram for sFlashFS_t:



Data Fields

- bool **IsMounted**
- lfs_t **fs**
- lfs_file_t **fileHandles** [eFS_MAX]

4.27.1 Detailed Description

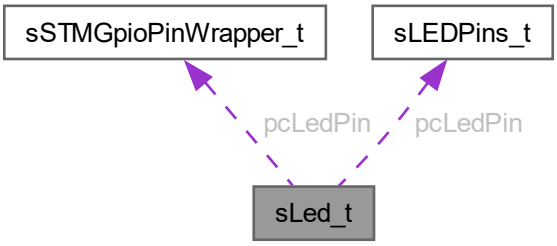
The documentation for this struct was generated from the following file:

- AppFlash_API.h

4.28 sLed_t Struct Reference

```
#include <TriColorLED.h>
```

Collaboration diagram for sLed_t:



Data Fields

- const sSTMGpioPinWrapper_t * **pcLedPin**
- eLEDState_t **ledState**
- const sLEDPins_t * **pcLedPin**

4.28.1 Detailed Description

The documentation for this struct was generated from the following files:

- AppIndication/TriColorLED/TriColorLED.h
- TriColorLED/TriColorLED.h

4.29 sLEDPins_t Struct Reference

`#include <TriColorLED.h>`

Data Fields

- GPIO_TypeDef * **LEDPort**
- uint16_t **LEDPin**

4.29.1 Detailed Description

The documentation for this struct was generated from the following file:

- TriColorLED/TriColorLED.h

4.30 sSDFS_t Struct Reference

`#include <AppSD_API.h>`

Data Fields

- bool **IsMounted**
- FATFS **fs**
- FIL **fileHandles** [eFS_MAX]

4.30.1 Detailed Description

The documentation for this struct was generated from the following file:

- AppSD_API.h

4.31 sSoftTimer_t Struct Reference

`#include <SoftTimer.h>`

Data Fields

- bool **IsArmed**
- bool **IsPeriodic**
- uint32_t **currentTicks**
- uint32_t **setTicks**
- pfCallBack_t **pfCallBack**

4.31.1 Detailed Description

The documentation for this struct was generated from the following file:

- SoftTimer.h

4.32 sSTMGpioPinWrapper_t Struct Reference

`#include <CommonDefinitions.h>`

Data Fields

- GPIO_TypeDef * **GpioPort**
- uint16_t **GpioPin**

4.32.1 Detailed Description

The documentation for this struct was generated from the following file:

- CommonDefinitions.h

4.33 sTriColorIndication_t Struct Reference

`#include <TriColorLED.h>`

Data Fields

- bool **IsBlinkNeeded**
- eTriColorLEDStates_t **ledState**
- eTriColorLEDBlinkPeriod_t **ledBlinkPeriod**

4.33.1 Detailed Description

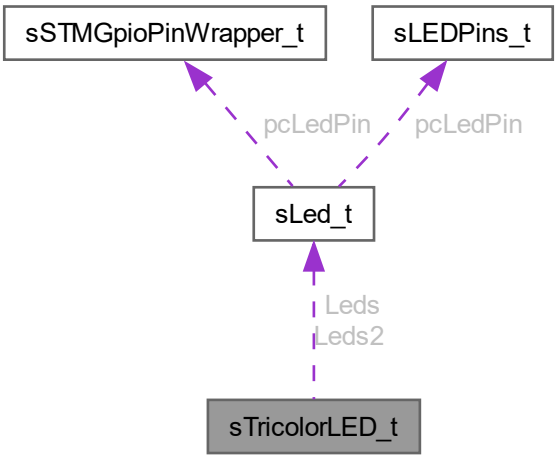
The documentation for this struct was generated from the following files:

- AppIndication/TriColorLED/TriColorLED.h
- TriColorLED/TriColorLED.h

4.34 sTricolorLED_t Struct Reference

`#include <TriColorLED.h>`

Collaboration diagram for sTricolorLED_t:



Data Fields

- bool **IsInitialized**
- bool **IsBlinkEnabled**
- size_t **blinkTicksSet**
- size_t **blinkTicksCurrent**
- eTriColorLEDStates_t **triColorLEDState**
- eTriColorLEDStates_t **prevOnLEDState**
- sLed_t **Leds** [eLED_MAX]
- sLed_t Leds2 [eLED_MAX]

4.34.1 Detailed Description

4.34.2 Field Documentation

4.34.2.1 Leds2

```
sLed_t sTricolorLED_t::Leds2
```

Duplicate set of LED pins that shows the same indication.

Duplicate set of LED pins that shows the same indication

The documentation for this struct was generated from the following files:

- AppIndication/TriColorLED/TriColorLED.h
- TriColorLED/TriColorLED.h

4.35 sTriColorLEDState_t Struct Reference

```
#include <TriColorLED.h>
```

Data Fields

- eLEDState_t **LedState** [eLED_MAX]

4.35.1 Detailed Description

The documentation for this struct was generated from the following files:

- AppIndication/TriColorLED/TriColorLED.h
- TriColorLED/TriColorLED.h

4.36 w25qxx_t Struct Reference

Data Fields

- W25QXX_ID_t **ID**
- uint8_t **UniqID** [8]
- uint16_t **PageSize**
- uint32_t **PageCount**
- uint32_t **SectorSize**
- uint32_t **SectorCount**
- uint32_t **BlockSize**
- uint32_t **BlockCount**
- uint32_t **CapacityInKiloByte**
- uint8_t **StatusRegister1**
- uint8_t **StatusRegister2**
- uint8_t **StatusRegister3**
- uint8_t **Lock**

The documentation for this struct was generated from the following file:

- W25Qxx.h

Chapter 5

File Documentation

5.1 AppCommon.c File Reference

```
#include <stdint.h>
#include ¨stm32f1xx_hal.h¨
#include ¨AppCommon.h¨
#include ¨Version.h¨
#include ¨AppIndication.h¨
#include ¨Console.h¨
#include ¨AppConfiguration.h¨
```

Functions

- __attribute__ ((unused))
- static void AppCommon_PreResetRoutine ()
- eDeviceErrorCode_t AppCommon_GetErrorCode ()
- void AppCommon_AccumlateErrorCode (eDeviceErrorCode_t err)
- void AppCommon_ResetErrorCode ()
- void AppCommon_DetermineResetCause ()
- eDeviceResetCause_t AppCommon_GetCachedResetCause ()
- void AppCommon_HALErrorHandler ()
- void AppCommon_STMFaultHandler ()
- void __assert_func (const char *file, int line, const char *func, const char *failedexpr)
- void AppCommon_PrintDelimiter ()
- void AppCommon_PrintLineBreak ()
- void AppCommon_StartUpMessagePrint ()
- const char *const AppCommon_GetStatusString (int status)

Variables

- static eDeviceResetCause_t gDeviceResetCause = eRESET_CAUSE_UNKNOWN
- static eDeviceErrorCode_t gErrorCode = eERR_NO_ERRORS
- static const char * gcStatusStringHelper [eSTATUS_STRING_MAX]

5.1.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-06-03

Copyright

5.1.2 Function Documentation

5.1.2.1 __attribute__()

```
__attribute__ (
             (unused)  )
```

convert eDeviceResetCause_t enum to strings for printing

Delete File.

Write to a file in FatFS.

Parameters

pMe	FatFS wrapper instance
fileEnum	File to be written into
IsAppend	Pass true to write to file in append mode, false to write at the beginning
pInWriteBuf	data to be written is passed here
bufSize	size of buffer passed for writing

Returns

eStorageFSStatus_t

Parameters

pMe	FatFS wrapper instance
fileEnum	File to be deleted

Returns

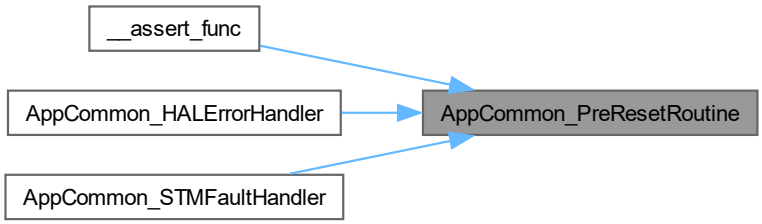
eStorageFSStatus_t

5.1.2.2 AppCommon_PreResetRoutine()

```
static void AppCommon_PreResetRoutine ( )  [static]
```

Function to be invoked prior to resetting the system to recover from an hard fault/error.

< Device has been reset so set up-time back to ZeroHere is the caller graph for this function:



5.1.2.3 AppCommon_GetErrorCode()

```
eDeviceErrorCode_t AppCommon_GetErrorCode ( )
```

Setter for gErrorCode, previous values are unaffected.

Returns

eDeviceErrorCode_t

Here is the caller graph for this function:

5.1.2.4 AppCommon_AccumlateErrorCode()

```
void AppCommon_AccumlateErrorCode (
            eDeviceErrorCode_t err )
```

Setter for gErrorCode, previous values are unaffected.

Here is the caller graph for this function:

5.1.2.5 AppCommon_ResetErrorCode()

```
void AppCommon_ResetErrorCode ( )
```

Reset error Code.

Here is the caller graph for this function:

5.1.2.6 AppCommon_DetermineResetCause()

```
void AppCommon_DetermineResetCause ( )
```

Function to determine why the device was reset, also sets global gDeviceResetCause.

Note

should be called before peripheral initialization

5.1.2.7 AppCommon_GetCachedResetCause()

```
eDeviceResetCause_t AppCommon_GetCachedResetCause ( )
```

Getter for gDeviceResetCause.

Note

Assumes AppCommon_DetermineResetCause is called at startup

Returns

eDeviceResetCause_t

5.1.2.8 AppCommon_HALErrorHandler()

```
void AppCommon_HALErrorHandler ( )
```

STM HAL error handler.

Warning

This function resets the device

Here is the call graph for this function:

5.1.2.9 AppCommon_STMFaultHandler()

```
void AppCommon_STMFaultHandler ( )
```

STM Fault error handler.

Warning

This function resets the device

Here is the call graph for this function:

5.1.2.10 __assert_func()

```
void __assert_func (
            const char * file,
            int line,
            const char * func,
            const char * failedexpr )
```

Assert failure error handler.

Warning

This function resets the device

Here is the call graph for this function:

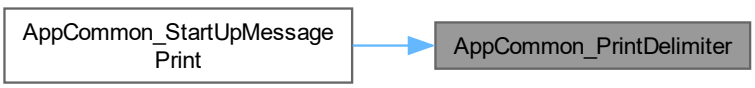
5.1.2.11 AppCommon_PrintDelimiter()

```
void AppCommon_PrintDelimiter ( )
```

Utility function to print delimiter used in the console.

Here is the call graph for this function:

Here is the caller graph for this function:



5.1.2.12 AppCommon_PrintLineBreak()

```
void AppCommon_PrintLineBreak ( )
```

Utility function to print delimiter used in the console.

Here is the call graph for this function:

5.1.2.13 AppCommon_StartUpMessagePrint()

```
void AppCommon_StartUpMessagePrint ( )
```

Start up message print.

Here is the call graph for this function:

5.1.2.14 AppCommon_GetStatusString()

```
const char *const AppCommon_GetStatusString (
            int status )
```

Utility function returns ¨Success¨ or ¨failure¨ based on status code passed.

See also

gcStatusStringHelper

Parameters

status	0 is success across all modules in the code-base , non 0 is treated as failure

Returns

const char* const

Here is the caller graph for this function:

5.1.3 Variable Documentation

5.1.3.1 gDeviceResetCause

```
eDeviceResetCause_t gDeviceResetCause = eRESET_CAUSE_UNKNOWN  [static]
```

Global that maintains reset cause

5.1.3.2 gErrorCode

```
eDeviceErrorCode_t gErrorCode = eERR_NO_ERRORS  [static]
```

Global that maintains error cause

5.1.3.3 gcStatusStringHelper

```
const char* gcStatusStringHelper[eSTATUS_STRING_MAX]  [static]
```

Initial value:

```
=
{
        [eSUCCESS_STRING] = "Success",
        [eFAILURE_STRING] = "Failure"
}
```

Map eStatusPrintHelperEnum_t to string. Used by AppCommon_GetStatusString.

5.2 AppCommon.h File Reference

Macros

- #define LOOP_FOREVER {while(1);}

Enumerations

- enum eStatusPrintHelperEnum_t { **eSUCCESS_STRING** , **eFAILURE_STRING** , **eSTATUS_STRING_MAX** }
- enum eDeviceResetCause_t {
  eRESET_CAUSE_UNKNOWN = 0 , **eRESET_CAUSE_WAKEUP** , **eRESET_CAUSE_LOW_POWER_RESET** , **eRESET_CAUSE_WINDOW_WATCHDOG_RESET** ,
  eRESET_CAUSE_INDEPENDENT_WATCHDOG_RESET , **eRESET_CAUSE_SOFTWARE_RESET** , **eRESET_CAUSE_POWER_ON_POWER_DOWN_RESET** , **eRESET_CAUSE_EXTERNAL_RESET_PIN_RESET** ,
  eRESET_CAUSE_BROWNOUT_RESET , **eRESET_CAUSE_MAX** }
- enum eDeviceErrorCode_t {
  eERR_NO_ERRORS = 0x0000 , **eERR_SDCARD_NOT_FOUND** = 0x0001 , **eERR_SDCARD_FILE_NOT_FOUND** = 0x0002 , **eERR_FLASH_NOT_FOUND** = 0x0004 ,
  eERR_FLASH_TRANSFER_FAILURE = 0x0008 , **eERR_CRC_FAILURE** = 0x0010 , **eERR_UNUSED_1** = 0x0020 , **eERR_UNUSED_2** = 0x0040 ,
  eERR_UNUSED_3 = 0x0080 , **eERR_UNUSED_4** = 0x0100 , **eERR_UNUSED_5** = 0X0200 , **eERR_UNUSED_6** = 0x0400 ,
  eERR_UNUSED_7 = 0x0800 , **eERR_STM_HAL_FAILURE** = 0x1000 , **eERR_ARM_FAULT** = 0x2000 , **eERR_ASSERTION_FAILURE** = 0x4000 ,
  eERR_SLEEP_FAILURE = 0x8000 }

Functions

- void AppCommon_PrintDelimiter ()
- void AppCommon_PrintLineBreak ()
- void AppCommon_StartUpMessagePrint ()
- void AppCommon_HALErrorHandler ()
- void AppCommon_STMFaultHandler ()
- const char ∗const AppCommon_GetStatusString (int status)
- void __assert_func (const char ∗file, int line, const char ∗func, const char ∗failedexpr)
- void AppCommon_DetermineResetCause ()
- eDeviceResetCause_t AppCommon_GetCachedResetCause ()
- eDeviceErrorCode_t AppCommon_GetErrorCode ()
- void AppCommon_AccumlateErrorCode (eDeviceErrorCode_t err)
- void AppCommon_ResetErrorCode ()

5.2.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-06-03

Copyright

5.2.2 Macro Definition Documentation

5.2.2.1 LOOP_FOREVER

```
#define LOOP_FOREVER {while(1);}
```

Utility MACRO to loop forever

5.2.3 Enumeration Type Documentation

5.2.3.1 eStatusPrintHelperEnum_t

```
enum eStatusPrintHelperEnum_t
```

Enum for status helper.

5.2.3.2 eDeviceResetCause_t

```
enum eDeviceResetCause_t
```

Possible STM32 system reset causes.

5.2.3.3 eDeviceErrorCode_t

```
enum eDeviceErrorCode_t
```

Possible Errors in application.

5.2.4 Function Documentation

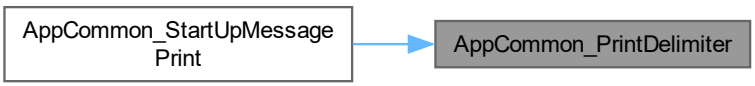
5.2.4.1 AppCommon_PrintDelimiter()

```
void AppCommon_PrintDelimiter ( )
```

Utility function to print delimiter used in the console.

Here is the call graph for this function:

Here is the caller graph for this function:



5.2.4.2 AppCommon_PrintLineBreak()

```
void AppCommon_PrintLineBreak ( )
```

Utility function to print delimiter used in the console.

Here is the call graph for this function:

5.2.4.3 AppCommon_StartUpMessagePrint()

```
void AppCommon_StartUpMessagePrint ( )
```

Start up message print.

Here is the call graph for this function:

5.2.4.4 AppCommon_HALErrorHandler()

```
void AppCommon_HALErrorHandler ( )
```

STM HAL error handler.

Warning

This function resets the device

Here is the call graph for this function:

5.2.4.5 AppCommon_STMFaultHandler()

```
void AppCommon_STMFaultHandler ( )
```

STM Fault error handler.

Warning

This function resets the device

Here is the call graph for this function:

5.2.4.6 AppCommon_GetStatusString()

```
const char *const AppCommon_GetStatusString (
            int status )
```

Utility function returns ¨Success¨ or ¨failure¨ based on status code passed.

See also

gcStatusStringHelper

Parameters

status	0 is success across all modules in the code-base , non 0 is treated as failure

Returns

const char* const

5.2.4.7 __assert_func()

```
void __assert_func (
            const char * file,
            int line,
            const char * func,
            const char * failedexpr )
```

Assert failure error handler.

Warning

This function resets the device

5.2.4.8 AppCommon_DetermineResetCause()

```
void AppCommon_DetermineResetCause ( )
```

Function to determine why the device was reset, also sets global gDeviceResetCause.

Note

should be called before peripheral initialization

5.2.4.9 AppCommon_GetCachedResetCause()

```
eDeviceResetCause_t AppCommon_GetCachedResetCause ( )
```

Getter for gDeviceResetCause.

Note

Assumes AppCommon_DetermineResetCause is called at startup

Returns

eDeviceResetCause_t

5.2.4.10 AppCommon_GetErrorCode()

```
eDeviceErrorCode_t AppCommon_GetErrorCode ( )
```

Setter for gErrorCode, previous values are unaffected.

Returns

eDeviceErrorCode_t

Returns

eDeviceErrorCode_t

5.2.4.11 AppCommon_AccumlateErrorCode()

```
void AppCommon_AccumlateErrorCode (
            eDeviceErrorCode_t err )
```

Setter for gErrorCode, previous values are unaffected.

5.2.4.12 AppCommon_ResetErrorCode()

```
void AppCommon_ResetErrorCode ( )
```

Reset error Code.

5.3 AppCommon.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCOMMON_APPCOMMON_H_
00015 #define APPCOMMON_APPCOMMON_H_
00016
00018
00019 #define LOOP_FOREVER {while(1);}
00022
00027 typedef enum
00028 {
00029     eSUCCESS_STRING,
00030     eFAILURE_STRING,
00031     eSTATUS_STRING_MAX
00032 }eStatusPrintHelperEnum_t;
00033
00038 typedef enum
00039 {
00040     eRESET_CAUSE_UNKNOWN = 0,
00041     eRESET_CAUSE_WAKEUP,
00042     eRESET_CAUSE_LOW_POWER_RESET,
00043     eRESET_CAUSE_WINDOW_WATCHDOG_RESET,
00044     eRESET_CAUSE_INDEPENDENT_WATCHDOG_RESET,
00045     eRESET_CAUSE_SOFTWARE_RESET,
00046     eRESET_CAUSE_POWER_ON_POWER_DOWN_RESET,
00047     eRESET_CAUSE_EXTERNAL_RESET_PIN_RESET,
00048     eRESET_CAUSE_BROWNOUT_RESET,
00049     eRESET_CAUSE_MAX
00050 } eDeviceResetCause_t;
00051
00052
00057 typedef enum
00058 {
00059     eERR_NO_ERRORS                  = 0x0000,
00060
00061     eERR_SDCARD_NOT_FOUND           = 0x0001,
00062     eERR_SDCARD_FILE_NOT_FOUND      = 0x0002,
00063
00064     eERR_FLASH_NOT_FOUND            = 0x0004,
00065     eERR_FLASH_TRANSFER_FAILURE     = 0x0008,
00066
00067     eERR_CRC_FAILURE                = 0x0010,
00068
00069     eERR_UNUSED_1                   = 0x0020,
00070     eERR_UNUSED_2                   = 0x0040,
00071     eERR_UNUSED_3                   = 0x0080,
00072     eERR_UNUSED_4                   = 0x0100,
00073     eERR_UNUSED_5                   = 0X0200,
00074     eERR_UNUSED_6                   = 0x0400,
00075     eERR_UNUSED_7                   = 0x0800,
00076
00077     eERR_STM_HAL_FAILURE            = 0x1000,
00078     eERR_ARM_FAULT                  = 0x2000,
00079     eERR_ASSERTION_FAILURE          = 0x4000,
00080     eERR_SLEEP_FAILURE              = 0x8000,
00081
00082 }eDeviceErrorCode_t;
00083
00084
00086
00087
00088 void AppCommon_PrintDelimiter();
00089 void AppCommon_PrintLineBreak();
00090 void AppCommon_StartUpMessagePrint();
00091 void AppCommon_HALErrorHandler();
00092 void AppCommon_STMFaultHandler();
00093 const char* const AppCommon_GetStatusString(int status);
00094 void __assert_func(const char *file, int line, const char *func, const char *failedexpr);
00095
00096 void AppCommon_DetermineResetCause();
```

```
00097 eDeviceResetCause_t AppCommon_GetCachedResetCause();
00098
00099 eDeviceErrorCode_t AppCommon_GetErrorCode();
00100 void AppCommon_AccumlateErrorCode(eDeviceErrorCode_t err);
00101 void AppCommon_ResetErrorCode();
00102
00104
00105 #endif /* APPCOMMON_APPCOMMON_H_ */
```

5.4 AppConfiguration.c File Reference

`#include ¨AppConfiguration.h¨`

5.4.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

5.5 AppConfiguration.c File Reference

`#include ¨AppConfiguration.h¨`

5.5.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

5.6 AppConfiguration.h File Reference

5.6.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

5.7 AppCommon/AppConfiguration/AppConfiguration.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCONFIGURATION_APPCONFIGURATION_H_
00015 #define APPCONFIGURATION_APPCONFIGURATION_H_
00016
00018
00021 //#define ENABLE_DEBUG_PRINT               /**< Enabling this macro will redirect @ref DEBUG_PRINT to
      @ref Console_Print */
00022 //#define ENABLE_TESTS_DEFINITIONS         /**< If this is enabled then tests defined for individual
      modules are defined*/
00023 //#define FORCE_DISABLE_FILE_CRC_CHECK     /**< CRC of the SD card an Flash file copy will be
      computed and compared by default, define this variable to skip CRC check*/
00024
00025
00027
00028
00029 #endif /* APPCONFIGURATION_APPCONFIGURATION_H_ */
```

5.8 AppConfiguration.h File Reference

5.8.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

5.9 AppConfiguration/AppConfiguration.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCONFIGURATION_APPCONFIGURATION_H_
00015 #define APPCONFIGURATION_APPCONFIGURATION_H_
00016
00018
00020
00022 // #define ENABLE_DEBUG_PRINT
00023
00025 // #define ENABLE_TESTS_DEFINITIONS
00026
00028 // #define FORCE_DISABLE_FILE_CRC_CHECK
00029
00031
00032 #endif /* APPCONFIGURATION_APPCONFIGURATION_H_ */
```

5.10 Version.h File Reference

```
#include <stdio.h>
```

Macros

- #define **FW_VERSION_MAJOR** (1)
- #define **FW_VERSION_MINOR** (0)
- #define **HARDWARE_VERSION_MAJOR** (1)
- #define **HARDWARE_VERSION_MINOR** (1)
- #define **COMPILE_DATE** __DATE__
- #define **COMPILE_TIME** __TIME__

5.10.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

Copyright (c) 2023

5.11 AppCommon/AppConfiguration/Version.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCONFIGURATION_VERSION_H_
00015 #define APPCONFIGURATION_VERSION_H_
00016
00018
00019 #include <stdio.h>
00020
00022
00023 #define FW_VERSION_MAJOR        (1)
00024 #define FW_VERSION_MINOR        (0)
00025
00026 #define HARDWARE_VERSION_MAJOR  (1)
00027 #define HARDWARE_VERSION_MINOR  (1)
00028
00029 #define COMPILE_DATE            __DATE__
00030 #define COMPILE_TIME            __TIME__
00031
00033
00034 #endif /* APPCONFIGURATION_VERSION_H_ */
```

5.12 Version.h File Reference

Macros

- #define **FW_VERSION_MAJOR** (1)
- #define **FW_VERSION_MINOR** (1)
- #define **HARDWARE_VERSION_MAJOR** (1)
- #define **HARDWARE_VERSION_MINOR** (1)
- #define **COMPILE_DATE** __DATE__
- #define **COMPILE_TIME** __TIME__

5.12.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

5.13 AppConfiguration/Version.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCONFIGURATION_VERSION_H_
00015 #define APPCONFIGURATION_VERSION_H_
00016
00018
00019 #define FW_VERSION_MAJOR (1)
00020 #define FW_VERSION_MINOR (1)
00021
00022 #define HARDWARE_VERSION_MAJOR (1)
00023 #define HARDWARE_VERSION_MINOR (1)
00024
00025 #define COMPILE_DATE __DATE__
00026 #define COMPILE_TIME __TIME__
00027
00029
00030 #endif /* APPCONFIGURATION_VERSION_H_ */
```

5.14 AppIndication.c File Reference

```
#include <assert.h>
#include ¨AppIndication.h¨
#include ¨TriColorLED.h¨
```

Functions

- void AppIndicate_Init ()
- void AppIndicate_DeInit ()
- void AppIndicate_SetState (eAppIndicationStates_t state)
- void AppIndicate_RevertState ()

Variables

- static sTriColorIndication_t gcAppIndicationTable [eIND_MAX]
- static eAppIndicationStates_t gCurrentState = eIND_NONE
- static eAppIndicationStates_t gPrevState = eIND_NONE

5.14.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-05-09

Copyright

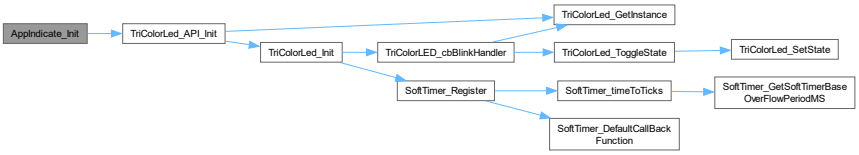
5.14.2 Function Documentation

5.14.2.1 AppIndicate_Init()

```
void AppIndicate_Init ( )
```

Wrapper around Indication medium initialization.

Here is the call graph for this function:



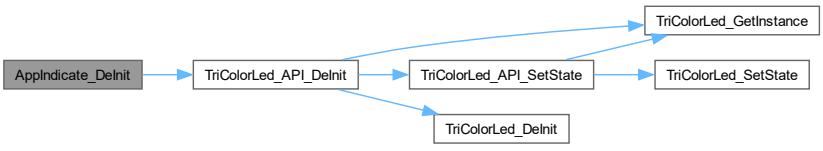
Here is the caller graph for this function:

5.14.2.2 AppIndicate_DeInit()

```
void AppIndicate_DeInit ( )
```

Wrapper around Indication medium De-initialization.

Here is the call graph for this function:



5.14.2.3 AppIndicate_SetState()

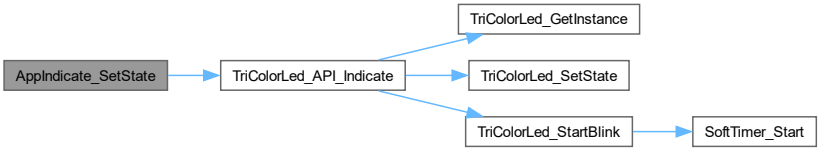
```
void AppIndicate_SetState (
            eAppIndicationStates_t state )
```

Set Indication medium state.

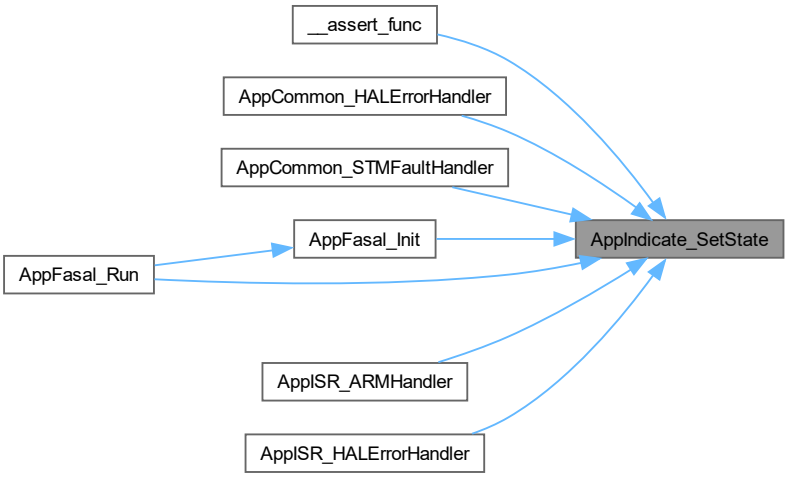
Parameters

state	Which application state to set to, gcAppIndicationTable

Here is the call graph for this function:



Here is the caller graph for this function:

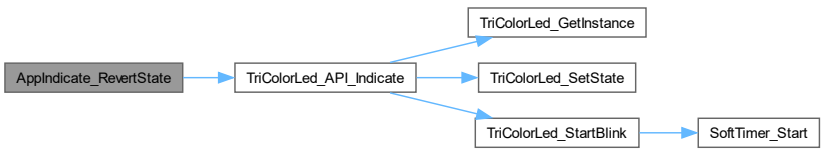


5.14.2.4 AppIndicate_RevertState()

```
void AppIndicate_RevertState ( )
```

Revert to previous indication state. For preserving previous Indication state.

Here is the call graph for this function:



5.14.3 Variable Documentation

5.14.3.1 gcAppIndicationTable

```
sTriColorIndication_t gcAppIndicationTable[eIND_MAX]  [static]
```

Configuration table that determine how application states correspond to TriColor LED states.

5.14.3.2 gCurrentState

```
eAppIndicationStates_t gCurrentState = eIND_NONE  [static]
```

Current Indication state maintained in this global

5.14.3.3 gPrevState

```
eAppIndicationStates_t gPrevState = eIND_NONE  [static]
```

Previous Indication state maintained here

5.15 AppIndication.c File Reference

```
#include <assert.h>
#include ¨AppIndication.h¨
#include ¨TriColorLED.h¨
```

Functions

- void AppIndicate_Init ()
- void AppIndicate_DeInit ()
- void AppIndicate_SetState (eAppIndicationStates_t state)
- void AppIndicate_RevertState ()

Variables

- static sTriColorIndication_t gcAppIndicationTable [eIND_MAX]
- static eAppIndicationStates_t **gCurrentState** = eIND_NONE
- static eAppIndicationStates_t **gPrevState** = eIND_NONE

5.15.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-05-09

Copyright

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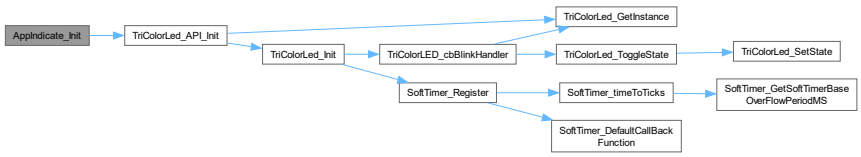
5.15.2 Function Documentation

5.15.2.1 AppIndicate_Init()

```
void AppIndicate_Init ( )
```

Wrapper around Indication medium initialization.

Here is the call graph for this function:

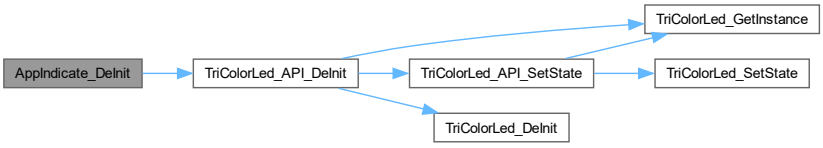


5.15.2.2 AppIndicate_DeInit()

```
void AppIndicate_DeInit ( )
```

Wrapper around Indication medium De-initialization.

Here is the call graph for this function:



5.15.2.3 AppIndicate_SetState()

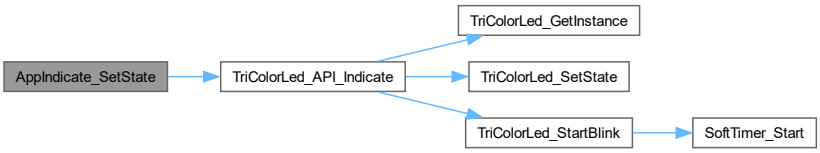
```
void AppIndicate_SetState (
            eAppIndicationStates_t state )
```

Set Indication medium state.

Parameters

state	Which application state to set to, gcAppIndicationTable

Here is the call graph for this function:

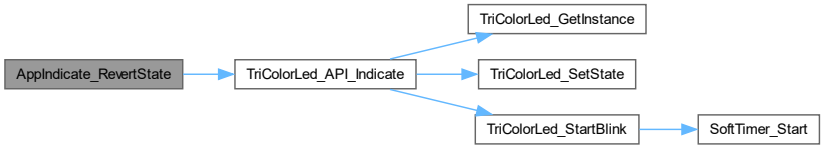


5.15.2.4 AppIndicate_RevertState()

```
void AppIndicate_RevertState ( )
```

Revert to previous indication state. For preserving previous Indication state.

Here is the call graph for this function:



5.15.3 Variable Documentation

5.15.3.1 gcAppIndicationTable

```
sTriColorIndication_t gcAppIndicationTable[eIND_MAX]  [static]
```

Configuration table that determine how application states correspond to TriColor LED states.

5.16 AppIndication.h File Reference

Enumerations

- enum eAppIndicationStates_t {
  eIND_NO_CHANGE , **eIND_NONE** , **eIND_RED_0** , **eIND_GREEN_0** ,
  eIND_BLUE_0 , **eIND_YELLOW_0** , **eIND_MAGENTA_0** , **eIND_CYAN_0** ,
  eIND_WHITE_0 , **eIND_RED_250MS** , **eIND_GREEN_250MS** , **eIND_BLUE_250MS** ,
  eIND_YELLOW_250MS , **eIND_MAGENTA_250MS** , **eIND_CYAN_250MS** , **eIND_WHITE_250MS** ,
  eIND_RED_500MS , **eIND_GREEN_500MS** , **eIND_BLUE_500MS** , **eIND_YELLOW_500MS** ,
  eIND_MAGENTA_500MS , **eIND_CYAN_500MS** , **eIND_WHITE_500MS** , **eIND_RED_1000MS** ,
  eIND_GREEN_1000MS , **eIND_BLUE_1000MS** , **eIND_YELLOW_1000MS** , **eIND_MAGENTA_1000MS** ,
  eIND_CYAN_1000MS , **eIND_WHITE_1000MS** , **eIND_RED_2000MS** , **eIND_GREEN_2000MS** ,
  eIND_BLUE_2000MS , **eIND_YELLOW_2000MS** , **eIND_MAGENTA_2000MS** , **eIND_CYAN_2000MS** ,
  eIND_WHITE_2000MS , **eIND_MAX** }

Functions

- void AppIndicate_Init ()
- void AppIndicate_SetState (eAppIndicationStates_t state)
- void AppIndicate_RevertState ()

5.16.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-05-09

Copyright

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5.16.2 Enumeration Type Documentation

5.16.2.1 eAppIndicationStates_t

```
enum eAppIndicationStates_t
```

Application events that require LED indication are enumerated here gcAppIndicationTable.

5.16.3 Function Documentation

5.16.3.1 AppIndicate_Init()

```
void AppIndicate_Init ( )
```

Wrapper around Indication medium initialization.

5.16.3.2 AppIndicate_SetState()

```
void AppIndicate_SetState (
            eAppIndicationStates_t state )
```

Set Indication medium state.

Parameters

state	Which application state to set to, gcAppIndicationTable

Parameters

state	Which application state to set to, gcAppIndicationTable

5.16.3.3 AppIndicate_RevertState()

```
void AppIndicate_RevertState ( )
```

Revert to previous indication state. For preserving previous Indication state.

5.17 AppIndication.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCOMMON_APPINDICATION_H_
00015 #define APPCOMMON_APPINDICATION_H_
00016
00018
00023 typedef enum
00024 {
00025     eIND_NO_CHANGE,
00026     eIND_NONE,
00027
00028     eIND_RED_0,
00029     eIND_GREEN_0,
00030     eIND_BLUE_0,
00031     eIND_YELLOW_0,
00032     eIND_MAGENTA_0,
00033     eIND_CYAN_0,
00034     eIND_WHITE_0,
00035
00036     eIND_RED_250MS,
00037     eIND_GREEN_250MS,
00038     eIND_BLUE_250MS,
00039     eIND_YELLOW_250MS,
00040     eIND_MAGENTA_250MS,
00041     eIND_CYAN_250MS,
00042     eIND_WHITE_250MS,
00043
00044     eIND_RED_500MS,
00045     eIND_GREEN_500MS,
00046     eIND_BLUE_500MS,
00047     eIND_YELLOW_500MS,
00048     eIND_MAGENTA_500MS,
00049     eIND_CYAN_500MS,
00050     eIND_WHITE_500MS,
00051
00052     eIND_RED_1000MS,
00053     eIND_GREEN_1000MS,
00054     eIND_BLUE_1000MS,
00055     eIND_YELLOW_1000MS,
00056     eIND_MAGENTA_1000MS,
00057     eIND_CYAN_1000MS,
00058     eIND_WHITE_1000MS,
00059
00060     eIND_RED_2000MS,
00061     eIND_GREEN_2000MS,
00062     eIND_BLUE_2000MS,
00063     eIND_YELLOW_2000MS,
00064     eIND_MAGENTA_2000MS,
00065     eIND_CYAN_2000MS,
00066     eIND_WHITE_2000MS,
00067
00068     eIND_MAX,
```

```
00069 }eAppIndicationStates_t;
00070
00072
00073 void AppIndicate_Init();
00074 void AppIndicate_SetState(eAppIndicationStates_t state);
00075 void AppIndicate_RevertState();
00076
00078
00079 #endif /* APPCOMMON_APPINDICATION_H_ */
```

5.18 AppIndication.h File Reference

Enumerations

- enum eAppIndicationStates_t {
  eIND_NO_CHANGE , **eIND_NONE** , **eIND_RED_0** , **eIND_GREEN_0** ,
  eIND_BLUE_0 , **eIND_YELLOW_0** , **eIND_MAGENTA_0** , **eIND_CYAN_0** ,
  eIND_WHITE_0 , **eIND_RED_250MS** , **eIND_GREEN_250MS** , **eIND_BLUE_250MS** ,
  eIND_YELLOW_250MS , **eIND_MAGENTA_250MS** , **eIND_CYAN_250MS** , **eIND_WHITE_250MS** ,
  eIND_RED_500MS , **eIND_GREEN_500MS** , **eIND_BLUE_500MS** , **eIND_YELLOW_500MS** ,
  eIND_MAGENTA_500MS , **eIND_CYAN_500MS** , **eIND_WHITE_500MS** , **eIND_RED_1000MS** ,
  eIND_GREEN_1000MS , **eIND_BLUE_1000MS** , **eIND_YELLOW_1000MS** , **eIND_MAGENTA_1000MS** ,
  eIND_CYAN_1000MS , **eIND_WHITE_1000MS** , **eIND_RED_2000MS** , **eIND_GREEN_2000MS** ,
  eIND_BLUE_2000MS , **eIND_YELLOW_2000MS** , **eIND_MAGENTA_2000MS** , **eIND_CYAN_2000MS** ,
  eIND_WHITE_2000MS , **eIND_MAX** }

Functions

- void AppIndicate_Init ()
- void AppIndicate_SetState (eAppIndicationStates_t state)
- void AppIndicate_RevertState ()

5.18.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-05-09

Copyright

5.18.2 Enumeration Type Documentation

5.18.2.1 eAppIndicationStates_t

```
enum eAppIndicationStates_t
```

Application events that require LED indication are enumerated here gcAppIndicationTable.

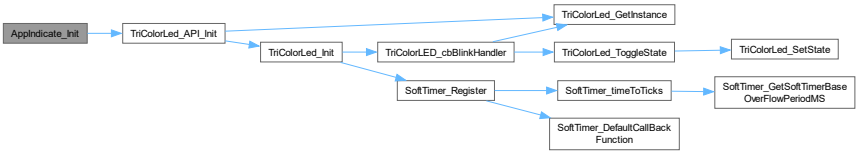
5.18.3 Function Documentation

5.18.3.1 AppIndicate_Init()

```
void AppIndicate_Init ( )
```

Wrapper around Indication medium initialization.

Here is the call graph for this function:



Here is the caller graph for this function:

5.18.3.2 AppIndicate_SetState()

```
void AppIndicate_SetState (
            eAppIndicationStates_t state )
```

Set Indication medium state.

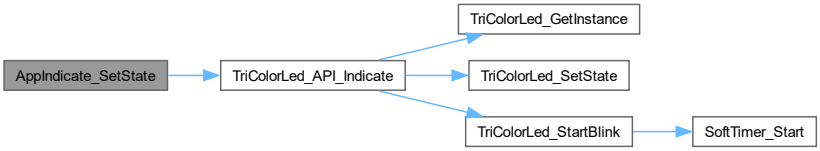
Parameters

state	Which application state to set to, gcAppIndicationTable

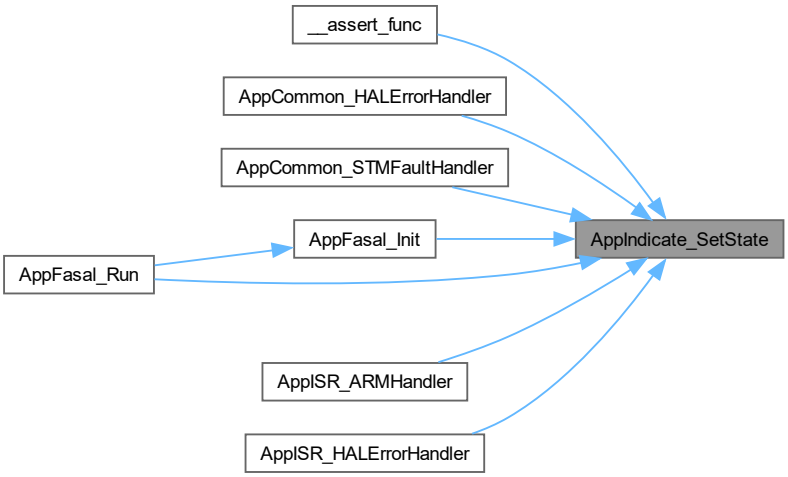
Parameters

state	Which application state to set to, gcAppIndicationTable

Here is the call graph for this function:



Here is the caller graph for this function:

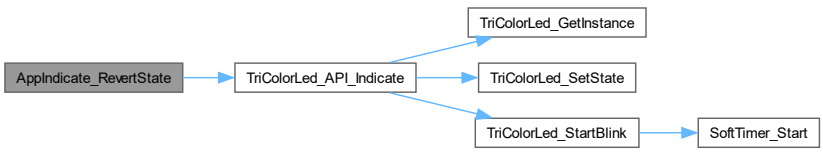


5.18.3.3 AppIndicate_RevertState()

```
void AppIndicate_RevertState ( )
```

Revert to previous indication state. For preserving previous Indication state.

Here is the call graph for this function:



5.19 AppIndication/AppIndication.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCOMMON_APPINDICATION_H_
00015 #define APPCOMMON_APPINDICATION_H_
00016
00018
00023 typedef enum
00024 {
00025     eIND_NO_CHANGE,
00026     eIND_NONE,
00027
00028     eIND_RED_0,
00029     eIND_GREEN_0,
00030     eIND_BLUE_0,
00031     eIND_YELLOW_0,
00032     eIND_MAGENTA_0,
00033     eIND_CYAN_0,
00034     eIND_WHITE_0,
00035
00036     eIND_RED_250MS,
00037     eIND_GREEN_250MS,
00038     eIND_BLUE_250MS,
00039     eIND_YELLOW_250MS,
00040     eIND_MAGENTA_250MS,
00041     eIND_CYAN_250MS,
00042     eIND_WHITE_250MS,
00043
00044     eIND_RED_500MS,
00045     eIND_GREEN_500MS,
00046     eIND_BLUE_500MS,
00047     eIND_YELLOW_500MS,
00048     eIND_MAGENTA_500MS,
00049     eIND_CYAN_500MS,
00050     eIND_WHITE_500MS,
00051
00052     eIND_RED_1000MS,
00053     eIND_GREEN_1000MS,
00054     eIND_BLUE_1000MS,
00055     eIND_YELLOW_1000MS,
00056     eIND_MAGENTA_1000MS,
00057     eIND_CYAN_1000MS,
00058     eIND_WHITE_1000MS,
00059
00060     eIND_RED_2000MS,
00061     eIND_GREEN_2000MS,
00062     eIND_BLUE_2000MS,
00063     eIND_YELLOW_2000MS,
00064     eIND_MAGENTA_2000MS,
00065     eIND_CYAN_2000MS,
00066     eIND_WHITE_2000MS,
00067
00068     eIND_MAX,
00069 } eAppIndicationStates_t;
00070
00072
00073 void AppIndicate_Init();
00074 void AppIndicate_SetState(eAppIndicationStates_t state);
00075 void AppIndicate_RevertState();
00076
00078
00079 #endif /* APPCOMMON_APPINDICATION_H_ */
```

5.20 TriColorLED.c File Reference

```
#include <assert.h>
#include ¨SoftTimer.h¨
#include ¨TriColorLED.h¨
```

Functions

- static void TriColorLed_ToggleState (volatile sTricolorLED_t ∗const pMe)

- static volatile sTricolorLED_t *const TriColorLed_GetInstance ()
- static void TriColorLED_cbBlinkHandler ()
- static void TriColorLed_SetState (volatile sTricolorLED_t *const pMe, eTriColorLEDStates_t triColor↩ LEDState)
- static void TriColorLed_Init (volatile sTricolorLED_t *const pMe)
- static void TriColorLed_DeInit (volatile sTricolorLED_t *const pMe)
- static void TriColorLed_StartBlink (volatile sTricolorLED_t *const pMe, eTriColorLEDBlinkPeriod_t eBlink↩ Period, bool IsStartNeeded)
- void TriColorLed_API_Init ()
- void TriColorLed_API_DeInit ()
- void TriColorLed_API_SetState (eTriColorLEDStates_t state)
- void TriColorLed_API_Indicate (eTriColorLEDStates_t state, eTriColorLEDBlinkPeriod_t eBlinkPeriod, bool IsBlinkNeeded)

Variables

- static const GPIO_PinState gcLEDEnumtoGPIOStateConverter [eLED_STATE_MAX]
- static const sSTMGpioPinWrapper_t gcLEDPins [eLED_MAX]
- static const sSTMGpioPinWrapper_t gcLEDPins2 [eLED_MAX]
- static const sTriColorLEDState_t gcLEDStateHelperTable [eLED_COLOR_STATE_MAX]
- static const uint32_t gcBlinkPeriodToCountHelper [eLED_BLINK_PERIOD_MAX]
- static volatile sTricolorLED_t gvTricolorLed

5.20.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-09-28

Copyright

5.20.2 Function Documentation

5.20.2.1 TriColorLed_ToggleState()

```
static void TriColorLed_ToggleState (
            volatile sTricolorLED_t *const pMe )  [static]
```

Toggle TriColor LED.

Parameters

pMe	TriColor LED instance

Here is the call graph for this function:

Here is the caller graph for this function:

5.20.2.2 TriColorLed_GetInstance()

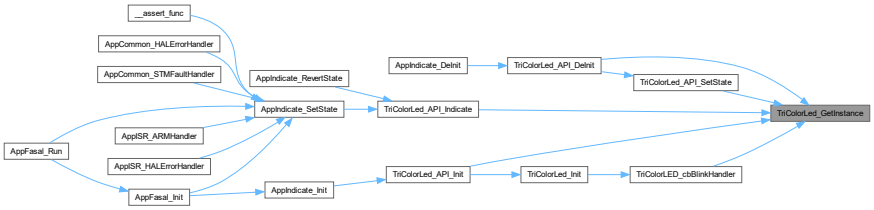
```
static volatile sTricolorLED_t *const TriColorLed_GetInstance ( )  [static]
```

Get tricolor LED instance.

Returns

volatile* const

Here is the caller graph for this function:

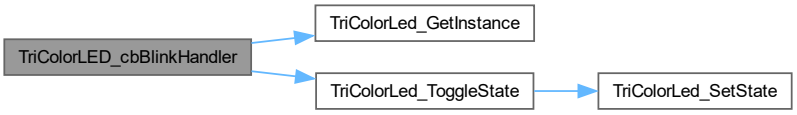


5.20.2.3 TriColorLED_cbBlinkHandler()

```
static void TriColorLED_cbBlinkHandler ( )  [static]
```

Call back function registered with softtimer.

Here is the call graph for this function:



Here is the caller graph for this function:

5.20.2.4 TriColorLed_SetState()

```
static void TriColorLed_SetState (
            volatile sTricolorLED_t *const pMe,
            eTriColorLEDStates_t triColorLEDState )  [static]
```

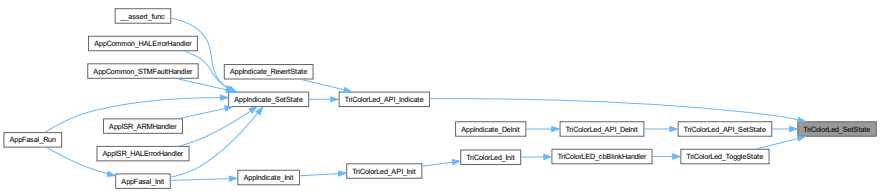
Set TriColor LED state.

Parameters

pMe	TriColor LED instance
triColorLEDState	

< Set Primary set of LED

< Duplicate set of LED follows primary LED indicationHere is the caller graph for this function:



5.20.2.5 TriColorLed_Init()

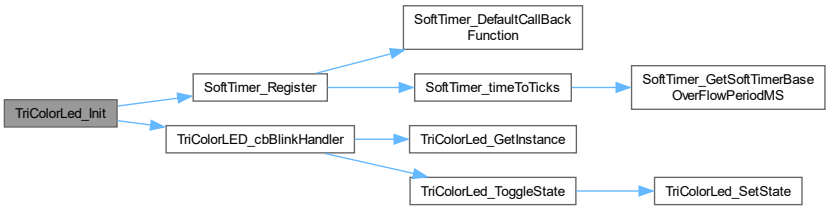
```
static void TriColorLed_Init (
            volatile sTricolorLED_t *const pMe )  [static]
```

Initialize Tricolor LED.

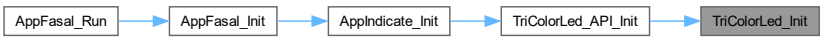
Parameters

pMe	TriColor LED instance

Here is the call graph for this function:



Here is the caller graph for this function:



5.20.2.6 TriColorLed_DeInit()

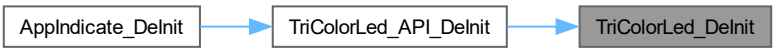
```
static void TriColorLed_DeInit (
            volatile sTricolorLED_t *const pMe )  [static]
```

DeInitialize Tricolor LED.

Parameters

pMe	TriColor LED instance

Here is the caller graph for this function:



5.20.2.7 TriColorLed_StartBlink()

```
static void TriColorLed_StartBlink (
            volatile sTricolorLED_t *const pMe,
            eTriColorLEDBlinkPeriod_t eBlinkPeriod,
            bool IsStartNeeded )  [static]
```

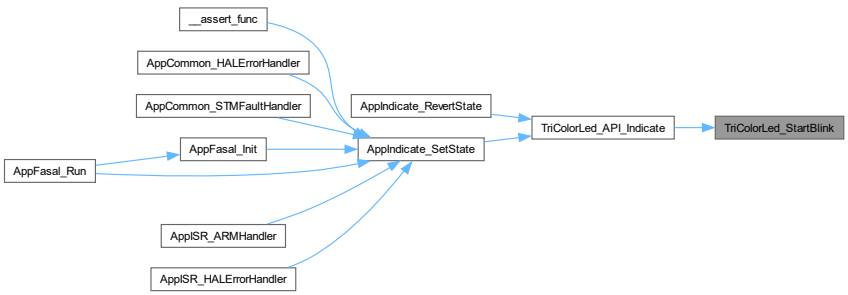
Enable/Disable LED blink routine. The Blink will be executed though TriColorLED_cbBlinkHandler.

Parameters

pMe	TriColor LED instance
eBlinkPeriod	
IsStartNeeded	Pass true if blink needed, false to cancel blink

Here is the call graph for this function:

Here is the caller graph for this function:

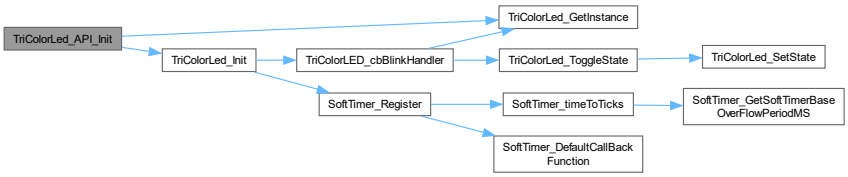


5.20.2.8 TriColorLed_API_Init()

```
void TriColorLed_API_Init ( )
```

Initialize tricolor LED API.

Here is the call graph for this function:



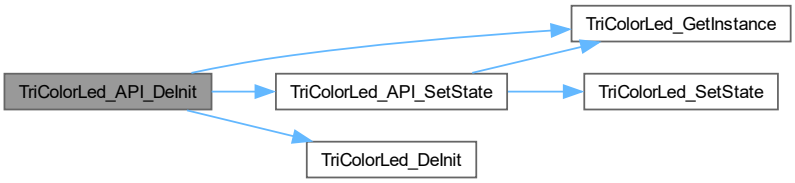
Here is the caller graph for this function:

5.20.2.9 TriColorLed_API_DeInit()

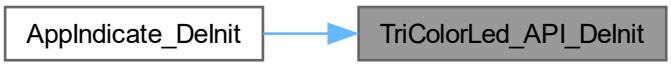
```
void TriColorLed_API_DeInit ( )
```

DeInitialize tricolor LED API.

Here is the call graph for this function:



Here is the caller graph for this function:



5.20.2.10 TriColorLed_API_SetState()

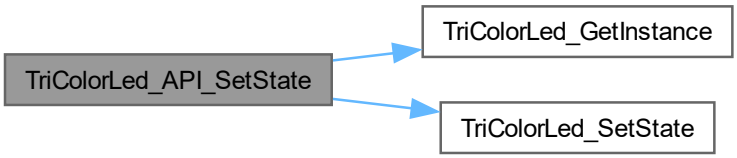
```
void TriColorLed_API_SetState (
            eTriColorLEDStates_t state )
```

API to set Tricolor LED state.

Parameters

state	state to set TriColor LED

Here is the call graph for this function:



Here is the caller graph for this function:

5.20.2.11 TriColorLed_API_Indicate()

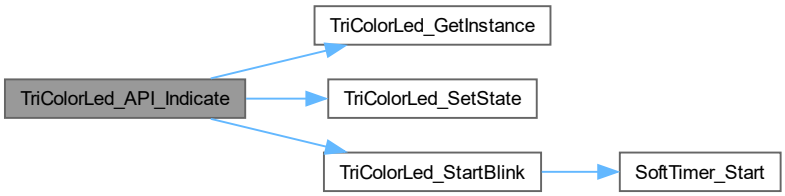
```
void TriColorLed_API_Indicate (
            eTriColorLEDStates_t state,
            eTriColorLEDBlinkPeriod_t eBlinkPeriod,
            bool IsBlinkNeeded )
```

Tricolor LED to indicate various Application states.

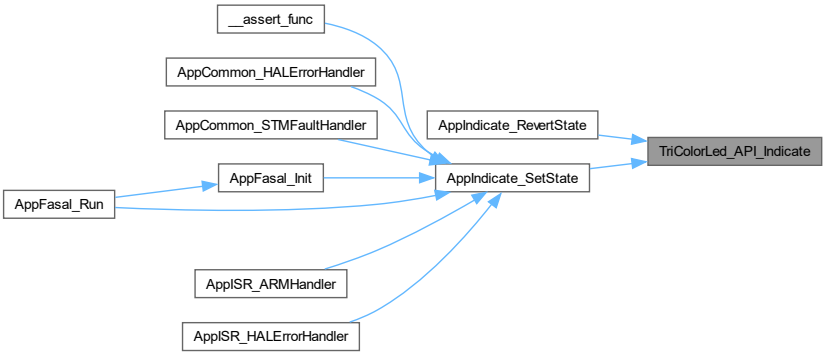
Parameters

state	TriColor LED color to set
eBlinkPeriod	Period to blink LED over
IsBlinkNeeded	Set to true if blinking is needed, else the LED will only light up

Here is the call graph for this function:



Here is the caller graph for this function:



5.20.3 Variable Documentation

5.20.3.1 gcLEDEnumtoGPIOStateConverter

```
const GPIO_PinState gcLEDEnumtoGPIOStateConverter[eLED_STATE_MAX]  [static]
```

Initial value:

```
=
{
        [eLED_OFF] = GPIO_PIN_RESET,
        [eLED_ON] = GPIO_PIN_SET,
}
```

Helper table that maps LED state to GPIO state.

5.20.3.2 gcLEDPins

```
const sSTMGpioPinWrapper_t gcLEDPins[eLED_MAX]  [static]
```

Initial value:

```
=
{
        [eLED_RED]      = {.GpioPort = LED_R_GPIO_Port, .GpioPin = LED_R_Pin },
        [eLED_GREEN]    = {.GpioPort = LED_G_GPIO_Port, .GpioPin = LED_G_Pin },
        [eLED_BLUE]     = {.GpioPort = LED_B_GPIO_Port, .GpioPin = LED_B_Pin },
}
```

LED pin mapping helper table.

5.20.3.3 gcLEDPins2

```
const sSTMGpioPinWrapper_t gcLEDPins2[eLED_MAX]  [static]
```

Initial value:

```
=
{
        [eLED_BLUE]     = {.GpioPort = LED_R1_GPIO_Port, .GpioPin = LED_R1_Pin },
        [eLED_GREEN]    = {.GpioPort = LED_G1_GPIO_Port, .GpioPin = LED_G1_Pin },
        [eLED_RED]      = {.GpioPort = LED_B1_GPIO_Port, .GpioPin = LED_B1_Pin },
}
```

LED pin mapping helper table.

- warning eLED_BLUE and eLED_RED are swapped in Rev 1A Hardware

5.20.3.4 gcLEDStateHelperTable

```
const sTriColorLEDState_t gcLEDStateHelperTable[eLED_COLOR_STATE_MAX]  [static]
```

Helper structure to map TriColor LED enumerations to Individual LED combinations.

5.20.3.5 gcBlinkPeriodToCountHelper

```
const uint32_t gcBlinkPeriodToCountHelper[eLED_BLINK_PERIOD_MAX]  [static]
```

Initial value:

```
=
{
        [eTRICOLORLED_BLINK_NONE]       = UINT32_MAX,
        [eTRICOLORLED_BLINK_250MS]      = 250/TRICOLOR_LED_BLINK_TIME_BASE_MS,
        [eTRICOLORLED_BLINK_500MS]      = 500/TRICOLOR_LED_BLINK_TIME_BASE_MS,
        [eTRICOLORLED_BLINK_1000MS]     = 1000/TRICOLOR_LED_BLINK_TIME_BASE_MS,
        [eTRICOLORLED_BLINK_2000MS]     = 2000/TRICOLOR_LED_BLINK_TIME_BASE_MS
}
```

Utility table that saves LED timings to ticks based on TRICOLOR_LED_BLINK_TIME_BASE_MS.

5.20.3.6 gvTricolorLed

```
volatile sTricolorLED_t gvTricolorLed  [static]
```

Tricolor LED instance.

5.21 TriColorLED.c File Reference

```
#include <assert.h>
#include ¨TriColorLED.h¨
#include ¨SoftTimer.h¨
```

Functions

- static void TriColorLed_ToggleState (volatile sTricolorLED_t ∗const pMe)
- static volatile sTricolorLED_t ∗const TriColorLed_GetInstance ()
- static void TriColorLED_cbBlinkHandler ()
- static void TriColorLed_SetState (volatile sTricolorLED_t ∗const pMe, eTriColorLEDStates_t triColor↩ LEDState)
- static void TriColorLed_Init (volatile sTricolorLED_t ∗const pMe)
- static void TriColorLed_DeInit (volatile sTricolorLED_t ∗const pMe)
- static void TriColorLed_StartBlink (volatile sTricolorLED_t ∗const pMe, eTriColorLEDBlinkPeriod_t eBlink↩ Period, bool IsStartNeeded)
- void TriColorLed_API_Init ()
- void TriColorLed_API_DeInit ()
- void TriColorLed_API_SetState (eTriColorLEDStates_t state)
- void TriColorLed_API_Indicate (eTriColorLEDStates_t state, eTriColorLEDBlinkPeriod_t eBlinkPeriod, bool IsBlinkNeeded)

Variables

- static const GPIO_PinState gcLEDEnumtoGPIOStateConverter [eLED_STATE_MAX]
- static const sLEDPins_t gcLEDPins [eLED_MAX]
- static const sLEDPins_t gcLEDPins2 [eLED_MAX]
- static const sTriColorLEDState_t gcLEDStateHelperTable [eLED_COLOR_STATE_MAX]
- static const uint32_t gcBlinkPeriodToCountHelper [eLED_BLINK_PERIOD_MAX]
- static volatile sTricolorLED_t gvTricolorLed

5.21.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-09-28

Copyright

5.21.2 Function Documentation

5.21.2.1 TriColorLed_ToggleState()

```
static void TriColorLed_ToggleState (
            volatile sTricolorLED_t *const pMe )  [static]
```

Toggle TriColor LED.

Parameters

pMe	TriColor LED instance

Here is the call graph for this function:

Here is the caller graph for this function:

5.21.2.2 TriColorLed_GetInstance()

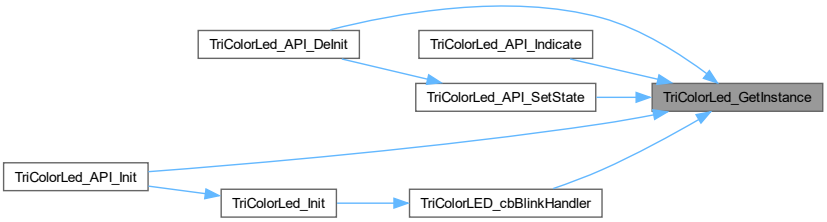
```
static volatile sTricolorLED_t *const TriColorLed_GetInstance ( )  [static]
```

Get tricolor LED instance.

Returns

volatile* const

Here is the caller graph for this function:

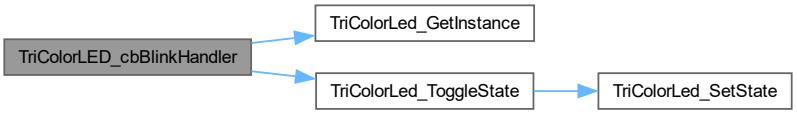


5.21.2.3 TriColorLED_cbBlinkHandler()

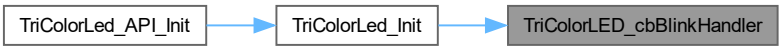
```
static void TriColorLED_cbBlinkHandler ( )  [static]
```

Call back function registered with softtimer.

Here is the call graph for this function:



Here is the caller graph for this function:



5.21.2.4 TriColorLed_SetState()

```
static void TriColorLed_SetState (
            volatile sTricolorLED_t *const pMe,
            eTriColorLEDStates_t triColorLEDState )  [static]
```

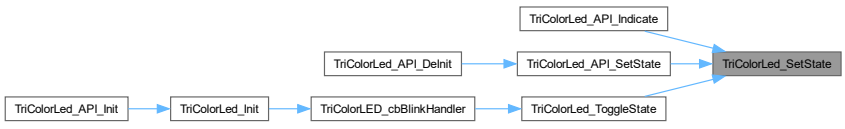
Set TriColor LED state.

Parameters

pMe	TriColor LED instance
triColorLEDState	

< Set Primary set of LED

< Duplicate set of LED follows primary LED indicationHere is the caller graph for this function:



5.21.2.5 TriColorLed_Init()

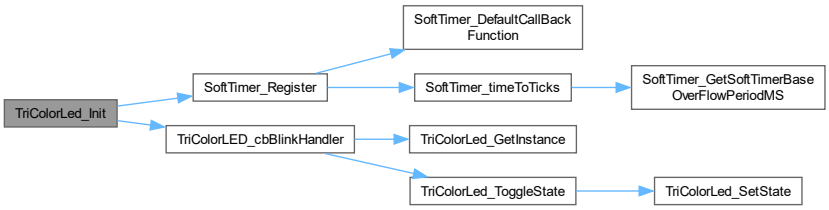
```
static void TriColorLed_Init (
            volatile sTricolorLED_t *const pMe )  [static]
```

Initialize Tricolor LED.

Parameters

pMe	TriColor LED instance

Here is the call graph for this function:



Here is the caller graph for this function:

5.21.2.6 TriColorLed_DeInit()

```
static void TriColorLed_DeInit (
            volatile sTricolorLED_t *const pMe )  [static]
```

DeInitialize Tricolor LED.

Parameters

pMe	TriColor LED instance

Here is the caller graph for this function:

5.21.2.7 TriColorLed_StartBlink()

```
static void TriColorLed_StartBlink (
            volatile sTricolorLED_t *const pMe,
            eTriColorLEDBlinkPeriod_t eBlinkPeriod,
            bool IsStartNeeded )  [static]
```

Enable/Disable LED blink routine. The Blink will be executed though TriColorLED_cbBlinkHandler.

Parameters

pMe	TriColor LED instance
eBlinkPeriod	
IsStartNeeded	Pass true if blink needed, false to cancel blink

Here is the call graph for this function:

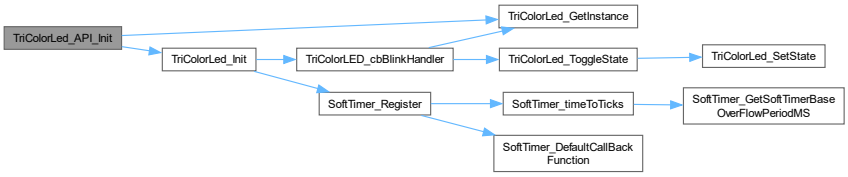
Here is the caller graph for this function:

5.21.2.8 TriColorLed_API_Init()

```
void TriColorLed_API_Init ( )
```

Initialize tricolor LED API.

Here is the call graph for this function:

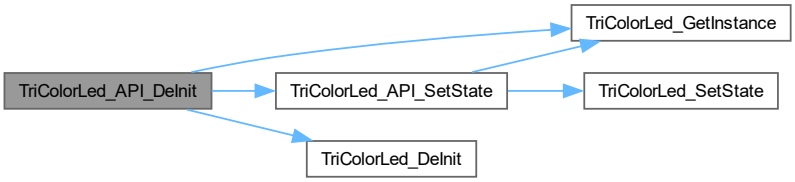


5.21.2.9 TriColorLed_API_DeInit()

```
void TriColorLed_API_DeInit ( )
```

DeInitialize tricolor LED API.

Here is the call graph for this function:



5.21.2.10 TriColorLed_API_SetState()

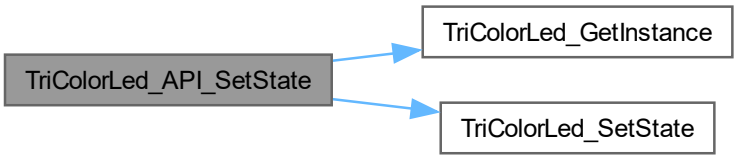
```
void TriColorLed_API_SetState (
            eTriColorLEDStates_t state )
```

API to set Tricolor LED state.

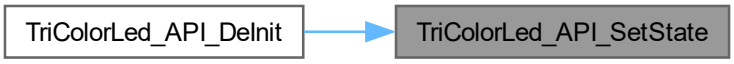
Parameters

state	state to set TriColor LED

Here is the call graph for this function:



Here is the caller graph for this function:



5.21.2.11 TriColorLed_API_Indicate()

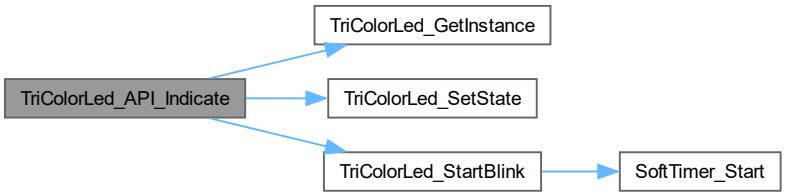
```
void TriColorLed_API_Indicate (
            eTriColorLEDStates_t state,
            eTriColorLEDBlinkPeriod_t eBlinkPeriod,
            bool IsBlinkNeeded )
```

Tricolor LED to indicate various Application states.

Parameters

state	TriColor LED color to set
eBlinkPeriod	Period to blink LED over
IsBlinkNeeded	Set to true if blinking is needed, else the LED will only light up

Here is the call graph for this function:



5.21.3 Variable Documentation

5.21.3.1 gcLEDEnumtoGPIOStateConverter

```
const GPIO_PinState gcLEDEnumtoGPIOStateConverter[eLED_STATE_MAX]  [static]
```

Initial value:

```
=
{
        [eLED_OFF] = GPIO_PIN_RESET,
        [eLED_ON] = GPIO_PIN_SET,
}
```

Helper table that maps LED state to GPIO state.

5.21.3.2 gcLEDPins

```
const sLEDPins_t gcLEDPins[eLED_MAX]  [static]
```

Initial value:

```
=
{
        [eLED_RED]      = {.LEDPort = LED_R_GPIO_Port, .LEDPin = LED_R_Pin },
        [eLED_GREEN]    = {.LEDPort = LED_G_GPIO_Port, .LEDPin = LED_G_Pin },
        [eLED_BLUE]     = {.LEDPort = LED_B_GPIO_Port, .LEDPin = LED_B_Pin },
}
```

LED pin mapping helper table.

5.21.3.3 gcLEDPins2

```
const sLEDPins_t gcLEDPins2[eLED_MAX]  [static]
```

Initial value:

```
=
{
        [eLED_BLUE]     = {.LEDPort = LED_R1_GPIO_Port, .LEDPin = LED_R1_Pin },
        [eLED_GREEN]    = {.LEDPort = LED_G1_GPIO_Port, .LEDPin = LED_G1_Pin },
        [eLED_RED]      = {.LEDPort = LED_B1_GPIO_Port, .LEDPin = LED_B1_Pin },
}
```

LED pin mapping helper table.

- warning eLED_BLUE and eLED_RED are swapped in Rev 1A Hardware

5.21.3.4 gcLEDStateHelperTable

```
const sTriColorLEDState_t gcLEDStateHelperTable[eLED_COLOR_STATE_MAX]  [static]
```

Helper structure to map TriColor LED enumerations to Individual LED combinations.

5.21.3.5 gcBlinkPeriodToCountHelper

```
const uint32_t gcBlinkPeriodToCountHelper[eLED_BLINK_PERIOD_MAX]  [static]
```

Initial value:

```
=
{
        [eTRICOLORLED_BLINK_NONE]       = UINT32_MAX,
        [eTRICOLORLED_BLINK_250MS]      = 250/TRICOLOR_LED_BLINK_TIME_BASE_MS,
        [eTRICOLORLED_BLINK_500MS]      = 500/TRICOLOR_LED_BLINK_TIME_BASE_MS,
        [eTRICOLORLED_BLINK_1000MS]     = 1000/TRICOLOR_LED_BLINK_TIME_BASE_MS,
        [eTRICOLORLED_BLINK_2000MS]     = 2000/TRICOLOR_LED_BLINK_TIME_BASE_MS
}
```

Utility table that saves LED timings to ticks based on TRICOLOR_LED_BLINK_TIME_BASE_MS.

5.21.3.6 gvTricolorLed

```
volatile sTricolorLED_t gvTricolorLed  [static]
```

Tricolor LED instance.

5.22 TriColorLED.h File Reference

```
#include <stdbool.h>
#include <stdint.h>
#include ¨gpio.h¨
#include ¨CommonDefinitions.h¨
```

Data Structures

- struct sTriColorLEDState_t
- struct sLed_t
- struct sTricolorLED_t
- struct sTriColorIndication_t

Macros

- #define **TRICOLOR_LED_BLINK_TIME_BASE_MS** (50u)

Enumerations

- enum eLEDId_t { **eLED_RED** , **eLED_GREEN** , **eLED_BLUE** , **eLED_MAX** }
- enum eLEDState_t { **eLED_OFF** , **eLED_ON** , **eLED_STATE_MAX** }
- enum eTriColorLEDBlinkPeriod_t { **eTRICOLORLED_BLINK_NONE** , **eTRICOLORLED_BLINK_250MS** , **eTRICOLORLED_BLINK_500MS** , **eTRICOLORLED_BLINK_1000MS** , **eTRICOLORLED_BLINK_2000MS** , **eLED_BLINK_PERIOD_MAX** }
- enum eTriColorLEDStates_t { **eLED_COLOR_ALL_OFF** , **eLED_COLOR_RED** , **eLED_COLOR_GREEN** , **eLED_COLOR_BLUE** , **eLED_COLOR_YELLOW** , **eLED_COLOR_MAGENTA** , **eLED_COLOR_CYAN** , **eLED_COLOR_WHITE** , **eLED_COLOR_STATE_MAX** }

Functions

- void TriColorLed_API_Init ()
- void TriColorLed_API_DeInit ()
- void TriColorLed_API_SetState (eTriColorLEDStates_t state)
- void TriColorLed_API_Indicate (eTriColorLEDStates_t state, eTriColorLEDBlinkPeriod_t eBlinkPeriod, bool IsBlinkNeeded)

5.22.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-09-28

Copyright

5.22.2 Enumeration Type Documentation

5.22.2.1 eLEDId_t

`enum eLEDId_t`

LEDs that make up tricolor LEDs.

5.22.2.2 eLEDState_t

```
enum eLEDState_t
```

LED state enumeration.

5.22.2.3 eTriColorLEDBlinkPeriod_t

```
enum eTriColorLEDBlinkPeriod_t
```

Possible LED blink periods.

5.22.2.4 eTriColorLEDStates_t

```
enum eTriColorLEDStates_t
```

TriColor LED state enumerations.

5.22.3 Function Documentation

5.22.3.1 TriColorLed_API_Init()

```
void TriColorLed_API_Init ( )
```

Initialize tricolor LED API.

5.22.3.2 TriColorLed_API_DeInit()

```
void TriColorLed_API_DeInit ( )
```

DeInitialize tricolor LED API.

5.22.3.3 TriColorLed_API_SetState()

```
void TriColorLed_API_SetState (
            eTriColorLEDStates_t state )
```

API to set Tricolor LED state.

Parameters

state	state to set TriColor LED

5.22.3.4 TriColorLed_API_Indicate()

```
void TriColorLed_API_Indicate (
            eTriColorLEDStates_t state,
            eTriColorLEDBlinkPeriod_t eBlinkPeriod,
            bool IsBlinkNeeded )
```

Tricolor LED to indicate various Application states.

Parameters

state	TriColor LED color to set
eBlinkPeriod	Period to blink LED over
IsBlinkNeeded	Set to true if blinking is needed, else the LED will only light up

5.23 AppIndication/TriColorLED/TriColorLED.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCOMMON_TRICOLORLED_TRICOLORLED_H_
00015 #define APPCOMMON_TRICOLORLED_TRICOLORLED_H_
00016
00018
00019 #include <stdbool.h>
00020 #include <stdint.h>
00021 #include "gpio.h"
00022 #include "CommonDefinitions.h"
00023
00025
00026 #define TRICOLOR_LED_BLINK_TIME_BASE_MS (50u)
00027
00029
00034 typedef enum
00035 {
00036     eLED_RED,
00037     eLED_GREEN,
00038     eLED_BLUE,
00039     eLED_MAX
00040 } eLEDId_t;
00041
00046 typedef enum
00047 {
00048     eLED_OFF,
00049     eLED_ON,
00050     eLED_STATE_MAX
00051 } eLEDState_t;
00052
00057 typedef enum
00058 {
00059     eTRICOLORLED_BLINK_NONE,
00060     eTRICOLORLED_BLINK_250MS,
00061     eTRICOLORLED_BLINK_500MS,
00062     eTRICOLORLED_BLINK_1000MS,
00063     eTRICOLORLED_BLINK_2000MS,
00064     eLED_BLINK_PERIOD_MAX
00065 } eTriColorLEDBlinkPeriod_t;
00066
00071 typedef enum
00072 {
00073     eLED_COLOR_ALL_OFF,
00074     eLED_COLOR_RED,
00075     eLED_COLOR_GREEN,
00076     eLED_COLOR_BLUE,
00077     eLED_COLOR_YELLOW,
00078     eLED_COLOR_MAGENTA,
00079     eLED_COLOR_CYAN,
00080     eLED_COLOR_WHITE,
00081     eLED_COLOR_STATE_MAX
00082 } eTriColorLEDStates_t;
00083
```

```
00085
00090 typedef struct
00091 {
00092     eLEDState_t LedState[eLED_MAX];
00093 } sTriColorLEDState_t;
00094
00099 typedef struct
00100 {
00101     const sSTMGpioPinWrapper_t* pcLedPin;
00102     eLEDState_t ledState;
00103 } sLed_t;
00104
00109 typedef struct
00110 {
00111     bool IsInitialized;
00112     bool IsBlinkEnabled;
00113     size_t blinkTicksSet;
00114     size_t blinkTicksCurrent;
00115     eTriColorLEDStates_t triColorLEDState;
00116     eTriColorLEDStates_t prevOnLEDState;
00117     sLed_t Leds[eLED_MAX];
00118     sLed_t Leds2[eLED_MAX];
00119 } sTricolorLED_t;
00120
00125 typedef struct
00126 {
00127     bool IsBlinkNeeded;
00128     eTriColorLEDStates_t ledState;
00129     eTriColorLEDBlinkPeriod_t ledBlinkPeriod;
00130 } sTriColorIndication_t;
00131
00133
00134 void TriColorLed_API_Init();
00135 void TriColorLed_API_DeInit();
00136 void TriColorLed_API_SetState(eTriColorLEDStates_t state);
00137 void TriColorLed_API_Indicate(eTriColorLEDStates_t state, eTriColorLEDBlinkPeriod_t eBlinkPeriod, bool
      IsBlinkNeeded);
00138
00140
00141 #endif /* APPCOMMON_TRICOLORLED_TRICOLORLED_H_ */
```

5.24 TriColorLED.h File Reference

```
#include <stdint.h>
#include <stddef.h>
#include <stdbool.h>
#include ¨gpio.h¨
```

Data Structures

- struct sTriColorLEDState_t
- struct sLEDPins_t
- struct sLed_t
- struct sTricolorLED_t
- struct sTriColorIndication_t

Macros

- #define **TRICOLOR_LED_BLINK_TIME_BASE_MS** (50u) /** < This is the timebase from which all LED blink period is composed of */

Enumerations

- enum eLEDId_t { **eLED_RED** , **eLED_GREEN** , **eLED_BLUE** , **eLED_MAX** }
- enum eLEDState_t { **eLED_OFF** , **eLED_ON** , **eLED_STATE_MAX** }
- enum eTriColorLEDBlinkPeriod_t { **eTRICOLORLED_BLINK_NONE** , **eTRICOLORLED_BLINK_250MS** , **eTRICOLORLED_BLINK_500MS** , **eTRICOLORLED_BLINK_1000MS** , **eTRICOLORLED_BLINK_2000MS** , **eLED_BLINK_PERIOD_MAX** }
- enum eTriColorLEDStates_t { **eLED_COLOR_ALL_OFF** , **eLED_COLOR_RED** , **eLED_COLOR_GREEN** , **eLED_COLOR_BLUE** , **eLED_COLOR_YELLOW** , **eLED_COLOR_MAGENTA** , **eLED_COLOR_CYAN** , **eLED_COLOR_WHITE** , **eLED_COLOR_STATE_MAX** }

Functions

- void TriColorLed_API_Init ()
- void TriColorLed_API_DeInit ()
- void TriColorLed_API_SetState (eTriColorLEDStates_t state)
- void TriColorLed_API_Indicate (eTriColorLEDStates_t state, eTriColorLEDBlinkPeriod_t eBlinkPeriod, bool IsBlinkNeeded)

5.24.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-09-28

Copyright

5.24.2 Enumeration Type Documentation

5.24.2.1 eLEDId_t

`enum eLEDId_t`

LEDs that make up tricolor LEDs.

5.24.2.2 eLEDState_t

```
enum eLEDState_t
```

LED state enumeration.

5.24.2.3 eTriColorLEDBlinkPeriod_t

```
enum eTriColorLEDBlinkPeriod_t
```

Possible LED blink periods.

5.24.2.4 eTriColorLEDStates_t

```
enum eTriColorLEDStates_t
```

TriColor LED state enumerations.

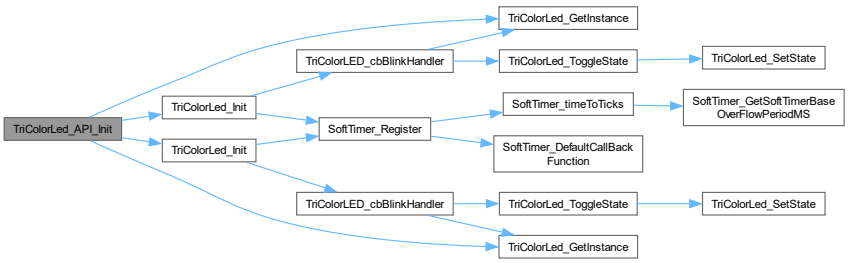
5.24.3 Function Documentation

5.24.3.1 TriColorLed_API_Init()

```
void TriColorLed_API_Init ( )
```

Initialize tricolor LED API.

Here is the call graph for this function:



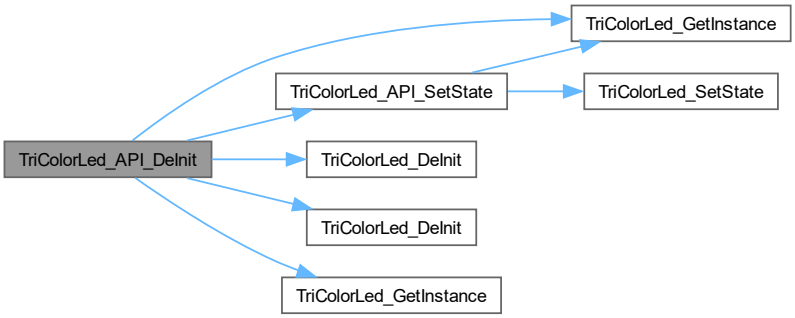
Here is the caller graph for this function:

5.24.3.2 TriColorLed_API_DeInit()

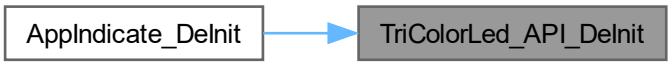
```
void TriColorLed_API_DeInit ( )
```

DeInitialize tricolor LED API.

Here is the call graph for this function:



Here is the caller graph for this function:



5.24.3.3 TriColorLed_API_SetState()

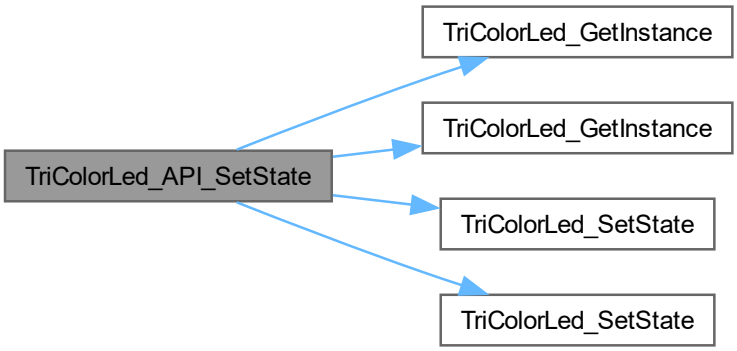
```
void TriColorLed_API_SetState (
            eTriColorLEDStates_t state )
```

API to set Tricolor LED state.

Parameters

state	state to set TriColor LED

Here is the call graph for this function:



Here is the caller graph for this function:

5.24.3.4 TriColorLed_API_Indicate()

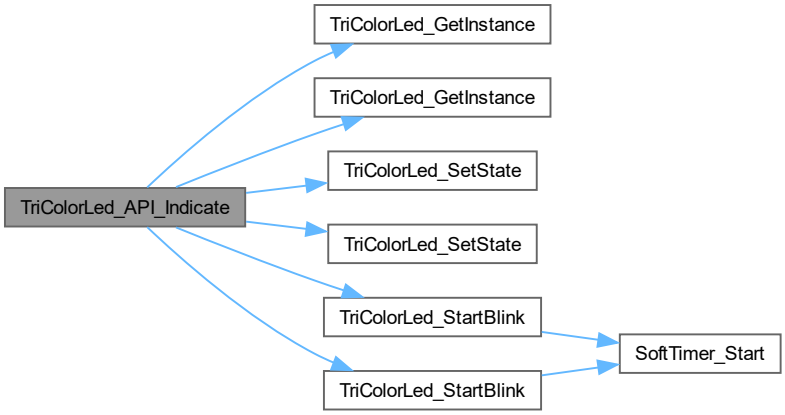
```
void TriColorLed_API_Indicate (
            eTriColorLEDStates_t state,
            eTriColorLEDBlinkPeriod_t eBlinkPeriod,
            bool IsBlinkNeeded )
```

Tricolor LED to indicate various Application states.

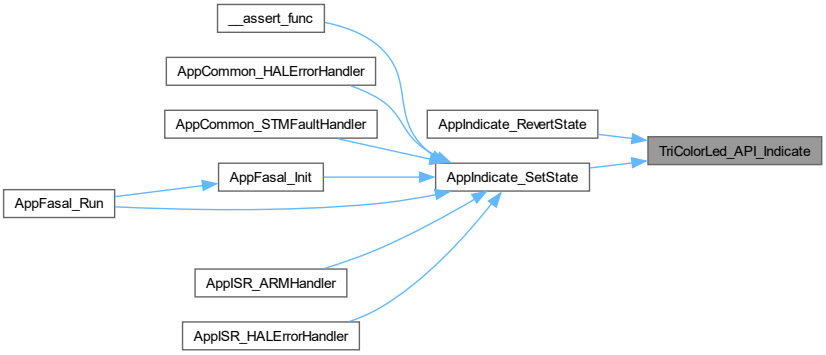
Parameters

state	TriColor LED color to set
eBlinkPeriod	Period to blink LED over
IsBlinkNeeded	Set to true if blinking is needed, else the LED will only light up

Here is the call graph for this function:



Here is the caller graph for this function:



5.25 TriColorLED/TriColorLED.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCOMMON_TRICOLORLED_TRICOLORLED_H_
00015 #define APPCOMMON_TRICOLORLED_TRICOLORLED_H_
00016
00018
00019 #include <stdint.h>
00020 #include <stddef.h>
00021 #include <stdbool.h>
00022 #include "gpio.h"
00023
00025
00026 #define TRICOLOR_LED_BLINK_TIME_BASE_MS (50u)
00029
00034 typedef enum
00035 {
00036     eLED_RED,
00037     eLED_GREEN,
```

```
00038     eLED_BLUE,
00039     eLED_MAX
00040 }eLEDId_t;
00041
00046 typedef enum
00047 {
00048     eLED_OFF,
00049     eLED_ON,
00050     eLED_STATE_MAX
00051 }eLEDState_t;
00052
00057 typedef enum
00058 {
00059     eTRICOLORLED_BLINK_NONE,
00060     eTRICOLORLED_BLINK_250MS,
00061     eTRICOLORLED_BLINK_500MS,
00062     eTRICOLORLED_BLINK_1000MS,
00063     eTRICOLORLED_BLINK_2000MS,
00064     eLED_BLINK_PERIOD_MAX
00065 }eTriColorLEDBlinkPeriod_t;
00066
00071 typedef enum
00072 {
00073     eLED_COLOR_ALL_OFF,
00074     eLED_COLOR_RED,
00075     eLED_COLOR_GREEN,
00076     eLED_COLOR_BLUE,
00077     eLED_COLOR_YELLOW,
00078     eLED_COLOR_MAGENTA,
00079     eLED_COLOR_CYAN,
00080     eLED_COLOR_WHITE,
00081     eLED_COLOR_STATE_MAX
00082 }eTriColorLEDStates_t;
00083
00085
00090 typedef struct
00091 {
00092     eLEDState_t LedState[eLED_MAX];
00093 }sTriColorLEDState_t;
00094
00099 typedef struct
00100 {
00101     GPIO_TypeDef* LEDPort;
00102     uint16_t LEDPin;
00103 }sLEDPins_t;
00104
00109 typedef struct
00110 {
00111     const sLEDPins_t* pcLedPin;
00112     eLEDState_t ledState;
00113 }sLed_t;
00114
00119 typedef struct
00120 {
00121     bool IsInitialized;
00122     bool IsBlinkEnabled;
00123     size_t blinkTicksSet;
00124     size_t blinkTicksCurrent;
00125     eTriColorLEDStates_t triColorLEDState;
00126     eTriColorLEDStates_t prevOnLEDState;
00127     sLed_t Leds[eLED_MAX];
00128     sLed_t Leds2[eLED_MAX];
00129 }sTricolorLED_t;
00130
00135 typedef struct
00136 {
00137     bool IsBlinkNeeded;
00138     eTriColorLEDStates_t ledState;
00139     eTriColorLEDBlinkPeriod_t ledBlinkPeriod;
00140 }sTriColorIndication_t;
00141
00143
00144 void TriColorLed_API_Init();
00145 void TriColorLed_API_DeInit();
00146 void TriColorLed_API_SetState(eTriColorLEDStates_t state);
00147 void TriColorLed_API_Indicate(eTriColorLEDStates_t state, eTriColorLEDBlinkPeriod_t eBlinkPeriod, bool
      IsBlinkNeeded);
00148
00150
00151 #endif /* APPCOMMON_TRICOLORLED_TRICOLORLED_H_ */
```

5.26 AppProfiler.c File Reference

```
#include ¨AppProfiler.h¨
#include ¨stm32f1xx_hal.h¨
```

Functions

- void **AppProfiler_StartExecutionTimeMeasurement** ()
- uint32_t AppProfiler_GetExecutionTimeMS ()

5.26.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-06-13

Copyright

5.26.2 Function Documentation

5.26.2.1 AppProfiler_GetExecutionTimeMS()

```
uint32_t AppProfiler_GetExecutionTimeMS ( )
```

Get function execution time.

Note

This function expects AppProfiler_StartExecutionTimeMeasurement to be invoked prior to use

Returns

execution time in milliseconds

5.27 AppProfiler.h File Reference

```
#include <stdint.h>
```

Functions

- void **AppProfiler_StartExecutionTimeMeasurement** ()
- uint32_t AppProfiler_GetExecutionTimeMS ()

5.27.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-06-13

Copyright

Copyright (c) 2023

5.27.2 Function Documentation

5.27.2.1 AppProfiler_GetExecutionTimeMS()

```
uint32_t AppProfiler_GetExecutionTimeMS ( )
```

Get function execution time.

Note

This function expects AppProfiler_StartExecutionTimeMeasurement to be invoked prior to use

Returns

execution time in milliseconds

5.28 AppProfiler.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCOMMON_APPUTILITY_APPPROFILER_APPPROFILER_H_
00015 #define APPCOMMON_APPUTILITY_APPPROFILER_APPPROFILER_H_
00016
00018
00019 #include <stdint.h>
00020
00022
00023 void AppProfiler_StartExecutionTimeMeasurement();
00024 uint32_t AppProfiler_GetExecutionTimeMS();
00025
00027
00028 #endif /* APPCOMMON_APPUTILITY_APPPROFILER_APPPROFILER_H_ */
```

5.29 DebugPrint.h File Reference

```
#include ¨AppConfiguration.h¨
#include ¨Console.h¨
```

Macros

- #define **DEBUG_PRINT**(...)

5.29.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

Copyright (c) 2023

5.30 AppCommon/AppUtility/DebugPrint.h

Go to the documentation of this file.

```
00001
00012 #ifndef UTILITY_DEBUGPRINT_H_
00013 #define UTILITY_DEBUGPRINT_H_
00014
00016
00017 #include "AppConfiguration.h"
00018 #include "Console.h"
00019
00021
00022 #ifdef ENABLE_DEBUG_PRINT
00023 #define DEBUG_PRINT Console_Print
00024 #else
00025 #define DEBUG_PRINT(...)
00026 #endif
00027
00028
00029 #endif /* UTILITY_DEBUGPRINT_H_ */
```

5.31 DebugPrint.h File Reference

```
#include ¨AppConfiguration.h¨
#include ¨Console.h¨
```

Macros

- #define **DEBUG_PRINT**(...)

5.31.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

Copyright (c) 2023

5.32 AppInterface/Console/DebugPrint.h

Go to the documentation of this file.

```
00001
00012 #ifndef UTILITY_DEBUGPRINT_H_
00013 #define UTILITY_DEBUGPRINT_H_
00014
00016
00017 #include "AppConfiguration.h"
00018 #include "Console.h"
00019
00021
00022 #ifdef ENABLE_DEBUG_PRINT
00023 #define DEBUG_PRINT Console_Print
00024 #else
00025 #define DEBUG_PRINT(...)
00026 #endif
00027
00028 #endif /* UTILITY_DEBUGPRINT_H_ */
```

5.33 SoftTimer.c File Reference

```
#include <assert.h>
#include ¨SoftTimer.h¨
```

Functions

- void SoftTimer_cbPeriodicCheck ()
- static void SoftTimer_DefaultCallBackFunction ()
- static uint32_t SoftTimer_GetSoftTimerBaseOverFlowPeriodMS ()
- static uint32_t SoftTimer_timeToTicks (uint32_t timeMs)
- void SoftTimer_Init ()
- __attribute__ ((unused))
- void SoftTimer_DeInit ()
- void SoftTimer_Register (eSoftTimerID_t id, uint32_t timeOut, bool IsPeriodic, pfCallBack_t callbackFunction)
- void SoftTimer_Start (eSoftTimerID_t id, bool IsStartNeeded)
- void SoftTimer_Pause (eSoftTimerID_t id, bool IsPauseNeeded)
- void SoftTimer_StartAll (bool IsStartNeeded)
- bool SofTimer_IsExpired (eSoftTimerID_t id)
- void SoftTimer_AperiodicTimerSet (uint32_t timeOut)
- bool SoftTimer_HasAperiodicTimerExpired ()
- void SoftTimer_DelayMS (uint32_t delayMS)
- uint32_t SoftTimer_GetCurrentMSTick ()

Variables

- static volatile sSoftTimer_t gvSoftTimers [eSOFT_TIMER_MAX]

5.33.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.2

Date

2023-05-28

Copyright

5.33.2 Function Documentation

5.33.2.1 SoftTimer_cbPeriodicCheck()

```
void SoftTimer_cbPeriodicCheck ( )
```

Periodic call from soft-timer that checks if any registered timers have expired and invokes the registered call back HAL_TIM_PeriodElapsedCallback.

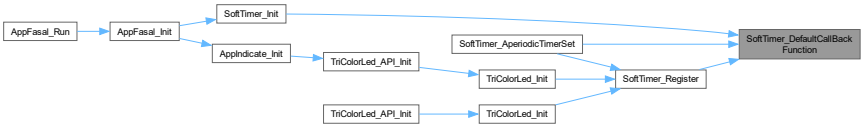
Here is the caller graph for this function:

5.33.2.2 SoftTimer_DefaultCallBackFunction()

```
static void SoftTimer_DefaultCallBackFunction ( )  [static]
```

Default call back for soft-timers without any executors.

Here is the caller graph for this function:



5.33.2.3 SoftTimer_GetSoftTimerBaseOverFlowPeriodMS()

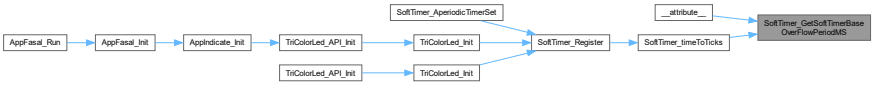
```
static uint32_t SoftTimer_GetSoftTimerBaseOverFlowPeriodMS ( )  [inline], [static]
```

Automatically determine the software overflow period from timer instance.

Note

Designed for 10ms for the current clock tree

Here is the caller graph for this function:



5.33.2.4 SoftTimer_timeToTicks()

```
static uint32_t SoftTimer_timeToTicks (
            uint32_t timeMs )  [static]
```

Utility function to convert timeinMS to soft-timer ticks count.

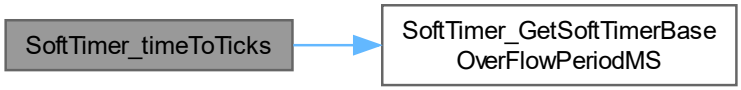
Parameters

timeMs	

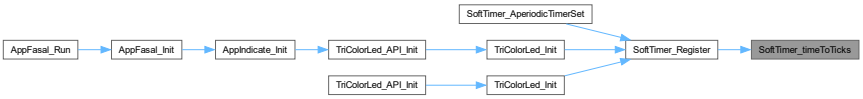
Returns

uint32_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.33.2.5 SoftTimer_Init()

```
void SoftTimer_Init ( )
```

Initialize soft-timer.

Here is the call graph for this function:

Here is the caller graph for this function:

5.33.2.6 __attribute__()

```
__attribute__ (
              (unused)  )
```

Get time left for expiry.

Parameters

id	ID of soft-timer

Returns

uint32_t time left for soft-timer expire in ms

Here is the call graph for this function:

5.33.2.7 SoftTimer_DeInit()

```
void SoftTimer_DeInit ( )
```

DeInitialize soft-timer.

5.33.2.8 SoftTimer_Register()

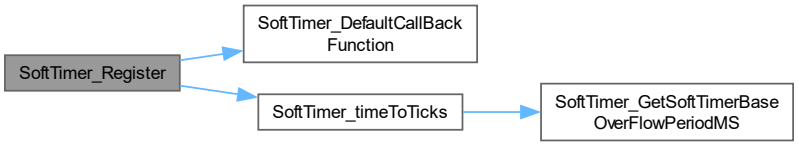
```
void SoftTimer_Register (
             eSoftTimerID_t id,
             uint32_t timeOut,
             bool IsPeriodic,
             pfCallBack_t callbackFunction )
```

Register a periodic function with soft-timer, create ID under eSoftTimerID and attach call-back and time period through this function.

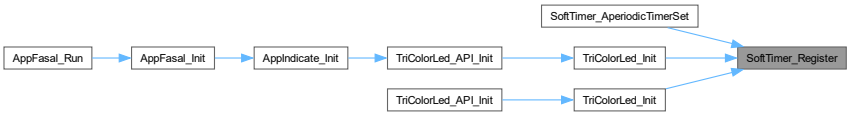
Parameters

id	
timeOut	
IsPeriodic	
callbackFunction	

Here is the call graph for this function:



Here is the caller graph for this function:



5.33.2.9 SoftTimer_Start()

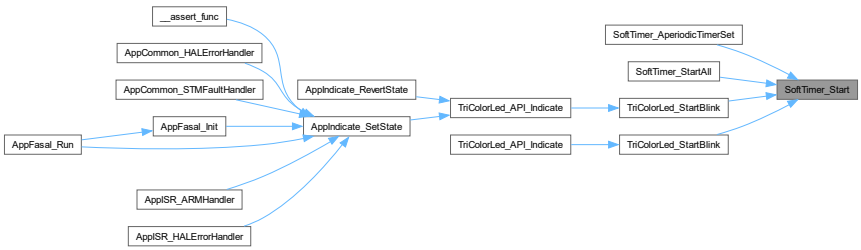
```
void SoftTimer_Start (
            eSoftTimerID_t id,
            bool IsStartNeeded )
```

Start soft-timer.

Parameters

id	
IsStartNeeded	

Here is the caller graph for this function:



5.33.2.10 SoftTimer_Pause()

```
void SoftTimer_Pause (
```

```
            eSoftTimerID_t id,
            bool IsPauseNeeded )
```

Function to disable and enable soft-timer for creating critical sections.

Parameters

id	ID of the soft-timer to pause/resume
IsPauseNeeded	Pause if true, resume if false

5.33.2.11 SoftTimer_StartAll()

```
void SoftTimer_StartAll (
            bool IsStartNeeded )
```

Collectively start all registered soft-timers.

Parameters

IsStartNeeded	

Here is the call graph for this function:

5.33.2.12 SofTimer_IsExpired()

```
bool SofTimer_IsExpired (
            eSoftTimerID_t id )
```

Check if soft-timers has expired.

Parameters

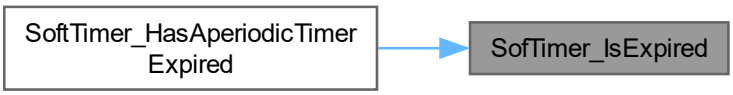
id	id of soft-timer to be checked

Returns

true if soft-timer has expired

false if V has not expired / has not been enabled

Here is the caller graph for this function:



5.33.2.13 SoftTimer_AperiodicTimerSet()

```
void SoftTimer_AperiodicTimerSet (
            uint32_t timeOut )
```

Set Aperiodic timer instance useful for setting timeouts.

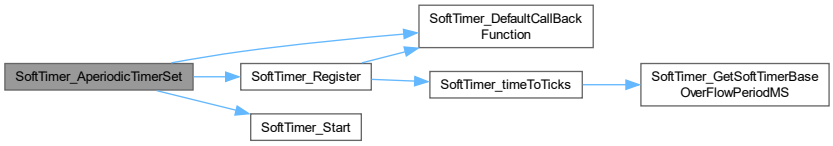
Note

eGENERIC_COUNT_DOWN_TIMER must not be used for other periodic timers

Parameters

timeOut	time out to set the aperiodic timer

Here is the call graph for this function:



5.33.2.14 SoftTimer_HasAperiodicTimerExpired()

```
bool SoftTimer_HasAperiodicTimerExpired ( )
```

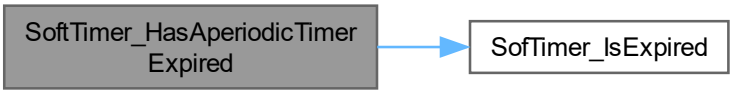
Check if aperiodic timer has expired.

Returns

true if timer has expired

false if timer has not expired/ Registered / started

Here is the call graph for this function:



5.33.2.15 SoftTimer_DelayMS()

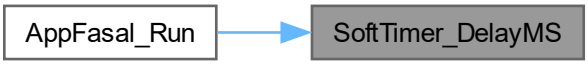
```
void SoftTimer_DelayMS (
            uint32_t delayMS )
```

Wrapper around HAL_Delay delay.

Parameters

delayMS	delay needed in mS

Here is the caller graph for this function:



5.33.2.16 SoftTimer_GetCurrentMSTick()

```
uint32_t SoftTimer_GetCurrentMSTick ( )
```

Wrapper around HAL tick count.

Returns

tick count from HAL

5.33.3 Variable Documentation

5.33.3.1 gvSoftTimers

```
volatile sSoftTimer_t gvSoftTimers[eSOFT_TIMER_MAX]  [static]
```

All soft-timer instances are captured here.

5.34 SoftTimer.h File Reference

```
#include <stdint.h>
#include <stdbool.h>
#include "tim.h"
```

Data Structures

- struct sSoftTimer_t

Macros

- #define **SOFTTIMER_TIMER_INSTANCE** (&htim6)
- #define **SOFTTIMER_IRQ** (TIM6_IRQn)

Typedefs

- typedef void(* **pfCallBack_t**) (void)

Enumerations

- enum eSoftTimerID_t { **eGENERIC_COUNT_DOWN_TIMER** , **ePUSH_BUTTON_TIMER** , **eDEBUG_LED_SOFT_TIMER** , **eSOFT_TIMER_MAX** }

Functions

- void SoftTimer_Register (eSoftTimerID_t id, uint32_t timeOut, bool IsPeriodic, pfCallBack_t callbackFunction)
- void SoftTimer_Init ()
- void SoftTimer_DeInit ()
- void SoftTimer_Start (eSoftTimerID_t id, bool IsStartNeeded)
- void SoftTimer_Pause (eSoftTimerID_t id, bool IsPauseNeeded)
- void SoftTimer_AperiodicTimerSet (uint32_t timeOut)
- void SoftTimer_DelayMS (uint32_t delayMS)
- void SoftTimer_cbPeriodicCheck ()
- bool SofTimer_IsExpired (eSoftTimerID_t id)
- bool SoftTimer_HasAperiodicTimerExpired ()
- uint32_t SoftTimer_GetCurrentMSTick ()

5.34.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.2

Date

2023-05-28

Copyright

5.34.2 Enumeration Type Documentation

5.34.2.1 eSoftTimerID_t

```
enum eSoftTimerID_t
```

soft-timer IDs

5.34.3 Function Documentation

5.34.3.1 SoftTimer_Register()

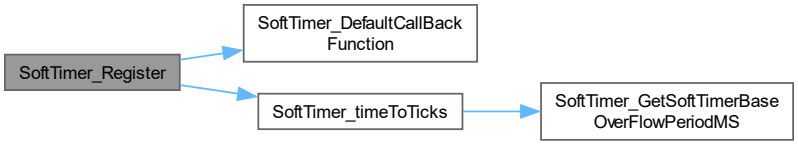
```
void SoftTimer_Register (
            eSoftTimerID_t id,
            uint32_t timeOut,
            bool IsPeriodic,
            pfCallBack_t callbackFunction )
```

Register a periodic function with soft-timer, create ID under eSoftTimerID and attach call-back and time period through this function.

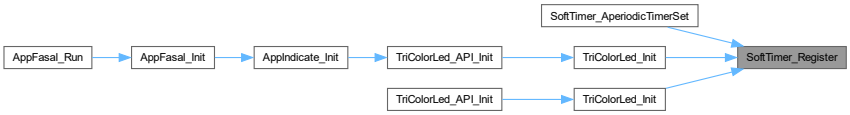
Parameters

id	
timeOut	
IsPeriodic	
callbackFunction	

Here is the call graph for this function:



Here is the caller graph for this function:



5.34.3.2 SoftTimer_Init()

```
void SoftTimer_Init ( )
```

Initialize soft-timer.

Here is the call graph for this function:

Here is the caller graph for this function:

5.34.3.3 SoftTimer_DeInit()

```
void SoftTimer_DeInit ( )
```

DeInitialize soft-timer.

5.34.3.4 SoftTimer_Start()

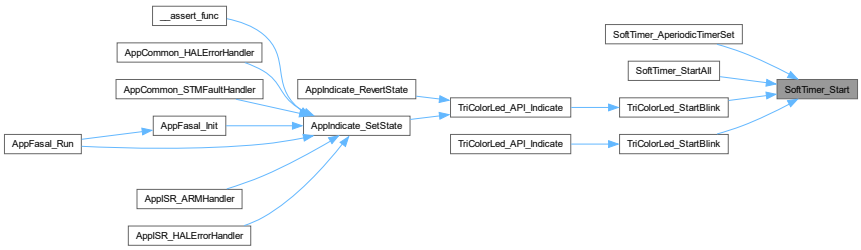
```
void SoftTimer_Start (
            eSoftTimerID_t id,
            bool IsStartNeeded )
```

Start soft-timer.

Parameters

id	
IsStartNeeded	

Here is the caller graph for this function:



5.34.3.5 SoftTimer_Pause()

```
void SoftTimer_Pause (
            eSoftTimerID_t id,
            bool IsPauseNeeded )
```

Function to disable and enable soft-timer for creating critical sections.

Parameters

id	ID of the soft-timer to pause/resume
IsPauseNeeded	Pause if true, resume if false

5.34.3.6 SoftTimer_AperiodicTimerSet()

```
void SoftTimer_AperiodicTimerSet (
```

```
            uint32_t timeOut )
```

Set Aperiodic timer instance useful for setting timeouts.

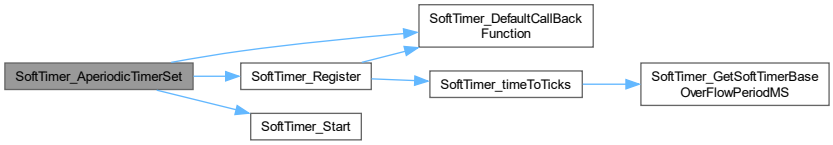
Note

eGENERIC_COUNT_DOWN_TIMER must not be used for other periodic timers

Parameters

timeOut	time out to set the aperiodic timer

Here is the call graph for this function:



5.34.3.7 SoftTimer_DelayMS()

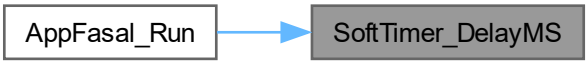
```
void SoftTimer_DelayMS (
            uint32_t delayMS )
```

Wrapper around HAL_Delay delay.

Parameters

delayMS	delay needed in mS

Here is the caller graph for this function:



5.34.3.8 SoftTimer_cbPeriodicCheck()

```
void SoftTimer_cbPeriodicCheck ( )
```

Periodic call from soft-timer that checks if any registered timers have expired and invokes the registered call back HAL_TIM_PeriodElapsedCallback.

Here is the caller graph for this function:

5.34.3.9 SofTimer_IsExpired()

```
bool SofTimer_IsExpired (
            eSoftTimerID_t id )
```

Check if soft-timers has expired.

Parameters

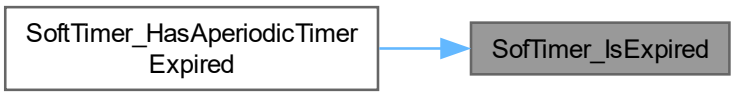
id	id of soft-timer to be checked

Returns

true if soft-timer has expired

false if V has not expired / has not been enabled

Here is the caller graph for this function:



5.34.3.10 SoftTimer_HasAperiodicTimerExpired()

```
bool SoftTimer_HasAperiodicTimerExpired ( )
```

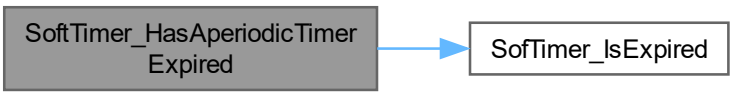
Check if aperiodic timer has expired.

Returns

true if timer has expired

false if timer has not expired/ Registered / started

Here is the call graph for this function:



5.34.3.11 SoftTimer_GetCurrentMSTick()

```
uint32_t SoftTimer_GetCurrentMSTick ( )
```

Wrapper around HAL tick count.

Returns

tick count from HAL

5.35 SoftTimer.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef UTILITY_SOFTTIMER_SOFTTIMER_H_
00015 #define UTILITY_SOFTTIMER_SOFTTIMER_H_
00016
00018
00019 #include <stdint.h>
00020 #include <stdbool.h>
00021 #include "tim.h"
00022
00024
00025 #define SOFTTIMER_TIMER_INSTANCE (&htim6)
00026 #define SOFTTIMER_IRQ (TIM6_IRQn)
00027
00029
00030 typedef void (*pfCallBack_t)(void);
00031
00033
00038 typedef enum
00039 {
00040     eGENERIC_COUNT_DOWN_TIMER,
00041     ePUSH_BUTTON_TIMER,
00042     eDEBUG_LED_SOFT_TIMER,
00043     eSOFT_TIMER_MAX
00044 } eSoftTimerID_t;
00045
00050 typedef struct
00051 {
00052     bool IsArmed;
00053     bool IsPeriodic;
00054     uint32_t currentTicks;
00055     uint32_t setTicks;
00056     pfCallBack_t pfCallBack;
```

```
00057 } sSoftTimer_t;
00058
00060
00061 void SoftTimer_Register(eSoftTimerID_t id, uint32_t timeOut, bool IsPeriodic, pfCallBack_t
      callbackFunction);
00062 void SoftTimer_Init();
00063 void SoftTimer_DeInit();
00064 void SoftTimer_Start(eSoftTimerID_t id, bool IsStartNeeded);
00065 void SoftTimer_Pause(eSoftTimerID_t id, bool IsPauseNeeded);
00066 void SoftTimer_AperiodicTimerSet(uint32_t timeOut);
00067 void SoftTimer_DelayMS(uint32_t delayMS);
00068 void SoftTimer_cbPeriodicCheck();
00069 bool SofTimer_IsExpired(eSoftTimerID_t id);
00070 bool SoftTimer_HasAperiodicTimerExpired();
00071 uint32_t SoftTimer_GetCurrentMSTick();
00072
00074
00075 #endif /* UTILITY_SOFTTIMER_SOFTTIMER_H_ */
```

5.36 Utility.h File Reference

```
#include <stdint.h>
#include <stdbool.h>
#include <stddef.h>
```

Macros

- #define **BUFFER_SIZE_4** (4u)
- #define **BUFFER_SIZE_8** (8u)
- #define **BUFFER_SIZE_16** (16u)
- #define **BUFFER_SIZE_32** (32u)
- #define **BUFFER_SIZE_64** (64u)
- #define **BUFFER_SIZE_128** (128u)
- #define **BUFFER_SIZE_256** (256u)
- #define **BUFFER_SIZE_512** (512u)
- #define **BUFFER_SIZE_1024** (1024u)

5.36.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

5.37 Utility.h

Go to the documentation of this file.

```
00001
00012 #ifndef UTILITY_UTILITY_H_
00013 #define UTILITY_UTILITY_H_
00014
00016
00017 #include <stdint.h>
00018 #include <stdbool.h>
00019 #include <stddef.h>
00020
00022
00023 #define BUFFER_SIZE_4           (4u)
00024 #define BUFFER_SIZE_8           (8u)
00025 #define BUFFER_SIZE_16          (16u)
00026 #define BUFFER_SIZE_32          (32u)
00027 #define BUFFER_SIZE_64          (64u)
00028 #define BUFFER_SIZE_128         (128u)
00029 #define BUFFER_SIZE_256         (256u)
00030 #define BUFFER_SIZE_512         (512u)
00031 #define BUFFER_SIZE_1024        (1024u)
00032
00034
00035
00036 #endif /* UTILITY_UTILITY_H_ */
```

5.38 CommonDefinitions.h File Reference

```
#include <stdbool.h>
#include <stdint.h>
#include ¨gpio.h¨
```

Data Structures

- struct sSTMGpioPinWrapper_t

Macros

- #define **BUFFER_SIZE_4** (4u)
- #define **BUFFER_SIZE_8** (8u)
- #define **BUFFER_SIZE_16** (16u)
- #define **BUFFER_SIZE_32** (32u)
- #define **BUFFER_SIZE_64** (64u)
- #define **BUFFER_SIZE_128** (128u)
- #define **BUFFER_SIZE_256** (256u)
- #define **BUFFER_SIZE_512** (512u)
- #define **BUFFER_SIZE_1024** (1024u)

5.38.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-10-24

Copyright

5.39 CommonDefinitions.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCOMMON_COMMONDEFINITIONS_COMMONDEFINITIONS_H_
00015 #define APPCOMMON_COMMONDEFINITIONS_COMMONDEFINITIONS_H_
00016
00018
00019 #include <stdbool.h>
00020 #include <stdint.h>
00021 #include "gpio.h"
00022
00024
00025 #define BUFFER_SIZE_4 (4u)
00026 #define BUFFER_SIZE_8 (8u)
00027 #define BUFFER_SIZE_16 (16u)
00028 #define BUFFER_SIZE_32 (32u)
00029 #define BUFFER_SIZE_64 (64u)
00030 #define BUFFER_SIZE_128 (128u)
00031 #define BUFFER_SIZE_256 (256u)
00032 #define BUFFER_SIZE_512 (512u)
00033 #define BUFFER_SIZE_1024 (1024u)
00034
00036
00041 typedef struct
00042 {
00043     GPIO_TypeDef* GpioPort;
00044     uint16_t GpioPin;
00045 } sSTMGpioPinWrapper_t;
00046
00048
00049 #endif /* APPCOMMON_COMMONDEFINITIONS_COMMONDEFINITIONS_H_ */
```

5.40 CommonInterrupts.c File Reference

```
#include ¨usart.h¨
#include ¨Console.h¨
#include ¨softTimer.h¨
#include ¨CommonInterrupts.h¨
```

Functions

- void HAL_UART_RxCpltCallback (UART_HandleTypeDef *huart)
- void HAL_TIM_PeriodElapsedCallback (TIM_HandleTypeDef *htim)

5.40.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-28

Copyright

5.40.2 Function Documentation

5.40.2.1 HAL_UART_RxCpltCallback()

```
void HAL_UART_RxCpltCallback (
            UART_HandleTypeDef * huart )
```

STM HAL reception callback.

Parameters

huart	

Here is the call graph for this function:

5.40.2.2 HAL_TIM_PeriodElapsedCallback()

```
void HAL_TIM_PeriodElapsedCallback (
            TIM_HandleTypeDef * htim )
```

Timer callback associated with soft-timer, part of STM HAL.

Parameters

htim	timer instance

Here is the call graph for this function:

5.41 CommonInterrupts.h

```
00001 /*
00002  * CommonInterrupts.h
00003  *
00004  *  Created on: Jun 5, 2024
00005  *      Author: Fasal
00006  */
00007
00008 #ifndef COMMONINTERRUPTS_H_
00009 #define COMMONINTERRUPTS_H_
00010
00011
00012
00013 #endif /* COMMONINTERRUPTS_H_ */
```

5.42 ConfigSetting.c File Reference

```
#include ¨ConfigSetting.h¨
#include ¨GPIO.h¨
```

Functions

- eConfigSettingMode_t ConfigSetting_GetCurrentSetting ()

Variables

- static const eConfigSettingMode_t gcConfigSettingHelper [2][2]

5.42.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-04-04

Copyright

Copyright (c) 2024

5.42.2 Function Documentation

5.42.2.1 ConfigSetting_GetCurrentSetting()

```
eConfigSettingMode_t ConfigSetting_GetCurrentSetting ( )
```

Get the existing config Setting.

Returns

eConfigSettingMode_t current config setting

5.42.3 Variable Documentation

5.42.3.1 gcConfigSettingHelper

```
const eConfigSettingMode_t gcConfigSettingHelper[2][2]  [static]
```

Initial value:

```
=
{
        [GPIO_PIN_RESET][GPIO_PIN_RESET]    = eCONFIG_SETTING_0,
        [GPIO_PIN_RESET][GPIO_PIN_SET]      = eCONFIG_SETTING_1,
        [GPIO_PIN_SET]  [GPIO_PIN_RESET]    = eCONFIG_SETTING_2,
        [GPIO_PIN_SET]  [GPIO_PIN_SET]      = eCONFIG_SETTING_3,
}
```

Utility table that maps GPIO levels with eConfigSettingMode_t.

5.43 ConfigSetting.h File Reference

Enumerations

- enum eConfigSettingMode_t { **eCONFIG_SETTING_0** , **eCONFIG_SETTING_1** , **eCONFIG_SETTING_2** , **eCONFIG_SETTING_3** , **eCONFIG_SETTING_MAX** }

Functions

- eConfigSettingMode_t ConfigSetting_GetCurrentSetting ()

5.43.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-04-04

Copyright

Copyright (c) 2024

5.43.2 Function Documentation

5.43.2.1 ConfigSetting_GetCurrentSetting()

```
eConfigSettingMode_t ConfigSetting_GetCurrentSetting ( )
```

Get the existing config Setting.

Returns

eConfigSettingMode_t current config setting

5.44 ConfigSetting.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCOMMON_CONFIGSETTING_CONFIGSETTING_H_
00015 #define APPCOMMON_CONFIGSETTING_CONFIGSETTING_H_
00016
00018
00022 typedef enum
00023 {
00024     eCONFIG_SETTING_0,
00025     eCONFIG_SETTING_1,
00026     eCONFIG_SETTING_2,
00027     eCONFIG_SETTING_3,
00028     eCONFIG_SETTING_MAX,
00029 } eConfigSettingMode_t;
00030
00032
00033 eConfigSettingMode_t ConfigSetting_GetCurrentSetting();
00034
00036
00037 #endif /* APPCOMMON_CONFIGSETTING_CONFIGSETTING_H_ */
```

5.45 Console.c File Reference

```
#include <string.h>
#include <stdio.h>
#include <stdarg.h>
#include ¨usart.h¨
#include ¨Console.h¨
#include ¨SoftTimer.h¨
#include ¨AppConfiguration.h¨
```

Functions

- void Console_cbCommandReceived ()
- void Console_Init ()
- void Console_DeInit ()
- eConsolePrintStatus_t Console_TransmitChar (uint8_t data)
- eConsolePrintStatus_t Console_Print (eConsolePrintLevel_t currentLevel, char ∗format,...)
- eConsolePrintStatus_t Console_receive (uint8_t ∗pOutdata, uint16_t length)
- void Console_PrintProgressBar ()
- bool Console_IsCommandRaised (eConsoleCommandsEnum_t cmd)
- void Console_RaiseConsoleCmdRequest (eConsoleCommandsEnum_t cmd)
- void Console_Sync ()

Variables

- static char gDataBuffer [CONSOLE_BUFFER_SIZE] = {0}
- static sConsoleCommand_t gConsoleCommandHelperTable [eCONSOLE_MAX_COMMANDS]
- static volatile sElevatedPromptData_t gvElevatedPromptData = {.buf = {0}}

5.45.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

Copyright (c) 2023

5.45.2 Function Documentation

5.45.2.1 Console_cbCommandReceived()

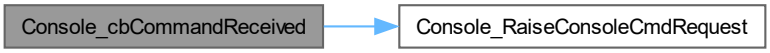
```
void Console_cbCommandReceived ( )
```

Call back function invoked by HAL_UART_RxCpltCallback upon receiving CONSOLE_COMMAND_TOKEN_SIZE characters.

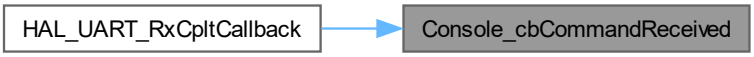
Note

CONSOLE_COMMAND_TOKEN_SIZE is set to strlen(gConsoleCommandHelperTable[eCONSOLE_↩ LEVEL2_ENABLE].CommandStr) when expecting Secret key for level 2 logs

< Check if required passkey is received, if not continue to listen on the console RxHere is the call graph for this function:



Here is the caller graph for this function:



5.45.2.2 Console_Init()

```
void Console_Init ( )
```

Initialize console.

Here is the caller graph for this function:

5.45.2.3 Console_DeInit()

```
void Console_DeInit ( )
```

DeInitialize console.

5.45.2.4 Console_TransmitChar()

```
eConsolePrintStatus_t Console_TransmitChar (
            uint8_t data )
```

Transmit character over Console.

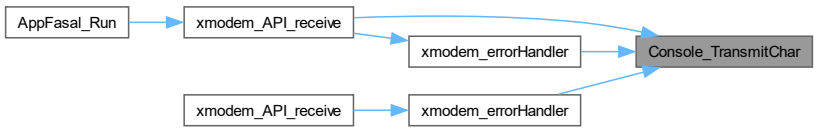
Parameters

data	char to be transmitted

Returns

eConsoleStatus_t

Here is the caller graph for this function:



5.45.2.5 Console_Print()

```
eConsolePrintStatus_t Console_Print (
            eConsolePrintLevel_t currentLevel,
            char * format,
             ...  )
```

Console print function.

Parameters

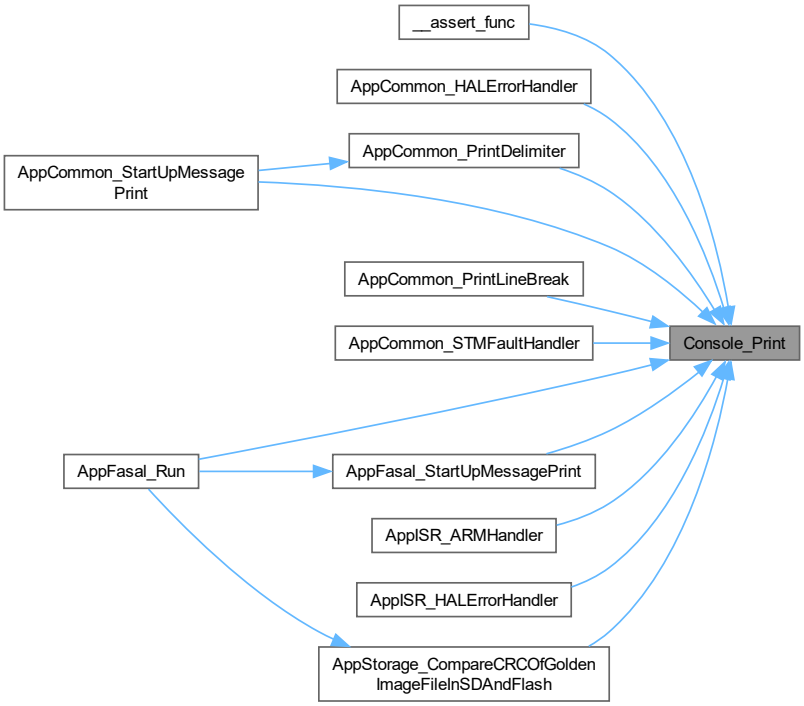
currentLevel	: What level to print the logs in
format	format string
...	

Returns

eConsolePrintStatus_t

< not supported in this Hardware

< For print to get through, print level should be sufficient, Hardware override works on levels below eCONSOLE↩
_PRINT_LVL2Here is the caller graph for this function:



5.45.2.6 Console_receive()

```
eConsolePrintStatus_t Console_receive (
            uint8_t * pOutdata,
            uint16_t length )
```

Receive data over console.

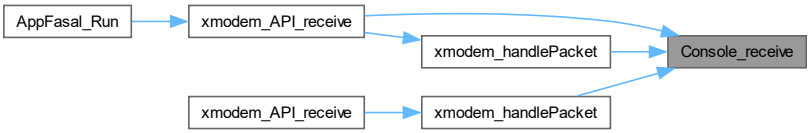
Parameters

pOutdata	Received data is saved here
length	Size of data read

Returns

eConsoleStatus_t

Here is the caller graph for this function:



5.45.2.7 Console_PrintProgressBar()

```
void Console_PrintProgressBar ( )
```

Thin wrapper around UART transmit for Progress Bar print on console.

Here is the caller graph for this function:

5.45.2.8 Console_IsCommandRaised()

```
bool Console_IsCommandRaised (
            eConsoleCommandsEnum_t cmd )
```

Utility function to check if any command request has been made.

Parameters

cmd	

Returns

true

false

5.45.2.9 Console_RaiseConsoleCmdRequest()

```
void Console_RaiseConsoleCmdRequest (
            eConsoleCommandsEnum_t cmd )
```

Raise a request to process a console command.

Here is the caller graph for this function:

5.45.2.10 Console_Sync()

```
void Console_Sync ( )
```

Deferred processing of commands received over console.

5.45.3 Variable Documentation

5.45.3.1 gDataBuffer

```
char gDataBuffer[CONSOLE_BUFFER_SIZE] = {0}  [static]
```

Global buffer used by Console print

5.45.3.2 gConsoleCommandHelperTable

```
sConsoleCommand_t gConsoleCommandHelperTable[eCONSOLE_MAX_COMMANDS]  [static]
```

Initial value:

```
=
{
}
```

Console commands, names and their flags are stored in this table.

5.45.3.3 gvElevatedPromptData

```
volatile sElevatedPromptData_t gvElevatedPromptData = {.buf = {0}}  [static]
```

Buffer for saving elevated prompt data

5.46 Console.c File Reference

```
#include <string.h>
#include <stdio.h>
#include <stdarg.h>
#include ¨Console.h¨
#include ¨AppConfiguration.h¨
#include ¨SoftTimer.h¨
#include ¨usart.h¨
```

Functions

- void Console_cbCommandReceived ()
- void Console_Init ()
- void Console_DeInit ()
- eConsolePrintStatus_t Console_TransmitChar (uint8_t data)
- eConsolePrintStatus_t Console_Print (eConsolePrintLevel_t currentLevel, char *format,...)
- eConsolePrintStatus_t Console_receive (uint8_t *pOutdata, uint16_t length)
- void Console_PrintProgressBar ()
- void Console_PrintLineBreak ()
- void Console_PrintDelimiter ()
- bool Console_IsCommandRaised (eConsoleCommandsEnum_t cmd)
- void Console_RaiseConsoleCmdRequest (eConsoleCommandsEnum_t cmd)
- void Console_Sync ()

Variables

- static char **gDataBuffer** [CONSOLE_BUFFER_SIZE] = {0}
- static sConsoleCommand_t gConsoleCommandHelperTable [eCONSOLE_MAX_COMMANDS] = {}
- static volatile sElevatedPromptData_t **gvElevatedPromptData** = {.buf = {0}}

5.46.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

5.46.2 Function Documentation

5.46.2.1 Console_cbCommandReceived()

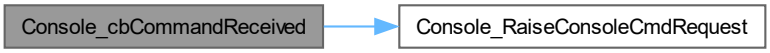
```
void Console_cbCommandReceived ( )
```

Call back function invoked by HAL_UART_RxCpltCallback upon receiving CONSOLE_COMMAND_TOKEN_SIZE characters.

Note

CONSOLE_COMMAND_TOKEN_SIZE is set to strlen(gConsoleCommandHelperTable[eCONSOLE_↩ LEVEL2_ENABLE].CommandStr) when expecting Secret key for level 2 logs

< Check if required pass-key is received, if not continue to listen on the console RxHere is the call graph for this function:



5.46.2.2 Console_Init()

```
void Console_Init ( )
```

Initialize console.

5.46.2.3 Console_DeInit()

```
void Console_DeInit ( )
```

DeInitialize console.

5.46.2.4 Console_TransmitChar()

```
eConsolePrintStatus_t Console_TransmitChar (
            uint8_t data )
```

Transmit character over Console.

Parameters

data	char to be transmitted

Returns

eConsoleStatus_t

< Make available the UART module.

5.46.2.5 Console_Print()

```
eConsolePrintStatus_t Console_Print (
            eConsolePrintLevel_t currentLevel,
```

```
            char * format,
             ...  )
```

Console print function.

Parameters

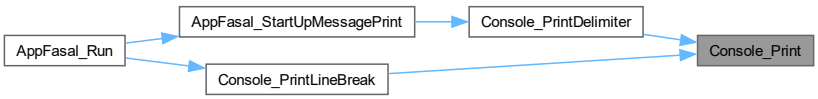
currentLevel	: What level to print the logs in
format	format string
...	

Returns

eConsolePrintStatus_t

< not supported in this Hardware

< For print to get through, print level should be sufficient, Hardware override works on levels below eCONSOLE↩ _PRINT_LVL2Here is the caller graph for this function:



5.46.2.6 Console_receive()

```
eConsolePrintStatus_t Console_receive (
            uint8_t * pOutdata,
            uint16_t length )
```

Receive data over console.

Parameters

pOutdata	Received data is saved here
length	Size of data read

Returns

eConsoleStatus_t

< Make available the UART module.

5.46.2.7 Console_PrintProgressBar()

```
void Console_PrintProgressBar ( )
```

Thin wrapper around UART transmit for Progress Bar print on console.

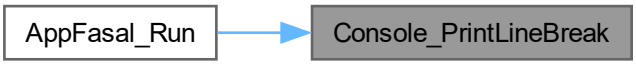
5.46.2.8 Console_PrintLineBreak()

```
void Console_PrintLineBreak ( )
```

Utility function to print delimiter used in the console.

Here is the call graph for this function:

Here is the caller graph for this function:

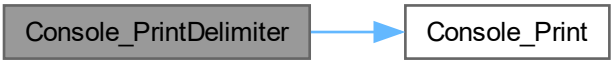


5.46.2.9 Console_PrintDelimiter()

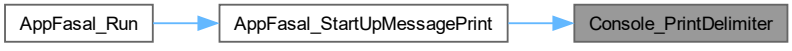
```
void Console_PrintDelimiter ( )
```

Utility function to print delimiter used in the console.

Here is the call graph for this function:



Here is the caller graph for this function:



5.46.2.10 Console_IsCommandRaised()

```
bool Console_IsCommandRaised (
            eConsoleCommandsEnum_t cmd )
```

Utility function to check if any command request has been made.

Parameters

cmd	

Returns

true

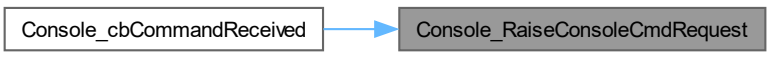
false

5.46.2.11 Console_RaiseConsoleCmdRequest()

```
void Console_RaiseConsoleCmdRequest (
            eConsoleCommandsEnum_t cmd )
```

Raise a request to process a console command.

Here is the caller graph for this function:



5.46.2.12 Console_Sync()

```
void Console_Sync ( )
```

Deferred processing of commands received over console.

5.46.3 Variable Documentation

5.46.3.1 gConsoleCommandHelperTable

```
sConsoleCommand_t gConsoleCommandHelperTable[eCONSOLE_MAX_COMMANDS] = {}  [static]
```

Console commands, names and their flags are stored in this table.

5.47 Console.h File Reference

```
#include <stdbool.h>
#include "Utility.h"
```

Data Structures

- struct sElevatedPromptData_t
- struct sConsoleCommand_t

Macros

- #define CONSOLE_BUFFER_SIZE (512u)
- #define **CONSOLE_UART_TIMEOUT_MS** (1000u)
- #define **CONSOLE_PROMPT_BUFFER_SIZE** (8u)
- #define **CONSOLE_COMMAND_TOKEN_SIZE** (3u)
- #define **CONSOLE_UART_HANDLE** (&huart1)

Typedefs

- typedef void(* pfConsoleCommandActor_t) (void)

Enumerations

- enum eConsolePrintStatus_t { **eCONSOLE_SUCCESS** , **eCONSOLE_FAIL** , **eCONSOLE_STATUS_MAX** }
- enum eConsoleCommandsEnum_t { **eCONSOLE_LEVEL1_ENABLE** , **eCONSOLE_LEVEL2_REQUEST** , **eCONSOLE_LEVEL2_ENABLE** , **eCONSOLE_ERASE_FLASH_REQUEST** , **eCONSOLE_SENSOR_TEST_REQUEST** , **eCONSOLE_MAX_COMMANDS** }
- enum eConsolePrintLevel_t { **eCONSOLE_PRINT_LVL0** , **eCONSOLE_PRINT_LVL1** , **eCONSOLE_PRINT_LVL2** , **eCONSOLE_PRINT_LVL_MAX** }

Functions

- void Console_Init ()
- void Console_DeInit ()
- eConsolePrintStatus_t Console_TransmitChar (uint8_t data)
- eConsolePrintStatus_t Console_Print (eConsolePrintLevel_t currentLevel, char *format,...)
- eConsolePrintStatus_t Console_receive (uint8_t *pOutdata, uint16_t length)
- void Console_cbCommandReceived ()
- bool Console_IsCommandRaised (eConsoleCommandsEnum_t cmd)
- void Console_RaiseConsoleCmdRequest (eConsoleCommandsEnum_t cmd)
- void Console_Sync ()
- void Console_PrintProgressBar ()

5.47.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

5.47.2 Macro Definition Documentation

5.47.2.1 CONSOLE_BUFFER_SIZE

```
#define CONSOLE_BUFFER_SIZE (512u)
```

Console buffer size, To be increased if larger strings are to be printed.

Maximum expected response is just under 200 bytes

5.47.3 Typedef Documentation

5.47.3.1 pfConsoleCommandActor_t

```
typedef void(* pfConsoleCommandActor_t) (void)
```

Console command handlers.

5.47.4 Enumeration Type Documentation

5.47.4.1 eConsolePrintStatus_t

```
enum eConsolePrintStatus_t
```

Enum that maintains Console status.

5.47.4.2 eConsoleCommandsEnum_t

```
enum eConsoleCommandsEnum_t
```

Command names are enumerated here.

5.47.4.3 eConsolePrintLevel_t

```
enum eConsolePrintLevel_t
```

Console print level.

5.47.5 Function Documentation

5.47.5.1 Console_Init()

```
void Console_Init ( )
```

Initialize console.

5.47.5.2 Console_DeInit()

```
void Console_DeInit ( )
```

DeInitialize console.

5.47.5.3 Console_TransmitChar()

```
eConsolePrintStatus_t Console_TransmitChar (
            uint8_t data )
```

Transmit character over Console.

Parameters

data	char to be transmitted

Returns

eConsoleStatus_t

< Make available the UART module.

5.47.5.4 Console_Print()

```
eConsolePrintStatus_t Console_Print (
            eConsolePrintLevel_t currentLevel,
            char * format,
             ... )
```

Console print function.

Parameters

currentLevel	: What level to print the logs in
format	format string
...	

Returns

eConsolePrintStatus_t

Parameters

currentLevel	: What level to print the logs in
format	format string
...	

Returns

eConsolePrintStatus_t

< not supported in this Hardware

< For print to get through, print level should be sufficient, Hardware override works on levels below eCONSOLE↩ _PRINT_LVL2

< not supported in this Hardware

< For print to get through, print level should be sufficient, Hardware override works on levels below eCONSOLE↩ _PRINT_LVL2

5.47.5.5 Console_receive()

```
eConsolePrintStatus_t Console_receive (
            uint8_t * pOutdata,
            uint16_t length )
```

Receive data over console.

Parameters

pOutdata	Received data is saved here
length	Size of data read

Returns

eConsoleStatus_t

< Make available the UART module.

5.47.5.6 Console_cbCommandReceived()

```
void Console_cbCommandReceived ( )
```

Call back function invoked by HAL_UART_RxCpltCallback upon receiving CONSOLE_COMMAND_TOKEN_SIZE characters.

Note

CONSOLE_COMMAND_TOKEN_SIZE is set to strlen(gConsoleCommandHelperTable[eCONSOLE_↩ LEVEL2_ENABLE].CommandStr) when expecting Secret key for level 2 logs

< Check if required passkey is received, if not continue to listen on the console Rx

< Check if required pass-key is received, if not continue to listen on the console Rx

5.47.5.7 Console_IsCommandRaised()

```
bool Console_IsCommandRaised (
            eConsoleCommandsEnum_t cmd )
```

Utility function to check if any command request has been made.

Parameters

cmd	

Returns

true

false

Parameters

cmd	

Returns

true

false

5.47.5.8 Console_RaiseConsoleCmdRequest()

```
void Console_RaiseConsoleCmdRequest (
            eConsoleCommandsEnum_t cmd )
```

Raise a request to process a console command.

5.47.5.9 Console_Sync()

```
void Console_Sync ( )
```

Deferred processing of commands received over console.

5.47.5.10 Console_PrintProgressBar()

```
void Console_PrintProgressBar ( )
```

Thin wrapper around UART transmit for Progress Bar print on console.

5.48 AppCommon/Console/Console.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef CONSOLE_CONSOLE_H_
00015 #define CONSOLE_CONSOLE_H_
00016
00018
00019 #include <stdbool.h>
00020 #include "Utility.h"
00021
00023
00024
00029 #define CONSOLE_BUFFER_SIZE          (512u)
00030 #define CONSOLE_UART_TIMEOUT_MS      (1000u)
00031 #define CONSOLE_PROMPT_BUFFER_SIZE   (8u)
00032 #define CONSOLE_COMMAND_TOKEN_SIZE   (3u)
00033
00034 #define CONSOLE_UART_HANDLE          (&huart1)
00035
00037
00042 typedef void (*pfConsoleCommandActor_t)(void);
00043
00048 typedef enum
00049 {
00050     eCONSOLE_SUCCESS,
00051     eCONSOLE_FAIL,
00052     eCONSOLE_STATUS_MAX
00053 }eConsolePrintStatus_t;
00054
00059 typedef enum
00060 {
00061     eCONSOLE_LEVEL1_ENABLE,
00062     eCONSOLE_LEVEL2_REQUEST,
00063     eCONSOLE_LEVEL2_ENABLE,
00064     eCONSOLE_ERASE_FLASH_REQUEST,
00065     eCONSOLE_SENSOR_TEST_REQUEST,
00066     eCONSOLE_MAX_COMMANDS
00067 }eConsoleCommandsEnum_t;
00068
00073 typedef enum
00074 {
00075     eCONSOLE_PRINT_LVL0,
00076     eCONSOLE_PRINT_LVL1,
00077     eCONSOLE_PRINT_LVL2,
00078     eCONSOLE_PRINT_LVL_MAX,
00079 }eConsolePrintLevel_t;
00080
00082
00087 typedef struct
00088 {
00089     char buf[CONSOLE_PROMPT_BUFFER_SIZE];
00090 }sElevatedPromptData_t;
00091
00096 typedef struct
00097 {
00098     const char* CommandName;
00099     char CommandStr[BUFFER_SIZE_8];
00100     volatile bool IsCommandRaised;
00101     volatile bool IsCommandServiced;
```

```
00102     pfConsoleCommandActor_t pfConsoleCommandActor;
00103 }sConsoleCommand_t;
00104
00106
00107 void Console_Init();
00108 void Console_DeInit();
00109 eConsolePrintStatus_t Console_TransmitChar(uint8_t data);
00110 eConsolePrintStatus_t Console_Print(eConsolePrintLevel_t currentLevel, char *format,...);
00111 eConsolePrintStatus_t Console_receive(uint8_t* pOutdata, uint16_t length);
00112 void Console_cbCommandReceived();
00113 bool Console_IsCommandRaised(eConsoleCommandsEnum_t cmd);
00114 void Console_RaiseConsoleCmdRequest(eConsoleCommandsEnum_t cmd);
00115 void Console_Sync();
00116 void Console_PrintProgressBar();
00117
00119
00120 #endif /* CONSOLE_CONSOLE_H_ */
```

5.49 Console.h File Reference

```
#include <stdbool.h>
#include <stdint.h>
#include ¨CommonDefinitions.h¨
```

Data Structures

- struct sElevatedPromptData_t
- struct sConsoleCommand_t

Macros

- #define **CONSOLE_BUFFER_SIZE** (512u)
- #define **CONSOLE_UART_TIMEOUT_MS** (1000u)
- #define **CONSOLE_PROMPT_BUFFER_SIZE** (8u)
- #define **CONSOLE_COMMAND_TOKEN_SIZE** (3u)
- #define **CONSOLE_UART_HANDLE** (&huart1)

Typedefs

- typedef void(∗ pfConsoleCommandActor_t) (void)

Enumerations

- enum eConsolePrintStatus_t { **eCONSOLE_SUCCESS** , **eCONSOLE_FAIL** , **eCONSOLE_STATUS_MAX** }
- enum eConsoleCommandsEnum_t { **eCONSOLE_LEVEL1_ENABLE** , **eCONSOLE_LEVEL2_REQUEST** , **eCONSOLE_LEVEL2_ENABLE** , **eCONSOLE_ERASE_FLASH_REQUEST** , **eCONSOLE_SENSOR_TEST_REQUEST** , **eCONSOLE_MAX_COMMANDS** }
- enum eConsolePrintLevel_t { **eCONSOLE_PRINT_LVL0** , **eCONSOLE_PRINT_LVL1** , **eCONSOLE_↩ PRINT_LVL2** , **eCONSOLE_PRINT_LVL_MAX** }

Functions

- eConsolePrintStatus_t Console_TransmitChar (uint8_t data)
- eConsolePrintStatus_t Console_Print (eConsolePrintLevel_t currentLevel, char *format,...)
- eConsolePrintStatus_t Console_receive (uint8_t *pOutdata, uint16_t length)
- void Console_Init ()
- void Console_DeInit ()
- void Console_cbCommandReceived ()
- void Console_RaiseConsoleCmdRequest (eConsoleCommandsEnum_t cmd)
- void Console_Sync ()
- void Console_PrintProgressBar ()
- void Console_PrintDelimiter ()
- void Console_PrintLineBreak ()
- bool Console_IsCommandRaised (eConsoleCommandsEnum_t cmd)

5.49.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-05-24

Copyright

5.49.2 Typedef Documentation

5.49.2.1 pfConsoleCommandActor_t

```
typedef void(* pfConsoleCommandActor_t) (void)
```

Console command handlers.

5.49.3 Enumeration Type Documentation

5.49.3.1 eConsolePrintStatus_t

```
enum eConsolePrintStatus_t
```

Enum that maintains Console status.

5.49.3.2 eConsoleCommandsEnum_t

```
enum eConsoleCommandsEnum_t
```

Command names are enumerated here.

5.49.3.3 eConsolePrintLevel_t

```
enum eConsolePrintLevel_t
```

Console print level.

5.49.4 Function Documentation

5.49.4.1 Console_TransmitChar()

```
eConsolePrintStatus_t Console_TransmitChar (
            uint8_t data )
```

Transmit character over Console.

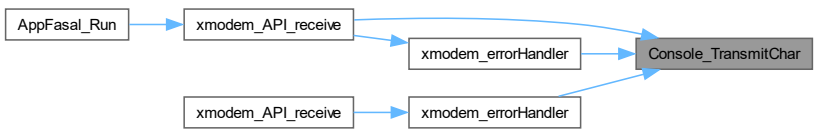
Parameters

data	char to be transmitted

Returns

eConsoleStatus_t

< Make available the UART module.Here is the caller graph for this function:



5.49.4.2 Console_Print()

```
eConsolePrintStatus_t Console_Print (
            eConsolePrintLevel_t currentLevel,
            char * format,
             ... )
```

Console print function.

Parameters

currentLevel	: What level to print the logs in
format	format string
...	

Returns

eConsolePrintStatus_t

Parameters

currentLevel	: What level to print the logs in
format	format string
...	

Returns

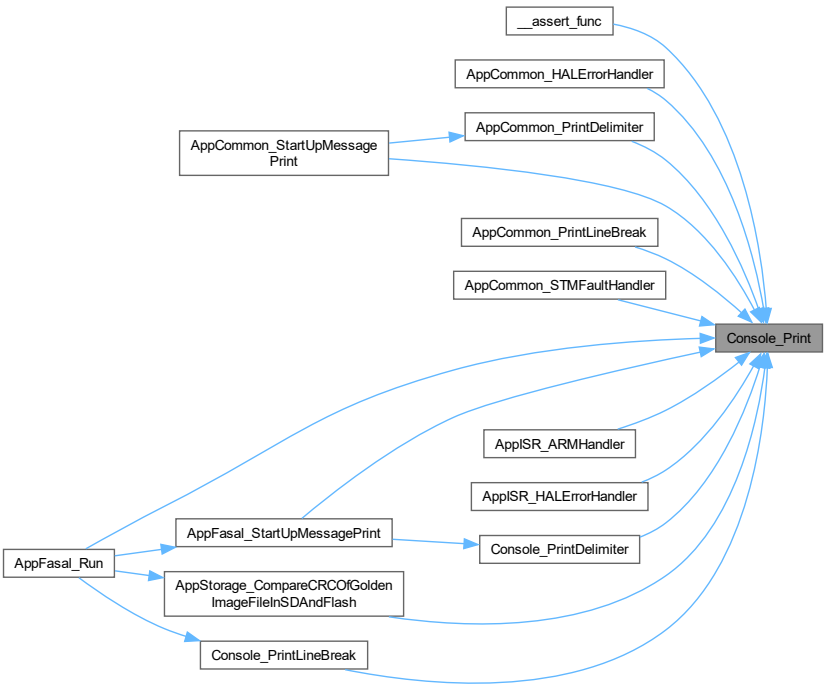
eConsolePrintStatus_t

< not supported in this Hardware

< For print to get through, print level should be sufficient, Hardware override works on levels below eCONSOLE↩ _PRINT_LVL2

< not supported in this Hardware

< For print to get through, print level should be sufficient, Hardware override works on levels below eCONSOLE↩ _PRINT_LVL2Here is the caller graph for this function:



5.49.4.3 Console_receive()

```
eConsolePrintStatus_t Console_receive (
            uint8_t * pOutdata,
            uint16_t length )
```

Receive data over console.

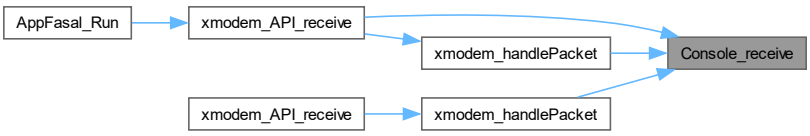
Parameters

pOutdata	Received data is saved here
length	Size of data read

Returns

eConsoleStatus_t

< Make available the UART module.Here is the caller graph for this function:



5.49.4.4 Console_Init()

```
void Console_Init ( )
```

Initialize console.

Here is the caller graph for this function:

5.49.4.5 Console_DeInit()

```
void Console_DeInit ( )
```

DeInitialize console.

5.49.4.6 Console_cbCommandReceived()

```
void Console_cbCommandReceived ( )
```

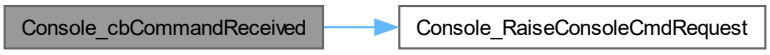
Call back function invoked by HAL_UART_RxCpltCallback upon receiving CONSOLE_COMMAND_TOKEN_SIZE characters.

Note

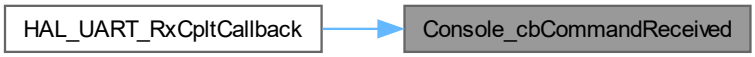
CONSOLE_COMMAND_TOKEN_SIZE is set to strlen(gConsoleCommandHelperTable[eCONSOLE_↩ LEVEL2_ENABLE].CommandStr) when expecting Secret key for level 2 logs

< Check if required passkey is received, if not continue to listen on the console Rx

< Check if required pass-key is received, if not continue to listen on the console RxHere is the call graph for this function:



Here is the caller graph for this function:

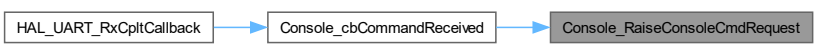


5.49.4.7 Console_RaiseConsoleCmdRequest()

```
void Console_RaiseConsoleCmdRequest (
            eConsoleCommandsEnum_t cmd )
```

Raise a request to process a console command.

Here is the caller graph for this function:



5.49.4.8 Console_Sync()

```
void Console_Sync ( )
```

Deferred processing of commands received over console.

5.49.4.9 Console_PrintProgressBar()

```
void Console_PrintProgressBar ( )
```

Thin wrapper around UART transmit for Progress Bar print on console.

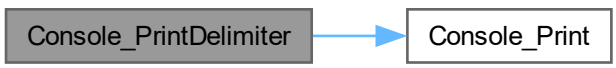
Here is the caller graph for this function:

5.49.4.10 Console_PrintDelimiter()

```
void Console_PrintDelimiter ( )
```

Utility function to print delimiter used in the console.

Here is the call graph for this function:



Here is the caller graph for this function:

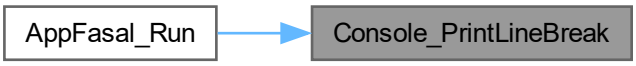
5.49.4.11 Console_PrintLineBreak()

```
void Console_PrintLineBreak ( )
```

Utility function to print delimiter used in the console.

Here is the call graph for this function:

Here is the caller graph for this function:



5.49.4.12 Console_IsCommandRaised()

```
bool Console_IsCommandRaised (
            eConsoleCommandsEnum_t cmd )
```

Utility function to check if any command request has been made.

Parameters

cmd	

Returns

true

false

Parameters

cmd	

Returns

true

false

5.50 AppInterface/Console/Console.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef CONSOLE_CONSOLE_H_
00015 #define CONSOLE_CONSOLE_H_
00016
00018
00019 #include <stdbool.h>
00020 #include <stdint.h>
00021 #include "CommonDefinitions.h"
00022
00024
00025 #define CONSOLE_BUFFER_SIZE (512u)
00026 #define CONSOLE_UART_TIMEOUT_MS (1000u)
00027 #define CONSOLE_PROMPT_BUFFER_SIZE (8u)
00028 #define CONSOLE_COMMAND_TOKEN_SIZE (3u)
00029 #define CONSOLE_UART_HANDLE (&huart1)
00030
00032
00037 typedef void (*pfConsoleCommandActor_t)(void);
00038
00043 typedef enum
00044 {
00045     eCONSOLE_SUCCESS,
00046     eCONSOLE_FAIL,
00047     eCONSOLE_STATUS_MAX
00048 } eConsolePrintStatus_t;
00049
00054 typedef enum
00055 {
00056     eCONSOLE_LEVEL1_ENABLE,
00057     eCONSOLE_LEVEL2_REQUEST,
00058     eCONSOLE_LEVEL2_ENABLE,
00059     eCONSOLE_ERASE_FLASH_REQUEST,
00060     eCONSOLE_SENSOR_TEST_REQUEST,
00061     eCONSOLE_MAX_COMMANDS
00062 } eConsoleCommandsEnum_t;
00063
00068 typedef enum
00069 {
00070     eCONSOLE_PRINT_LVL0,
00071     eCONSOLE_PRINT_LVL1,
00072     eCONSOLE_PRINT_LVL2,
00073     eCONSOLE_PRINT_LVL_MAX,
00074 } eConsolePrintLevel_t;
00075
00077
00082 typedef struct
00083 {
00084     char buf[CONSOLE_PROMPT_BUFFER_SIZE];
00085 } sElevatedPromptData_t;
00086
00091 typedef struct
00092 {
00093     const char* CommandName;
00094     char CommandStr[BUFFER_SIZE_8];
00095     volatile bool IsCommandRaised;
00096     volatile bool IsCommandServiced;
00097     pfConsoleCommandActor_t pfConsoleCommandActor;
00098 } sConsoleCommand_t;
00099
00101
00102 eConsolePrintStatus_t Console_TransmitChar(uint8_t data);
00103 eConsolePrintStatus_t Console_Print(eConsolePrintLevel_t currentLevel, char* format, ...);
00104 eConsolePrintStatus_t Console_receive(uint8_t* pOutdata, uint16_t length);
00105 void Console_Init();
00106 void Console_DeInit();
00107 void Console_cbCommandReceived();
00108 void Console_RaiseConsoleCmdRequest(eConsoleCommandsEnum_t cmd);
00109 void Console_Sync();
00110 void Console_PrintProgressBar();
00111 void Console_PrintDelimiter();
00112 void Console_PrintLineBreak();
```

```
00113 bool Console_IsCommandRaised(eConsoleCommandsEnum_t cmd);
00114
00116
00117 #endif /* CONSOLE_CONSOLE_H_ */
```

5.51 PushButton.c File Reference

```
#include <stdint.h>
#include <stdbool.h>
#include "gpio.h"
#include "PushButton.h"
```

Functions

- bool PushButton_IsFlashButtonPressed ()

5.51.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-04-04

Copyright

Copyright (c) 2024

5.51.2 Function Documentation

5.51.2.1 PushButton_IsFlashButtonPressed()

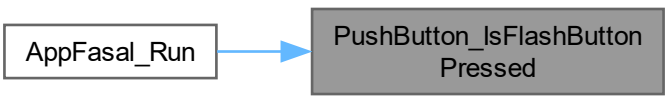
```
bool PushButton_IsFlashButtonPressed ( )
```

Check if Push Button is pressed.

Returns

true if button is pressed, false if not during function invocation

Here is the caller graph for this function:



5.52 PushButton.h File Reference

```
#include <stdbool.h>
```

Functions

- bool PushButton_IsFlashButtonPressed ()

5.52.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-04-04

Copyright

Copyright (c) 2024

5.52.2 Function Documentation

5.52.2.1 PushButton_IsFlashButtonPressed()

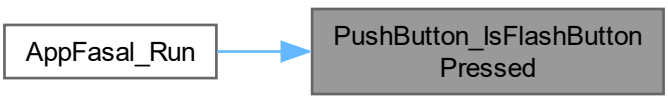
```
bool PushButton_IsFlashButtonPressed ( )
```

Check if Push Button is pressed.

Returns

true if button is pressed, false if not during function invocation

Here is the caller graph for this function:



5.53 PushButton.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCOMMON_PUSHBUTTON_PUSHBUTTON_H_
00015 #define APPCOMMON_PUSHBUTTON_PUSHBUTTON_H_
00016
00018
00019 #include <stdbool.h>
00020
00022
00023 bool PushButton_IsFlashButtonPressed();
00024
00026
00027 #endif /* APPCOMMON_PUSHBUTTON_PUSHBUTTON_H_ */
```

5.54 AppFasal.c File Reference

```
#include <assert.h>
#include <stdbool.h>
#include ¨AppConfiguration.h¨
#include ¨AppFasal.h¨
#include ¨AppFlash_API.h¨
#include ¨AppIndication.h¨
#include ¨AppProfiler.h¨
#include ¨AppResetAndError.h¨
#include ¨AppSD_API.h¨
#include ¨AppStorage.h¨
#include ¨AppStorageDataStructures.h¨
#include ¨ConfigSetting.h¨
#include ¨Console.h¨
#include ¨DebugPrint.h¨
#include ¨PushButton.h¨
#include ¨SoftTimer.h¨
#include ¨xmodem.h¨
#include ¨Version.h¨
```

Functions

- static void AppFasal_Init ()
- static void AppFasal_StartUpMessagePrint ()
- eAppFasalStates_t AppFasal_Run ()

Variables

- static const eAppIndicationStates_t gcIndicationToAppStateMap [eAPP_MAX_STATE]

5.54.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-06-23

Copyright

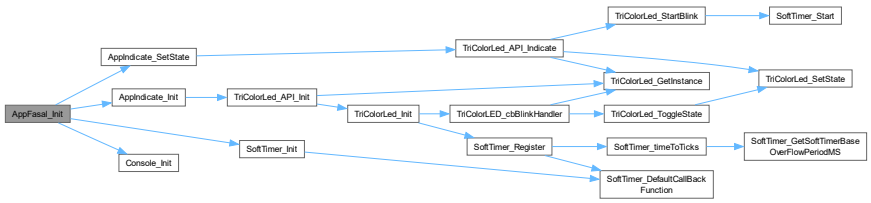
5.54.2 Function Documentation

5.54.2.1 AppFasal_Init()

```
static void AppFasal_Init ( )  [static]
```

Initialize all Application modules here.

Here is the call graph for this function:



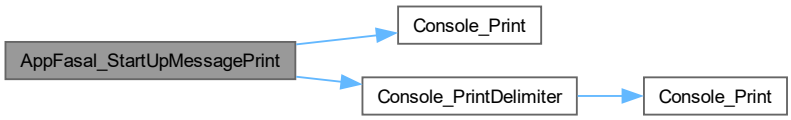
Here is the caller graph for this function:

5.54.2.2 AppFasal_StartUpMessagePrint()

```
static void AppFasal_StartUpMessagePrint ( )  [static]
```

Start up message print.

Here is the call graph for this function:



Here is the caller graph for this function:

5.54.2.3 AppFasal_Run()

```
eAppFasalStates_t AppFasal_Run ( )
```

Application Run.

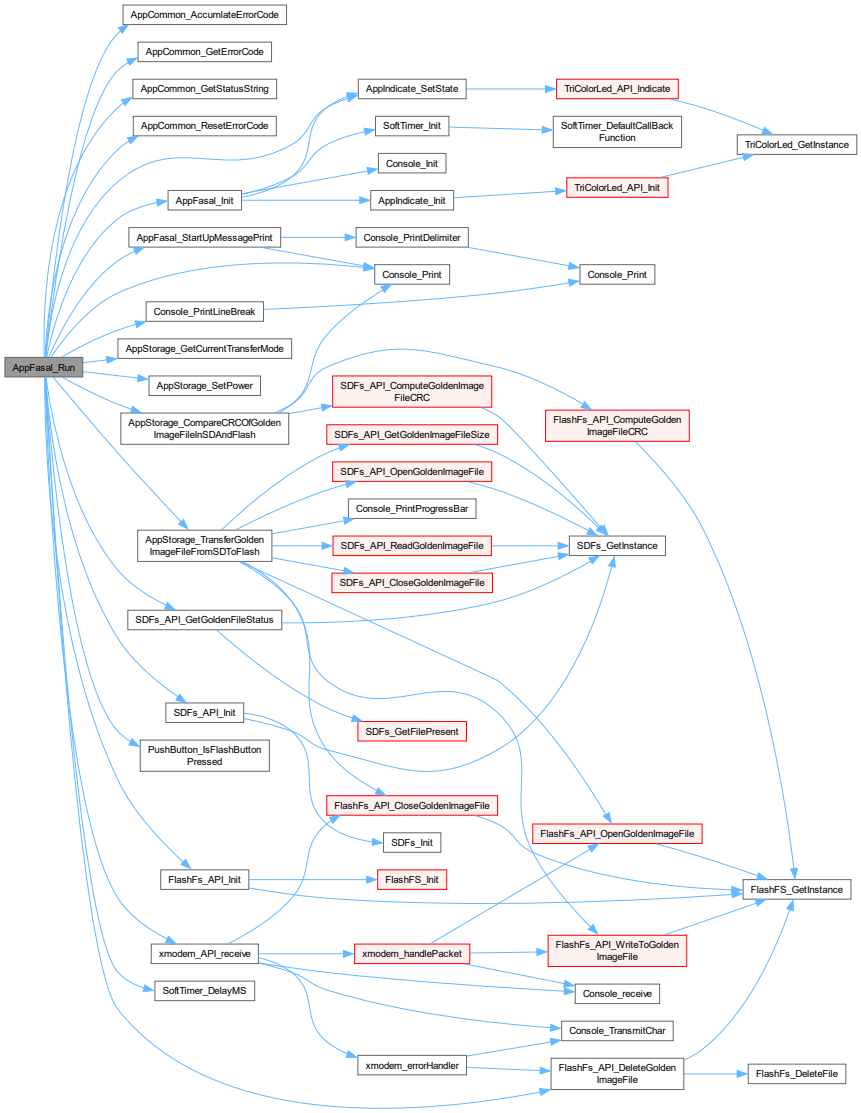
Returns

eAppFasalStates_t

< Set power to External Flash prior to Initializing the same

< Stop powering the external flash since transfer operation is complete

< Errors from previous run if any must be cleared hereHere is the call graph for this function:



5.54.3 Variable Documentation

5.54.3.1 gcIndicationToAppStateMap

```
const eAppIndicationStates_t gcIndicationToAppStateMap[eAPP_MAX_STATE]  [static]
```

Initial value:

```
=
{
        [eAPP_INIT]             = eIND_BLUE_0,
        [eAPP_STARTUP_MSG]      = eIND_BLUE_250MS,
        [eAPP_BUTTON_WAIT]      = eIND_BLUE_500MS,
        [eAPP_SD_INIT]          = eIND_BLUE_1000MS,
        [eAPP_SD_CHECK]         = eIND_BLUE_1000MS,
        [eAPP_FLASH_INIT]       = eIND_BLUE_1000MS,
        [eAPP_MODE_SELECTION]   = eIND_BLUE_1000MS,
        [eAPP_SD_FLASH_TRANSFER]= eIND_YELLOW_1000MS,
        [eAPP_XMODEM_TRANSFER]  = eIND_YELLOW_1000MS,
        [eAPP_CRC_COMPARE]      = eIND_YELLOW_1000MS,
```

```
        [eAPP_TRANSFER_SUCCESS] = eIND_GREEN_0,
        [eAPP_SD_FAIL]          = eIND_RED_250MS,
        [eAPP_SD_FILE_FAIL]     = eIND_RED_250MS,
        [eAPP_FLASH_FAIL]       = eIND_RED_250MS,
        [eAPP_TRANSFER_FAIL]    = eIND_RED_250MS,
        [eAPP_CRC_FAIL]         = eIND_RED_250MS,
        [eAPP_END]              = eIND_NO_CHANGE,
}
```

Map App state with indication state.

5.55 AppFasal.h File Reference

Enumerations

- enum eAppFasalStates_t {
  eAPP_INIT , **eAPP_STARTUP_MSG** , **eAPP_BUTTON_WAIT** , **eAPP_SD_INIT** ,
  eAPP_SD_CHECK , **eAPP_FLASH_INIT** , **eAPP_MODE_SELECTION** , **eAPP_SD_FLASH_TRANSFER** ,
  eAPP_XMODEM_TRANSFER , **eAPP_CRC_COMPARE** , **eAPP_TRANSFER_SUCCESS** , **eAPP_SD_↩ FAIL** ,
  eAPP_SD_FILE_FAIL , **eAPP_FLASH_FAIL** , **eAPP_TRANSFER_FAIL** , **eAPP_CRC_FAIL** ,
  eAPP_END , **eAPP_MAX_STATE** }

Functions

- eAppFasalStates_t AppFasal_Run ()

5.55.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-06-23

Copyright

5.55.2 Enumeration Type Documentation

5.55.2.1 eAppFasalStates_t

```
enum eAppFasalStates_t
```

Enumeration for various states of AppFasal_RunNova.

5.55.3 Function Documentation

5.55.3.1 AppFasal_Run()

```
eAppFasalStates_t AppFasal_Run ( )
```

Application Run.

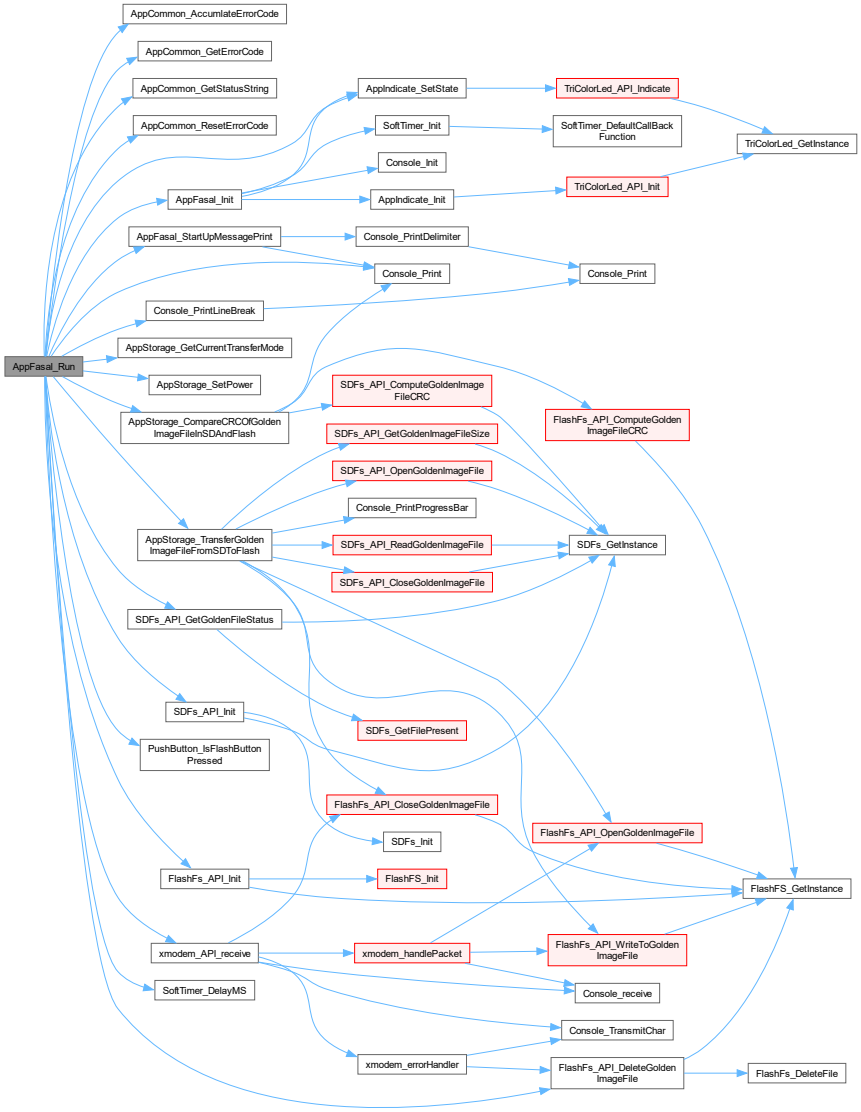
Returns

eAppFasalStates_t

< Set power to External Flash prior to Initializing the same

< Stop powering the external flash since transfer operation is complete

< Errors from previous run if any must be cleared hereHere is the call graph for this function:



5.56 AppFasal.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPFASAL_APPFASAL_H_
00015 #define APPFASAL_APPFASAL_H_
00016
00018
00023 typedef enum
00024 {
00025     eAPP_INIT,
00026     eAPP_STARTUP_MSG,
00027     eAPP_BUTTON_WAIT,
00028     eAPP_SD_INIT,
00029     eAPP_SD_CHECK,
00030     eAPP_FLASH_INIT,
00031     eAPP_MODE_SELECTION,
00032     eAPP_SD_FLASH_TRANSFER,
00033     eAPP_XMODEM_TRANSFER,
00034     eAPP_CRC_COMPARE,
00035     eAPP_TRANSFER_SUCCESS,
00036     eAPP_SD_FAIL,
00037     eAPP_SD_FILE_FAIL,
00038     eAPP_FLASH_FAIL,
00039     eAPP_TRANSFER_FAIL,
00040     eAPP_CRC_FAIL,
00041     eAPP_END,
00042     eAPP_MAX_STATE,
00043 } eAppFasalStates_t;
00044
00046
00047 eAppFasalStates_t AppFasal_Run();
00048
00050
00051 #endif /* APPFASAL_APPFASAL_H_ */
```

5.57 AppResetAndError.c File Reference

```
#include <stdint.h>
#include ¨AppResetAndError.h¨
#include ¨AppConfiguration.h¨
#include ¨AppIndication.h¨
#include ¨Console.h¨
```

Functions

- eDeviceErrorCode_t AppCommon_GetErrorCode ()
- void AppCommon_AccumlateErrorCode (eDeviceErrorCode_t err)
- void AppCommon_ResetErrorCode ()
- const char ∗const AppCommon_GetStatusString (int status)

Variables

- static eDeviceErrorCode_t **gErrorCode** = eERR_NO_ERRORS

5.57.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-10-24

Copyright

5.57.2 Function Documentation

5.57.2.1 AppCommon_GetErrorCode()

```
eDeviceErrorCode_t AppCommon_GetErrorCode ( )
```

Setter for gErrorCode, previous values are unaffected.

Returns

eDeviceErrorCode_t

5.57.2.2 AppCommon_AccumlateErrorCode()

```
void AppCommon_AccumlateErrorCode (
            eDeviceErrorCode_t err )
```

Setter for gErrorCode, previous values are unaffected.

5.57.2.3 AppCommon_ResetErrorCode()

```
void AppCommon_ResetErrorCode ( )
```

Reset error Code.

5.57.2.4 AppCommon_GetStatusString()

```
const char *const AppCommon_GetStatusString (
            int status )
```

Utility function returns ¨Success¨ or ¨failure¨ based on status code passed.

See also

gcStatusStringHelper

Parameters

status	0 is success across all modules in the code-base , non 0 is treated as failure

Returns

const char* const

5.58 AppResetAndError.h File Reference

Enumerations

- enum eDeviceResetCause_t {
  eRESET_CAUSE_UNKNOWN = 0 , **eRESET_CAUSE_WAKEUP** , **eRESET_CAUSE_LOW_POWER_RESET** , **eRESET_CAUSE_WINDOW_WATCHDOG_RESET** ,
  eRESET_CAUSE_INDEPENDENT_WATCHDOG_RESET , **eRESET_CAUSE_SOFTWARE_RESET** , **eRESET_CAUSE_POWER_ON_POWER_DOWN_RESET** , **eRESET_CAUSE_EXTERNAL_RESET_PIN_RESET** ,
  eRESET_CAUSE_BROWNOUT_RESET , **eRESET_CAUSE_MAX** }
- enum eDeviceErrorCode_t {
  eERR_NO_ERRORS = 0x0000 , **eERR_SDCARD_NOT_FOUND** = 0x0001 , **eERR_SDCARD_FILE_NOT_FOUND** = 0x0002 , **eERR_FLASH_NOT_FOUND** = 0x0004 ,
  eERR_FLASH_TRANSFER_FAILURE = 0x0008 , **eERR_CRC_FAILURE** = 0x0010 , **eERR_UNUSED_1** = 0x0020 , **eERR_UNUSED_2** = 0x0040 ,
  eERR_UNUSED_3 = 0x0080 , **eERR_UNUSED_4** = 0x0100 , **eERR_UNUSED_5** = 0X0200 , **eERR_UNUSED_6** = 0x0400 ,
  eERR_UNUSED_7 = 0x0800 , **eERR_STM_HAL_FAILURE** = 0x1000 , **eERR_ARM_FAULT** = 0x2000 , **eERR_ASSERTION_FAILURE** = 0x4000 ,
  eERR_SLEEP_FAILURE = 0x8000 }

Functions

- void AppCommon_AccumlateErrorCode (eDeviceErrorCode_t err)
- void AppCommon_ResetErrorCode ()
- const char *const AppCommon_GetStatusString (int status)
- eDeviceErrorCode_t AppCommon_GetErrorCode ()

5.58.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-10-24

Copyright

5.58.2 Enumeration Type Documentation

5.58.2.1 eDeviceResetCause_t

```
enum eDeviceResetCause_t
```

Possible STM32 system reset causes.

5.58.2.2 eDeviceErrorCode_t

```
enum eDeviceErrorCode_t
```

Possible Errors in application.

5.58.3 Function Documentation

5.58.3.1 AppCommon_AccumlateErrorCode()

```
void AppCommon_AccumlateErrorCode (
            eDeviceErrorCode_t err )
```

Setter for gErrorCode, previous values are unaffected.

Here is the caller graph for this function:

5.58.3.2 AppCommon_ResetErrorCode()

```
void AppCommon_ResetErrorCode ( )
```

Reset error Code.

Here is the caller graph for this function:

5.58.3.3 AppCommon_GetStatusString()

```
const char *const AppCommon_GetStatusString (
            int status )
```

Utility function returns ¨Success¨ or ¨failure¨ based on status code passed.

See also

gcStatusStringHelper

Parameters

status	0 is success across all modules in the code-base , non 0 is treated as failure

Returns

const char* const

Here is the caller graph for this function:

5.58.3.4 AppCommon_GetErrorCode()

```
eDeviceErrorCode_t AppCommon_GetErrorCode ( )
```

Setter for gErrorCode, previous values are unaffected.

Returns

eDeviceErrorCode_t

Returns

eDeviceErrorCode_t

Here is the caller graph for this function:

5.59 AppResetAndError.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPCOMMON_APPCOMMON_H_
00015 #define APPCOMMON_APPCOMMON_H_
00016
00018
00023 typedef enum
00024 {
00025     eRESET_CAUSE_UNKNOWN = 0,
00026     eRESET_CAUSE_WAKEUP,
00027     eRESET_CAUSE_LOW_POWER_RESET,
00028     eRESET_CAUSE_WINDOW_WATCHDOG_RESET,
00029     eRESET_CAUSE_INDEPENDENT_WATCHDOG_RESET,
00030     eRESET_CAUSE_SOFTWARE_RESET,
00031     eRESET_CAUSE_POWER_ON_POWER_DOWN_RESET,
00032     eRESET_CAUSE_EXTERNAL_RESET_PIN_RESET,
00033     eRESET_CAUSE_BROWNOUT_RESET,
00034     eRESET_CAUSE_MAX
00035 } eDeviceResetCause_t;
00036
00037 // clang format off
00038
00043 typedef enum
00044 {
00045     eERR_NO_ERRORS = 0x0000,
00046     eERR_SDCARD_NOT_FOUND = 0x0001,
00047     eERR_SDCARD_FILE_NOT_FOUND = 0x0002,
00048     eERR_FLASH_NOT_FOUND = 0x0004,
00049     eERR_FLASH_TRANSFER_FAILURE = 0x0008,
00050     eERR_CRC_FAILURE = 0x0010,
00051     eERR_UNUSED_1 = 0x0020,
00052     eERR_UNUSED_2 = 0x0040,
00053     eERR_UNUSED_3 = 0x0080,
00054     eERR_UNUSED_4 = 0x0100,
00055     eERR_UNUSED_5 = 0X0200,
00056     eERR_UNUSED_6 = 0x0400,
00057     eERR_UNUSED_7 = 0x0800,
00058     eERR_STM_HAL_FAILURE = 0x1000,
00059     eERR_ARM_FAULT = 0x2000,
00060     eERR_ASSERTION_FAILURE = 0x4000,
00061     eERR_SLEEP_FAILURE = 0x8000,
00062 } eDeviceErrorCode_t;
00063
00064 // clang-format on
00065
00067
00068 void AppCommon_AccumlateErrorCode(eDeviceErrorCode_t err);
00069 void AppCommon_ResetErrorCode();
00070 const char* const AppCommon_GetStatusString(int status);
00071 eDeviceErrorCode_t AppCommon_GetErrorCode();
00072
00074
00075 #endif /* APPCOMMON_APPCOMMON_H_ */
```

5.60 xmodem.c File Reference

```
#include <string.h>
#include ¨AppFlash_API.h¨
#include ¨Console.h¨
#include ¨xmodem.h¨
```

Functions

- static uint16_t xmodem_computeCRC (uint8_t *data, uint16_t length)
- static xmodem_status xmodem_handlePacket (uint8_t header)
- static xmodem_status xmodem_errorHandler (uint8_t *error_number, uint8_t max_error_number)
- xmodem_status xmodem_API_receive (void)

Variables

- static uint8_t **gxModemPacketNumber** = 1u
- static uint8_t **gxModemIsFirstPacket** = false

5.60.1 Detailed Description

Author

Vishal keshava Murthy

Version

0.1

Date

2024-02-22

Copyright

Copyright (c) 2024

Note

: Original module source https://github.com/ferenc-nemeth

5.60.2 Function Documentation

5.60.2.1 xmodem_computeCRC()

```
static uint16_t xmodem_computeCRC (
            uint8_t * data,
            uint16_t length )  [static]
```

Calculates the CRC-16 for the input package.

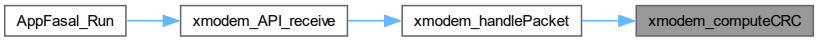
Parameters

∗data	Array of the data which we want to calculate.
length	Size of the data, either 128 or 1024 bytes.

Returns

status: The calculated CRC.

Here is the caller graph for this function:



5.60.2.2 xmodem_handlePacket()

```
static xmodem_status xmodem_handlePacket (
            uint8_t header )  [static]
```

This function handles the data packet we get from the xmodem protocol.

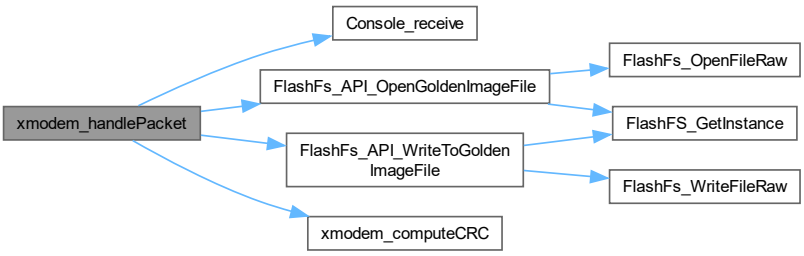
Parameters

header	SOH or STX.

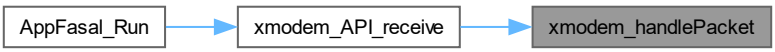
Returns

status: Report about the packet.

Here is the call graph for this function:



Here is the caller graph for this function:



5.60.2.3 xmodem_errorHandler()

```
static xmodem_status xmodem_errorHandler (
            uint8_t * error_number,
            uint8_t max_error_number )  [static]
```

Handles the xmodem error. Raises the error counter, then if the number of the errors reached critical, do a graceful abort, otherwise send a NAK.

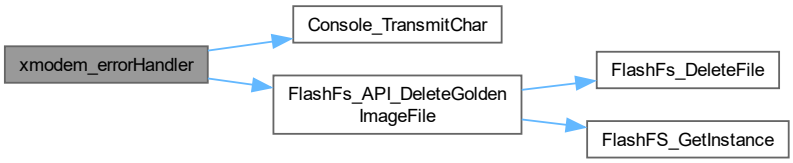
Parameters

∗error_number	Number of current errors (passed as a pointer).
max_error_number	Maximal allowed number of errors.

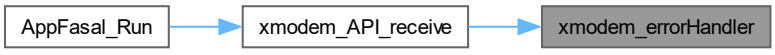
Returns

status: X_ERROR in case of too many errors, X_OK otherwise.

Here is the call graph for this function:



Here is the caller graph for this function:



5.60.2.4 xmodem_API_receive()

```
xmodem_status xmodem_API_receive (
            void  )
```

This function is the base of the Xmodem protocol. When we receive a header from UART, it decides what action it shall take.

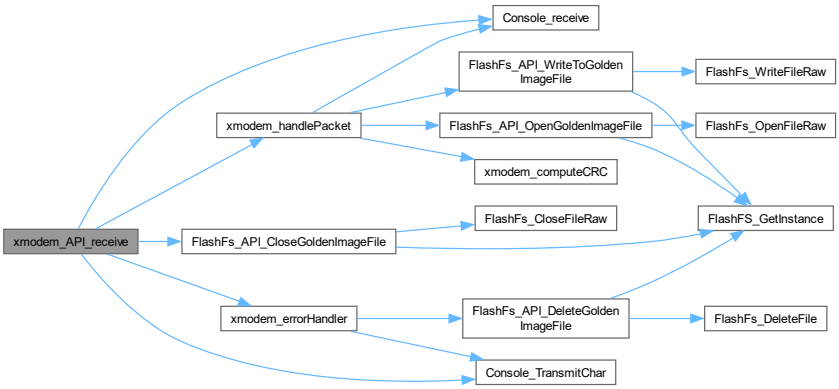
Parameters

void	

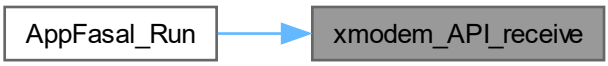
Returns

xmodem_status

Here is the call graph for this function:



Here is the caller graph for this function:



5.61 xmodem.c File Reference

```
#include <string.h>
#include ¨xmodem.h¨
#include ¨Console.h¨
#include ¨AppFlash_API.h¨
```

Functions

- static uint16_t xmodem_computeCRC (uint8_t ∗data, uint16_t length)
- static xmodem_status xmodem_handlePacket (uint8_t header)
- static xmodem_status xmodem_errorHandler (uint8_t ∗error_number, uint8_t max_error_number)
- xmodem_status xmodem_API_receive (void)

Variables

- static uint8_t gxModemPacketNumber = 1u
- static uint8_t gxModemIsFirstPacket = false

5.61.1 Detailed Description

Author

Vishal keshava Murthy

Version

0.1

Date

2024-02-22

Copyright

Copyright (c) 2024

Note

: Original module source https://github.com/ferenc-nemeth

5.61.2 Function Documentation

5.61.2.1 xmodem_computeCRC()

```
static uint16_t xmodem_computeCRC (
            uint8_t * data,
            uint16_t length )  [static]
```

Calculates the CRC-16 for the input package.

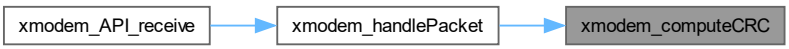
Parameters

∗data	Array of the data which we want to calculate.
length	Size of the data, either 128 or 1024 bytes.

Returns

status: The calculated CRC.

Here is the caller graph for this function:



5.61.2.2 xmodem_handlePacket()

```
static xmodem_status xmodem_handlePacket (
            uint8_t header )  [static]
```

This function handles the data packet we get from the xmodem protocol.

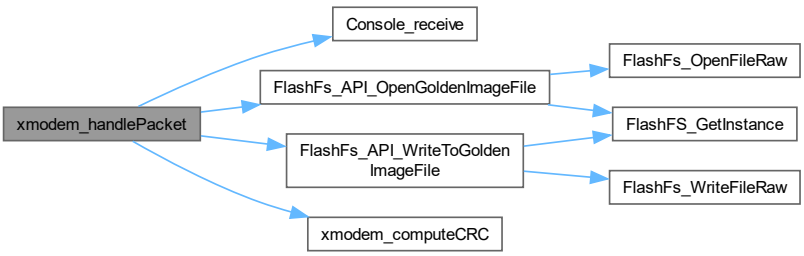
Parameters

header	SOH or STX.

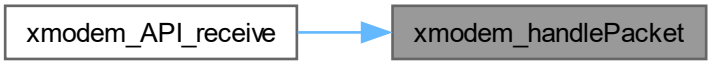
Returns

status: Report about the packet.

Here is the call graph for this function:



Here is the caller graph for this function:



5.61.2.3 xmodem_errorHandler()

```
static xmodem_status xmodem_errorHandler (
            uint8_t * error_number,
            uint8_t max_error_number )  [static]
```

Handles the xmodem error. Raises the error counter, then if the number of the errors reached critical, do a graceful abort, otherwise send a NAK.

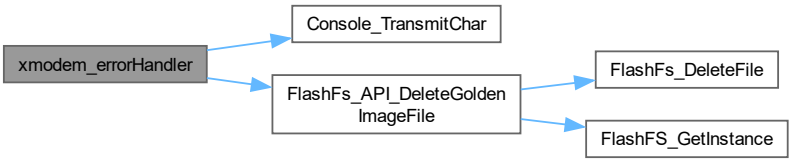
Parameters

∗error_number	Number of current errors (passed as a pointer).
max_error_number	Maximal allowed number of errors.

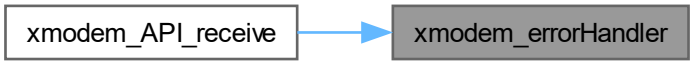
Returns

status: X_ERROR in case of too many errors, X_OK otherwise.

Here is the call graph for this function:



Here is the caller graph for this function:



5.61.2.4 xmodem_API_receive()

```
xmodem_status xmodem_API_receive (
            void  )
```

This function is the base of the Xmodem protocol. When we receive a header from UART, it decides what action it shall take.

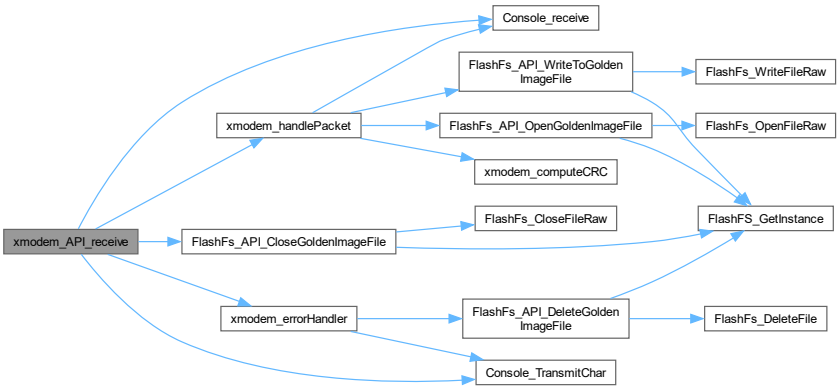
Parameters

void	

Returns

xmodem_status

Here is the call graph for this function:



5.61.3 Variable Documentation

5.61.3.1 gxModemPacketNumber

```
uint8_t gxModemPacketNumber = 1u  [static]
```

Packet number counter.

5.61.3.2 gxModemIsFirstPacket

```
uint8_t gxModemIsFirstPacket = false  [static]
```

First packet or not.

5.62 xmodem.h File Reference

```
#include ¨stdbool.h¨
```

Macros

- #define **X_MAX_ERRORS** ((uint8_t)3u)
- #define **X_PACKET_NUMBER_SIZE** ((uint16_t)2u)
- #define **X_PACKET_128_SIZE** ((uint16_t)128u)
- #define **X_PACKET_1024_SIZE** ((uint16_t)1024u)
- #define **X_PACKET_CRC_SIZE** ((uint16_t)2u)
- #define **X_PACKET_NUMBER_INDEX** ((uint16_t)0u)
- #define **X_PACKET_NUMBER_COMPLEMENT_INDEX** ((uint16_t)1u)
- #define **X_PACKET_CRC_HIGH_INDEX** ((uint16_t)0u)
- #define **X_PACKET_CRC_LOW_INDEX** ((uint16_t)1u)
- #define X_SOH ((uint8_t)0x01u)
- #define X_STX ((uint8_t)0x02u)
- #define X_EOT ((uint8_t)0x04u)
- #define X_ACK ((uint8_t)0x06u)
- #define X_NAK ((uint8_t)0x15u)
- #define X_CAN ((uint8_t)0x18u)
- #define X_C ((uint8_t)0x43u)

Enumerations

- enum xmodem_status {
  X_OK = 0x00u , X_ERROR_CRC = 0x01u , X_ERROR_NUMBER = 0x02u , X_ERROR_UART = 0x04u ,
  X_ERROR_FLASH = 0x08u , X_COMPLETE = 0x10u , X_ERROR = 0xFFu }

Functions

- xmodem_status xmodem_API_receive (void)

5.62.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-02-22

Copyright

Note

Original module source https://github.com/ferenc-nemeth

5.62.2 Macro Definition Documentation

5.62.2.1 X_SOH

```
#define X_SOH ((uint8_t)0x01u)
```

Start Of Header (128 bytes).

5.62.2.2 X_STX

```
#define X_STX ((uint8_t)0x02u)
```

Start Of Header (1024 bytes).

5.62.2.3 X_EOT

```
#define X_EOT ((uint8_t)0x04u)
```

End Of Transmission.

5.62.2.4 X_ACK

```
#define X_ACK ((uint8_t)0x06u)
```

Acknowledge.

5.62.2.5 X_NAK

```
#define X_NAK ((uint8_t)0x15u)
```

Not Acknowledge.

5.62.2.6 X_CAN

```
#define X_CAN ((uint8_t)0x18u)
```

Cancel.

5.62.2.7 X_C

```
#define X_C ((uint8_t)0x43u)
```

ASCII ¨C¨ to notify the host we want to use CRC16.

5.62.3 Enumeration Type Documentation

5.62.3.1 xmodem_status

```
enum xmodem_status
```

xModem status enums

Enumerator

X_OK	The action was successful.
X_ERROR_CRC	CRC calculation error.
X_ERROR_NUMBER	Packet number mismatch error.
X_ERROR_UART	UART communication error.
X_ERROR_FLASH	Flash related error.
X_COMPLETE	Generic error.
X_ERROR	Generic error.

5.62.4 Function Documentation

5.62.4.1 xmodem_API_receive()

```
xmodem_status xmodem_API_receive (
            void  )
```

This function is the base of the Xmodem protocol. When we receive a header from UART, it decides what action it shall take.

Parameters

void	

Returns

xmodem_status

5.63 AppInterface/xModem/xmodem.h

Go to the documentation of this file.

```
00001
00012 #ifndef XMODEM_H_
00013 #define XMODEM_H_
00014
00016
00017 #include "stdbool.h"
00018
00020
00021 /* Xmodem (128 bytes) packet format
00022  * Byte  0:       Header
00023  * Byte  1:       Packet number
00024  * Byte  2:       Packet number complement
00025  * Bytes 3-130:   Data
00026  * Bytes 131-132: CRC
00027  */
00028
00029 /* Xmodem (1024 bytes) packet format
00030  * Byte  0:         Header
00031  * Byte  1:         Packet number
00032  * Byte  2:         Packet number complement
00033  * Bytes 3-1026:    Data
00034  * Bytes 1027-1028: CRC
00035  */
00036
00037 /* Maximum allowed errors (user defined). */
00038 #define X_MAX_ERRORS ((uint8_t)3u)
00039
00040 /* Sizes of the packets. */
```

```
00041 #define X_PACKET_NUMBER_SIZE ((uint16_t)2u)
00042 #define X_PACKET_128_SIZE ((uint16_t)128u)
00043 #define X_PACKET_1024_SIZE ((uint16_t)1024u)
00044 #define X_PACKET_CRC_SIZE ((uint16_t)2u)
00045
00046 /* Indexes inside packets. */
00047 #define X_PACKET_NUMBER_INDEX ((uint16_t)0u)
00048 #define X_PACKET_NUMBER_COMPLEMENT_INDEX ((uint16_t)1u)
00049 #define X_PACKET_CRC_HIGH_INDEX ((uint16_t)0u)
00050 #define X_PACKET_CRC_LOW_INDEX ((uint16_t)1u)
00051
00052 /* Bytes defined by the protocol. */
00053 #define X_SOH ((uint8_t)0x01u)
00054 #define X_STX ((uint8_t)0x02u)
00055 #define X_EOT ((uint8_t)0x04u)
00056 #define X_ACK ((uint8_t)0x06u)
00057 #define X_NAK ((uint8_t)0x15u)
00058 #define X_CAN ((uint8_t)0x18u)
00059 #define X_C ((uint8_t)0x43u)
00062
00067 typedef enum
00068 {
00069     X_OK = 0x00u,
00070     X_ERROR_CRC = 0x01u,
00071     X_ERROR_NUMBER = 0x02u,
00072     X_ERROR_UART = 0x04u,
00073     X_ERROR_FLASH = 0x08u,
00074     X_COMPLETE = 0x10u,
00075     X_ERROR = 0xFFu
00076 } xmodem_status;
00077
00079
00080 xmodem_status xmodem_API_receive(void);
00081
00083
00084 #endif /* XMODEM_H_ */
```

5.64 xmodem.h File Reference

`#include ¨stdbool.h¨`

Macros

- #define **X_MAX_ERRORS** ((uint8_t)3u)
- #define **X_PACKET_NUMBER_SIZE** ((uint16_t)2u)
- #define **X_PACKET_128_SIZE** ((uint16_t)128u)
- #define **X_PACKET_1024_SIZE** ((uint16_t)1024u)
- #define **X_PACKET_CRC_SIZE** ((uint16_t)2u)
- #define **X_PACKET_NUMBER_INDEX** ((uint16_t)0u)
- #define **X_PACKET_NUMBER_COMPLEMENT_INDEX** ((uint16_t)1u)
- #define **X_PACKET_CRC_HIGH_INDEX** ((uint16_t)0u)
- #define **X_PACKET_CRC_LOW_INDEX** ((uint16_t)1u)
- #define X_SOH ((uint8_t)0x01u)
- #define X_STX ((uint8_t)0x02u)
- #define X_EOT ((uint8_t)0x04u)
- #define X_ACK ((uint8_t)0x06u)
- #define X_NAK ((uint8_t)0x15u)
- #define X_CAN ((uint8_t)0x18u)
- #define X_C ((uint8_t)0x43u)

Enumerations

- enum xmodem_status {
  X_OK = 0x00u , X_ERROR_CRC = 0x01u , X_ERROR_NUMBER = 0x02u , X_ERROR_UART = 0x04u ,
  X_ERROR_FLASH = 0x08u , X_COMPLETE = 0x10u , X_ERROR = 0xFFu }

Functions

- xmodem_status xmodem_API_receive (void)

5.64.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-02-22

Copyright

Copyright (c) 2024

Note

Original module source https://github.com/ferenc-nemeth

5.64.2 Macro Definition Documentation

5.64.2.1 X_SOH

```
#define X_SOH ((uint8_t)0x01u)
```

Start Of Header (128 bytes).

5.64.2.2 X_STX

```
#define X_STX ((uint8_t)0x02u)
```

Start Of Header (1024 bytes).

5.64.2.3 X_EOT

```
#define X_EOT ((uint8_t)0x04u)
```

End Of Transmission.

5.64.2.4 X_ACK

```
#define X_ACK ((uint8_t)0x06u)
```

Acknowledge.

5.64.2.5 X_NAK

```
#define X_NAK ((uint8_t)0x15u)
```

Not Acknowledge.

5.64.2.6 X_CAN

```
#define X_CAN ((uint8_t)0x18u)
```

Cancel.

5.64.2.7 X_C

```
#define X_C ((uint8_t)0x43u)
```

ASCII ¨C¨ to notify the host we want to use CRC16.

5.64.3 Enumeration Type Documentation

5.64.3.1 xmodem_status

```
enum xmodem_status
```

xModem status enums

Enumerator

X_OK	The action was successful.
X_ERROR_CRC	CRC calculation error.
X_ERROR_NUMBER	Packet number mismatch error.
X_ERROR_UART	UART communication error.
X_ERROR_FLASH	Flash related error.
X_COMPLETE	Generic error.
X_ERROR	Generic error.

5.64.4 Function Documentation

5.64.4.1 xmodem_API_receive()

```
xmodem_status xmodem_API_receive (
            void  )
```

This function is the base of the Xmodem protocol. When we receive a header from UART, it decides what action it shall take.

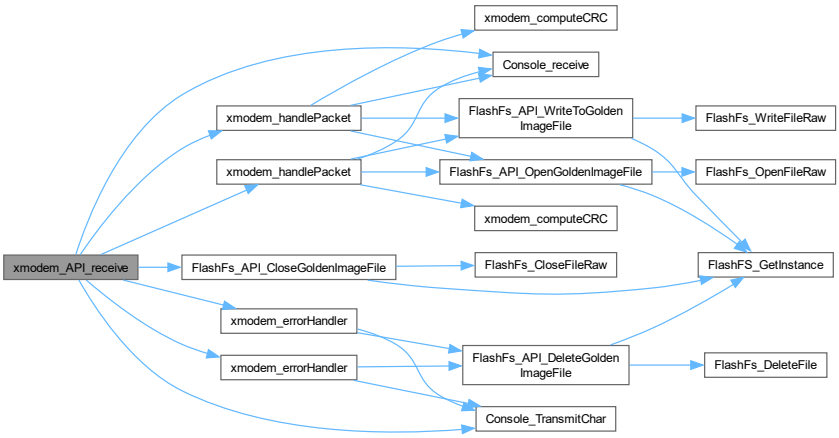
Parameters

void	

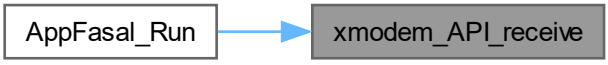
Returns

xmodem_status

Here is the call graph for this function:



Here is the caller graph for this function:



5.65 xModem/xmodem.h

Go to the documentation of this file.

```
00001
00012 #ifndef XMODEM_H_
00013 #define XMODEM_H_
00014
00016
00017 #include "stdbool.h"
00018
00020
00021 /* Xmodem (128 bytes) packet format
00022  * Byte  0:       Header
00023  * Byte  1:       Packet number
00024  * Byte  2:       Packet number complement
00025  * Bytes 3-130:   Data
00026  * Bytes 131-132: CRC
00027  */
00028
00029 /* Xmodem (1024 bytes) packet format
00030  * Byte  0:         Header
00031  * Byte  1:         Packet number
00032  * Byte  2:         Packet number complement
00033  * Bytes 3-1026:    Data
00034  * Bytes 1027-1028: CRC
00035  */
00036
00037 /* Maximum allowed errors (user defined). */
00038 #define X_MAX_ERRORS ((uint8_t)3u)
00039
00040 /* Sizes of the packets. */
00041 #define X_PACKET_NUMBER_SIZE  ((uint16_t)2u)
00042 #define X_PACKET_128_SIZE     ((uint16_t)128u)
00043 #define X_PACKET_1024_SIZE    ((uint16_t)1024u)
00044 #define X_PACKET_CRC_SIZE     ((uint16_t)2u)
00045
00046 /* Indexes inside packets. */
00047 #define X_PACKET_NUMBER_INDEX             ((uint16_t)0u)
00048 #define X_PACKET_NUMBER_COMPLEMENT_INDEX  ((uint16_t)1u)
00049 #define X_PACKET_CRC_HIGH_INDEX           ((uint16_t)0u)
00050 #define X_PACKET_CRC_LOW_INDEX            ((uint16_t)1u)
00051
00052
00053 /* Bytes defined by the protocol. */
00054 #define X_SOH ((uint8_t)0x01u)
00055 #define X_STX ((uint8_t)0x02u)
00056 #define X_EOT ((uint8_t)0x04u)
00057 #define X_ACK ((uint8_t)0x06u)
00058 #define X_NAK ((uint8_t)0x15u)
00059 #define X_CAN ((uint8_t)0x18u)
00060 #define X_C   ((uint8_t)0x43u)
00063
00068 typedef enum {
00069   X_OK            = 0x00u,
00070   X_ERROR_CRC     = 0x01u,
00071   X_ERROR_NUMBER  = 0x02u,
00072   X_ERROR_UART    = 0x04u,
00073   X_ERROR_FLASH   = 0x08u,
00074   X_COMPLETE      = 0x10u,
00075   X_ERROR         = 0xFFu
00076 } xmodem_status;
00077
00079
00080 xmodem_status xmodem_API_receive(void);
00081
00083
00084 #endif /* XMODEM_H_ */
```

5.66 AppInterrupts.c File Reference

```
#include ¨AppConfiguration.h¨
#include ¨AppIndication.h¨
#include ¨AppInterrupts.h¨
#include ¨Console.h¨
#include ¨softTimer.h¨
#include ¨usart.h¨
```

Functions

- static void AppISR_PreResetRoutine ()
- void AppISR_HALErrorHandler ()
- void AppISR_ARMHandler ()
- void __assert_func (const char *file, int line, const char *func, const char *failedexpr)
- void HAL_UART_RxCpltCallback (UART_HandleTypeDef *huart)
- void HAL_TIM_PeriodElapsedCallback (TIM_HandleTypeDef *htim)

5.66.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-10-24

Copyright

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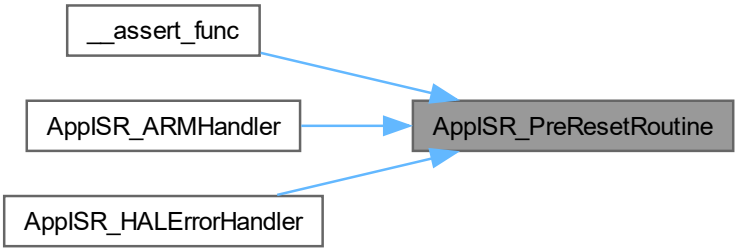
5.66.2 Function Documentation

5.66.2.1 AppISR_PreResetRoutine()

```
static void AppISR_PreResetRoutine ( )  [static]
```

Function to be invoked prior to resetting the system to recover from an hard fault/error.

Here is the caller graph for this function:



5.66.2.2 AppISR_HALErrorHandler()

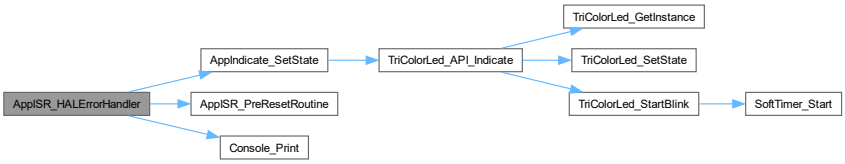
```
void AppISR_HALErrorHandler ( )
```

Error from STM HAL.

Warning

This function resets the device

Here is the call graph for this function:



5.66.2.3 AppISR_ARMHandler()

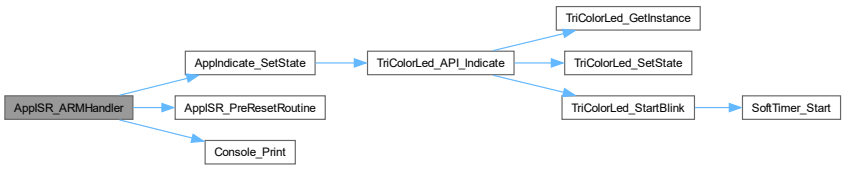
```
void AppISR_ARMHandler ( )
```

Handler for errors from ARM core.

Warning

This function resets the device

Here is the call graph for this function:



5.66.2.4 __assert_func()

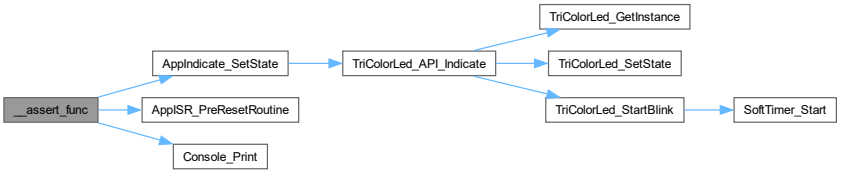
```
void __assert_func (
            const char * file,
            int line,
            const char * func,
            const char * failedexpr )
```

Assert failure error handler.

Warning

This function resets the device

Here is the call graph for this function:



5.66.2.5 HAL_UART_RxCpltCallback()

```
void HAL_UART_RxCpltCallback (
            UART_HandleTypeDef * huart )
```

STM HAL reception callback.

Parameters

huart	

Here is the call graph for this function:

5.66.2.6 HAL_TIM_PeriodElapsedCallback()

```
void HAL_TIM_PeriodElapsedCallback (
            TIM_HandleTypeDef * htim )
```

Timer callback associated with soft-timer, part of STM HAL.

Parameters

htim	timer instance

Here is the call graph for this function:

5.67 AppInterrupts.h File Reference

Macros

- #define LOOP_FOREVER

Functions

- void AppISR_HALErrorHandler ()
- void AppISR_ARMHandler ()
- void __assert_func (const char *file, int line, const char *func, const char *failedexpr)

5.67.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-10-24

Copyright

Copyright (c) 2024

5.67.2 Macro Definition Documentation

5.67.2.1 LOOP_FOREVER

```
#define LOOP_FOREVER
```

Value:

```
{
  \
    while (1)
  \
      ;
  \
}
```

Utility MACRO to loop forever ∗/.

5.67.3 Function Documentation

5.67.3.1 AppISR_HALErrorHandler()

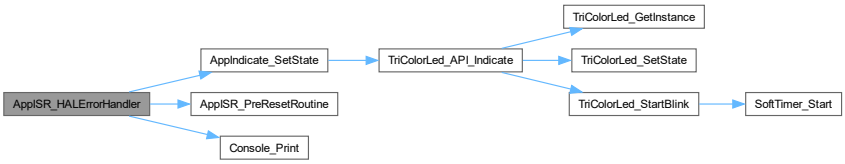
```
void AppISR_HALErrorHandler ( )
```

Error from STM HAL.

Warning

This function resets the device

Here is the call graph for this function:



5.67.3.2 AppISR_ARMHandler()

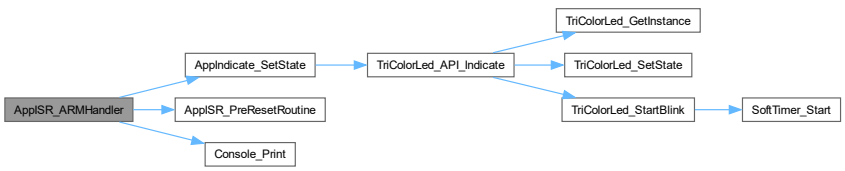
```
void AppISR_ARMHandler ( )
```

Handler for errors from ARM core.

Warning

This function resets the device

Here is the call graph for this function:



5.67.3.3 __assert_func()

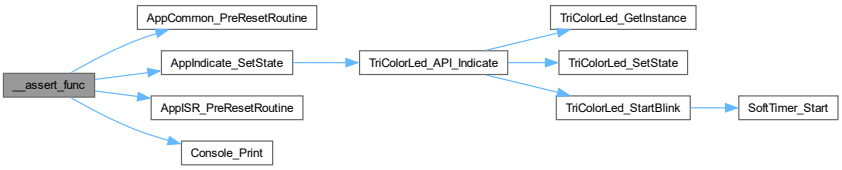
```
void __assert_func (
            const char * file,
            int line,
            const char * func,
            const char * failedexpr )
```

Assert failure error handler.

Warning

This function resets the device

Here is the call graph for this function:



5.68 AppInterrupts.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPINTERRUPTS_APPINTERRUPTS_H_
00015 #define APPINTERRUPTS_APPINTERRUPTS_H_
00016
00018
00019 #define LOOP_FOREVER
      \
00020     {
      \
00021         while (1)
      \
00022             ;
      \
00023     }
00024
00026
00027 void AppISR_HALErrorHandler();
00028 void AppISR_ARMHandler();
00029 void __assert_func(const char* file, int line, const char* func, const char* failedexpr);
00030
00032
00033 #endif /* APPINTERRUPTS_APPINTERRUPTS_H_ */
```

5.69 AppFlash_API.c File Reference

```
#include <assert.h>
#include <string.h>
#include ¨AppConfiguration.h¨
#include ¨AppFlash_API.h¨
#include ¨LittleFS_Wrapper.h¨
#include ¨W25Qxx.h¨
```

Functions

- static sFlashFS_t * FlashFS_GetInstance ()
- static eStorageFSStatus_t FlashFS_Init (sFlashFS_t *const pMe)
- __attribute__ ((unused))
- static eStorageFSStatus_t FlashFs_OpenFileRaw (sFlashFS_t *const pMe, eStorageFileNamesEnums_t fileEnum, eStorageFSOperationModes_t modeEnum)
- static eStorageFSStatus_t FlashFs_CloseFileRaw (sFlashFS_t *const pMe, eStorageFileNamesEnums_t fileEnum)
- static eStorageFSStatus_t FlashFs_ReadFileRaw (sFlashFS_t *const pMe, eStorageFileNamesEnums_t fileEnum, char *const pOutReadBuf, uint32_t bytesToRead, int32_t *const pOutBytesRead)
- static eStorageFSStatus_t FlashFs_WriteFileRaw (sFlashFS_t *const pMe, eStorageFileNamesEnums_t fileEnum, bool IsAppend, const char *const pInWriteBuf, size_t bufSize)
- static eStorageFSStatus_t FlashFs_DeleteFile (const sFlashFS_t *const pMe, eStorageFileNamesEnums_t fileEnum)
- static eStorageFSStatus_t FlashFs_ComputeFileCRC (sFlashFS_t *const pMe, eStorageFileNamesEnums_t fileEnum, uint8_t *const pInOutRamBuf, uint32_t RamBufSize, uint32_t *const pOutCRC)
- eStorageFSStatus_t FlashFs_API_Init ()
- eStorageFSStatus_t FlashFs_API_OpenGoldenImageFile ()
- eStorageFSStatus_t FlashFs_API_WriteToGoldenImageFile (const char *const pInWriteBuf, size_t bufSize)
- eStorageFSStatus_t FlashFs_API_CloseGoldenImageFile ()
- eStorageFSStatus_t FlashFs_API_DeleteGoldenImageFile ()
- eStorageFSStatus_t FlashFs_API_ComputeGoldenImageFileCRC (uint8_t *const pInOutRamBuf, uint32_t RamBufSize, uint32_t *const pOutCRC)

Variables

- static const int gcOpModeToLittleFSModeConverterTable [eFS_MODE_MAX]
- static const char *const gcFilesNamesTable [eFS_MAX] = {[eFS_GOLDEN_IMAGE] = "fallback.txt", [eFS_↩ FILE_2] = "file2.txt"}
- static sFlashFS_t **gsFlash** = {.IsMounted = false}

5.69.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-06-07

Copyright

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5.69.2 Function Documentation

5.69.2.1 FlashFS_GetInstance()

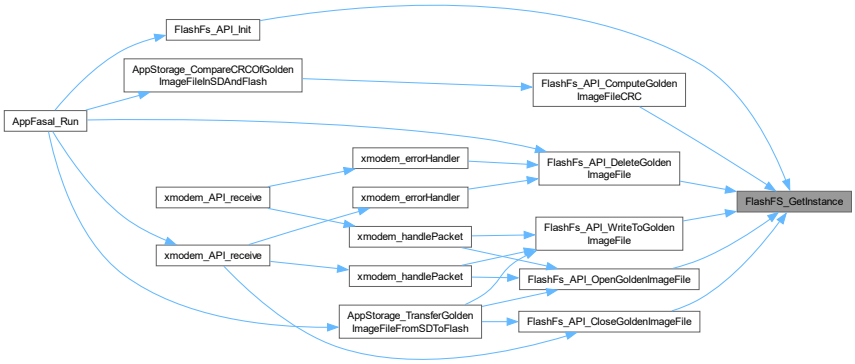
```
static sFlashFS_t * FlashFS_GetInstance ( )  [static]
```

Get FlashFS file instance.

Returns

sFlashFS_t*

Here is the caller graph for this function:



5.69.2.2 FlashFS_Init()

```
static eStorageFSStatus_t FlashFS_Init (
            sFlashFS_t *const pMe )  [static]
```

Initialize lfs.

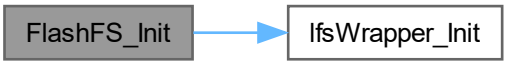
Parameters

pMe	lfs wrapper instance

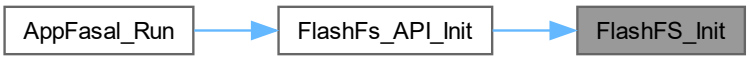
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.69.2.3 __attribute__()

```
__attribute__ (
              (unused)  )
```

Deinitialize LFS file system.

Parameters

pMe	lfs wrapper instance

Returns

eStorageFSStatus_t

5.69.2.4 FlashFs_OpenFileRaw()

```
static eStorageFSStatus_t FlashFs_OpenFileRaw (
             sFlashFS_t *const pMe,
             eStorageFileNamesEnums_t fileEnum,
             eStorageFSOperationModes_t modeEnum )  [static]
```

Open a file in lfs.

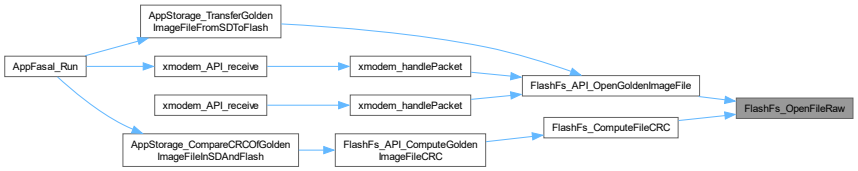
Parameters

pMe	lfs wrapper instance
fileEnum	File to be opened
modeEnum	Mode to open file In

Returns

eStorageFSStatus_t

Here is the caller graph for this function:



5.69.2.5 FlashFs_CloseFileRaw()

```
static eStorageFSStatus_t FlashFs_CloseFileRaw (
            sFlashFS_t *const pMe,
            eStorageFileNamesEnums_t fileEnum )  [static]
```

Close a file in lfs.

Parameters

pMe	lfs wrapper instance
fileEnum	file to be closed

Returns

eStorageFSStatus_t

Here is the caller graph for this function:

5.69.2.6 FlashFs_ReadFileRaw()

```
static eStorageFSStatus_t FlashFs_ReadFileRaw (
            sFlashFS_t *const pMe,
            eStorageFileNamesEnums_t fileEnum,
            char *const pOutReadBuf,
            uint32_t bytesToRead,
            int32_t *const pOutBytesRead )  [static]
```

read from a file in FlashFS

Parameters

pMe	FlashFS wrapper instance
fileEnum	File to be read from
pOutReadBuf	read data will be saved here
bytesToRead	number of bytes to read
pOutBytesRead	number of bytes read is saved here

Returns

eStorageFSStatus_t

Here is the caller graph for this function:

5.69.2.7 FlashFs_WriteFileRaw()

```
static eStorageFSStatus_t FlashFs_WriteFileRaw (
            sFlashFS_t *const pMe,
            eStorageFileNamesEnums_t fileEnum,
            bool IsAppend,
            const char *const pInWriteBuf,
            size_t bufSize )  [static]
```

Write to a file in LFS.

Parameters

pMe	lfs wrapper instance
fileEnum	File to be written into
IsAppend	Unused Kept for API consistency
pInWriteBuf	data to be written is passed here
bufSize	size of buffer passed for writing

Returns

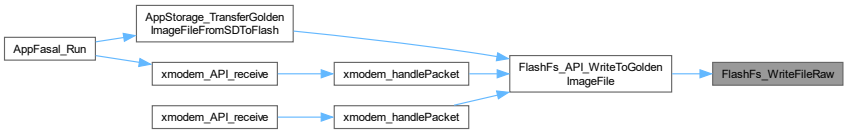
eStorageFSStatus_t

<

Warning

IsAppend Kept for API consistency, File should be opened in Append mode if Appending is needed

Here is the caller graph for this function:



5.69.2.8 FlashFs_DeleteFile()

```
static eStorageFSStatus_t FlashFs_DeleteFile (
            const sFlashFS_t *const pMe,
            eStorageFileNamesEnums_t fileEnum )  [static]
```

Delete File.

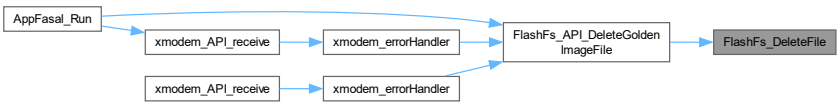
Parameters

pMe	lfs wrapper instance
fileEnum	File to be deleted

Returns

eStorageFSStatus_t

Here is the caller graph for this function:



5.69.2.9 FlashFs_ComputeFileCRC()

```
static eStorageFSStatus_t FlashFs_ComputeFileCRC (
            sFlashFS_t *const pMe,
```

```
eStorageFileNamesEnums_t fileEnum,
uint8_t *const pInOutRamBuf,
uint32_t RamBufSize,
uint32_t *const pOutCRC )  [static]
```

compute CRC of requested file

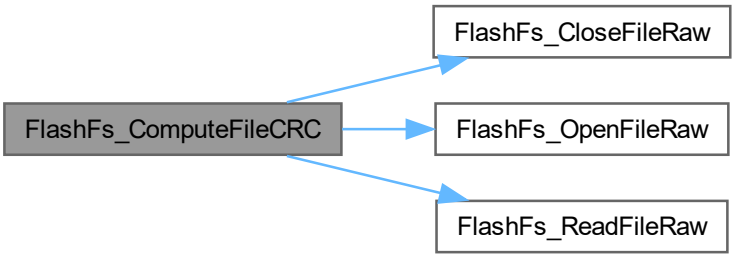
Parameters

pMe	lfs wrapper instance
fileEnum	File whose CRC needs to be computed
pInOutRamBuf	Ram buffer that will be used as temporary storage while computing CRC
RamBufSize	ram buffer size passed
pOutCRC	computed CRC will be saved here

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.69.2.10 FlashFs_API_Init()

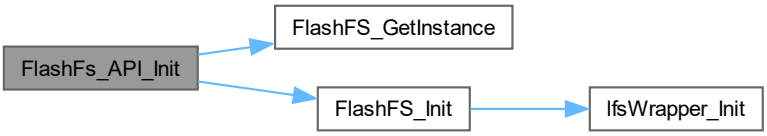
```
eStorageFSStatus_t FlashFs_API_Init ( )
```

API to initialize lfs.

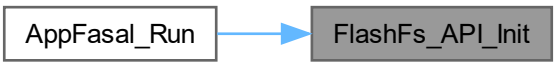
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.69.2.11 FlashFs_API_OpenGoldenImageFile()

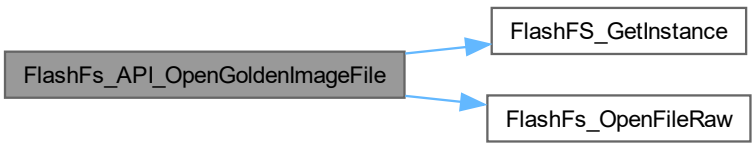
```
eStorageFSStatus_t FlashFs_API_OpenGoldenImageFile ( )
```

Open Golden Image file.

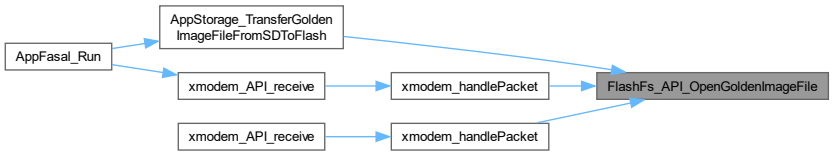
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.69.2.12 FlashFs_API_WriteToGoldenImageFile()

```
eStorageFSStatus_t FlashFs_API_WriteToGoldenImageFile (
            const char *const pInWriteBuf,
            size_t bufSize )
```

Write to Golden Image file.

Note

AppStorage_API_OpenGoldenFile must be invoked prior to using this function

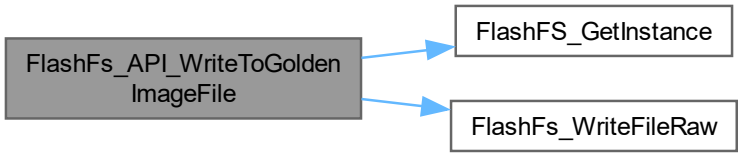
Parameters

pInWriteBuf	Contents to be written to Golden file is passed here
bufSize	write buffer size

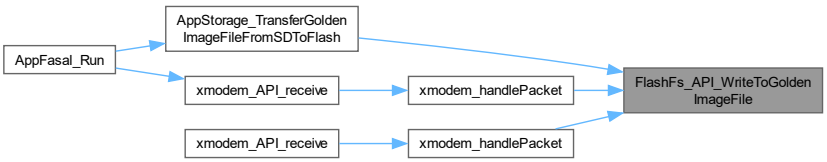
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.69.2.13 FlashFs_API_CloseGoldenImageFile()

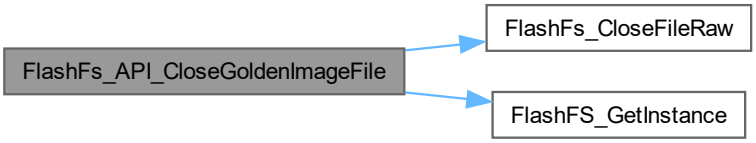
```
eStorageFSStatus_t FlashFs_API_CloseGoldenImageFile ( )
```

Close golden Image file.

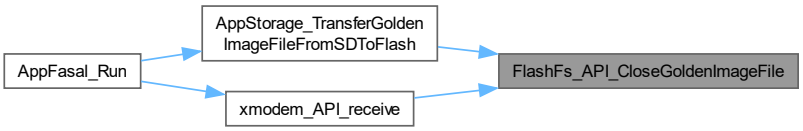
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.69.2.14 FlashFs_API_DeleteGoldenImageFile()

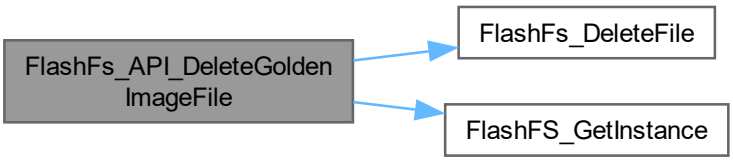
```
eStorageFSStatus_t FlashFs_API_DeleteGoldenImageFile ( )
```

Close golden Image file.

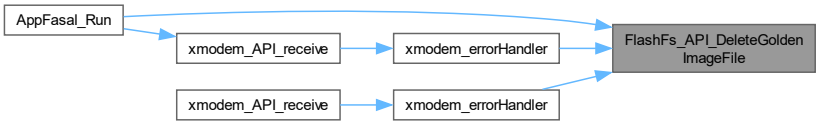
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.69.2.15 FlashFs_API_ComputeGoldenImageFileCRC()

```
eStorageFSStatus_t FlashFs_API_ComputeGoldenImageFileCRC (
            uint8_t *const pInOutRamBuf,
            uint32_t RamBufSize,
            uint32_t *const pOutCRC )
```

compute CRC of golden Image file

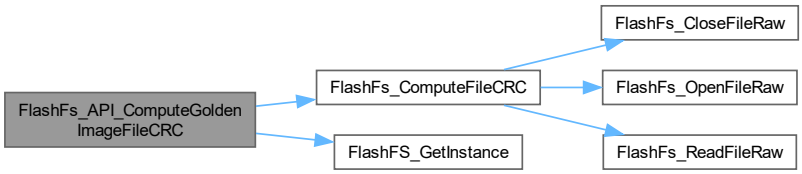
Parameters

pInOutRamBuf	Ram buffer that will be used as temporary storage while computing CRC
RamBufSize	ram buffer size passed
pOutCRC	computed CRC will be saved here

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.69.3 Variable Documentation

5.69.3.1 gcOpModeToLittleFSModeConverterTable

```
const int gcOpModeToLittleFSModeConverterTable[eFS_MODE_MAX]  [static]
```

Initial value:

```
= {
    [eFS_READONLY] = (LFS_O_RDONLY),
    [eFS_WRITEONLY] = (LFS_O_WRONLY),
    [eFS_READ_WRITE] = (LFS_O_RDWR),
    [eFS_READ_CREATE] = (LFS_O_RDONLY | LFS_O_CREAT),
    [eFS_WRITE_CREATE] = (LFS_O_WRONLY | LFS_O_CREAT),
    [eFS_WRITE_APPEND] = (LFS_O_WRONLY | LFS_O_CREAT | LFS_O_APPEND),
    [eFS_READ_WRITE_CREATE] = (LFS_O_RDWR | LFS_O_CREAT),
}
```

utility table that converts user file modes to littleFS file modes

5.69.3.2 gcFilesNamesTable

```
const char* const gcFilesNamesTable[eFS_MAX] = {[eFS_GOLDEN_IMAGE] = ¨fallback.txt¨, [eFS_↩
FILE_2] = ¨file2.txt¨}  [static]
```

Map Name enums to File name strings.

5.70 AppFlash_API.h File Reference

```
#include <stdbool.h>
#include ¨crc.h¨
#include ¨lfs.h¨
#include ¨AppStorageDataStructures.h¨
```

Data Structures

- struct sFlashFS_t

Macros

- #define **FLASHFS_CRC_INSTANCE** (&hcrc)

Functions

- eStorageFSStatus_t FlashFs_API_Init ()
- eStorageFSStatus_t FlashFs_API_OpenGoldenImageFile ()
- eStorageFSStatus_t FlashFs_API_WriteToGoldenImageFile (const char ∗const pInWriteBuf, size_t bufSize)
- eStorageFSStatus_t FlashFs_API_CloseGoldenImageFile ()
- eStorageFSStatus_t FlashFs_API_DeleteGoldenImageFile ()
- eStorageFSStatus_t FlashFs_API_ComputeGoldenImageFileCRC (uint8_t ∗const pInOutRamBuf, uint32_t RamBufSize, uint32_t ∗const pOutCRC)

5.70.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-06-07

Copyright

5.70.2 Function Documentation

5.70.2.1 FlashFs_API_Init()

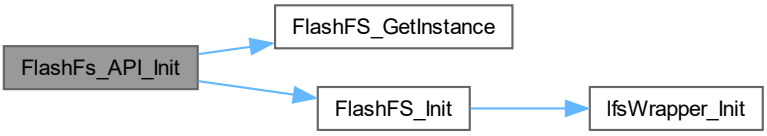
```
eStorageFSStatus_t FlashFs_API_Init ( )
```

API to initialize lfs.

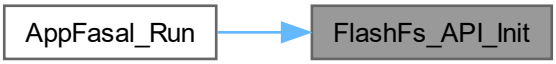
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.70.2.2 FlashFs_API_OpenGoldenImageFile()

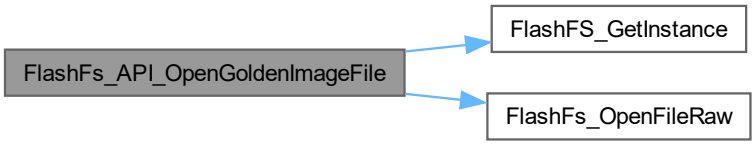
```
eStorageFSStatus_t FlashFs_API_OpenGoldenImageFile ( )
```

Open Golden Image file.

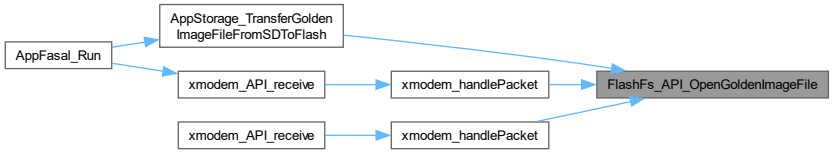
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.70.2.3 FlashFs_API_WriteToGoldenImageFile()

```
eStorageFSStatus_t FlashFs_API_WriteToGoldenImageFile (
            const char *const pInWriteBuf,
            size_t bufSize )
```

Write to Golden Image file.

Note

AppStorage_API_OpenGoldenFile must be invoked prior to using this function

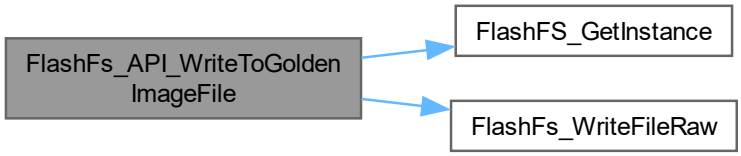
Parameters

pInWriteBuf	Contents to be written to Golden file is passed here
bufSize	write buffer size

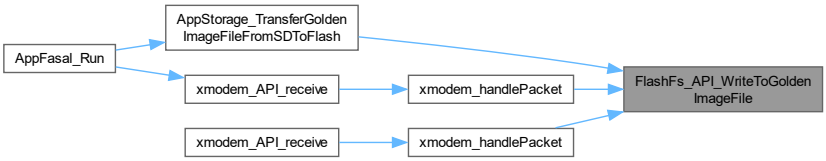
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.70.2.4 FlashFs_API_CloseGoldenImageFile()

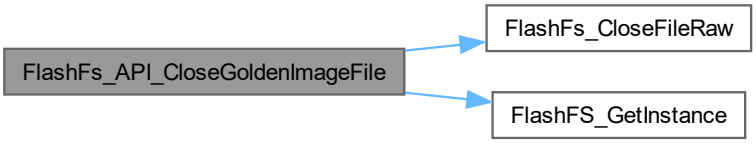
```
eStorageFSStatus_t FlashFs_API_CloseGoldenImageFile ( )
```

Close golden Image file.

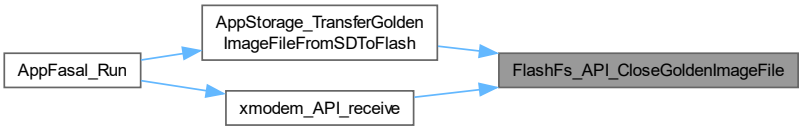
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.70.2.5 FlashFs_API_DeleteGoldenImageFile()

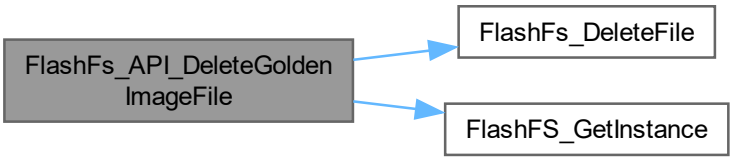
```
eStorageFSStatus_t FlashFs_API_DeleteGoldenImageFile ( )
```

Close golden Image file.

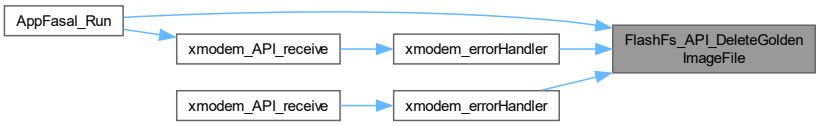
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.70.2.6 FlashFs_API_ComputeGoldenImageFileCRC()

```
eStorageFSStatus_t FlashFs_API_ComputeGoldenImageFileCRC (
            uint8_t *const pInOutRamBuf,
            uint32_t RamBufSize,
            uint32_t *const pOutCRC )
```

compute CRC of golden Image file

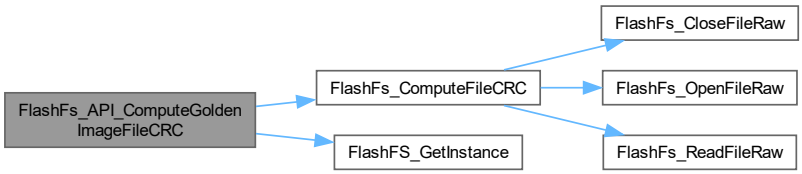
Parameters

pInOutRamBuf	Ram buffer that will be used as temporary storage while computing CRC
RamBufSize	ram buffer size passed
pOutCRC	computed CRC will be saved here

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.71 AppFlash_API.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPSTORAGE_APPFLASHFS_APPFLASH_API_H_
00015 #define APPSTORAGE_APPFLASHFS_APPFLASH_API_H_
00016
00018
00019 #include <stdbool.h>
00020 #include "crc.h"
00021 #include "lfs.h"
00022 #include "AppStorageDataStructures.h"
00023
00025
00026 #define FLASHFS_CRC_INSTANCE (&hcrc)
00027
00029
00034 typedef struct
00035 {
00036     bool IsMounted;
00037     lfs_t fs;
00038     lfs_file_t fileHandles[eFS_MAX];
00039 } sFlashFS_t;
00040
00042
00043 eStorageFSStatus_t FlashFs_API_Init();
```

```
00044 eStorageFSStatus_t FlashFs_API_OpenGoldenImageFile();
00045 eStorageFSStatus_t FlashFs_API_WriteToGoldenImageFile(const char* const pInWriteBuf, size_t bufSize);
00046 eStorageFSStatus_t FlashFs_API_CloseGoldenImageFile();
00047 eStorageFSStatus_t FlashFs_API_DeleteGoldenImageFile();
00048 eStorageFSStatus_t FlashFs_API_ComputeGoldenImageFileCRC(uint8_t* const pInOutRamBuf, uint32_t
      RamBufSize, uint32_t* const pOutCRC);
00049
00051
00052 #endif /* APPSTORAGE_APPFLASHFS_APPFLASH_API_H_ */
```

5.72 lfs.h

```
00001 /*
00002  * The little filesystem
00003  *
00004  * Copyright (c) 2022, The littlefs authors.
00005  * Copyright (c) 2017, Arm Limited. All rights reserved.
00006  * SPDX-License-Identifier: BSD-3-Clause
00007  */
00008 #ifndef LFS_H
00009 #define LFS_H
00010
00011 #include "lfs_util.h"
00012
00013 #ifdef __cplusplus
00014 extern "C"
00015 {
00016 #endif
00017
00019
00020 // Software library version
00021 // Major (top-nibble), incremented on backwards incompatible changes
00022 // Minor (bottom-nibble), incremented on feature additions
00023 #define LFS_VERSION 0x00020007
00024 #define LFS_VERSION_MAJOR (0xffff & (LFS_VERSION » 16))
00025 #define LFS_VERSION_MINOR (0xffff & (LFS_VERSION » 0))
00026
00027 // Version of On-disk data structures
00028 // Major (top-nibble), incremented on backwards incompatible changes
00029 // Minor (bottom-nibble), incremented on feature additions
00030 #define LFS_DISK_VERSION 0x00020001
00031 #define LFS_DISK_VERSION_MAJOR (0xffff & (LFS_DISK_VERSION » 16))
00032 #define LFS_DISK_VERSION_MINOR (0xffff & (LFS_DISK_VERSION » 0))
00033
00035
00036     // Type definitions
00037     typedef uint32_t lfs_size_t;
00038     typedef uint32_t lfs_off_t;
00039
00040     typedef int32_t lfs_ssize_t;
00041     typedef int32_t lfs_soff_t;
00042
00043     typedef uint32_t lfs_block_t;
00044
00045 // Maximum name size in bytes, may be redefined to reduce the size of the
00046 // info struct. Limited to <= 1022. Stored in superblock and must be
00047 // respected by other littlefs drivers.
00048 #ifndef LFS_NAME_MAX
00049 #define LFS_NAME_MAX 255
00050 #endif
00051
00052 // Maximum size of a file in bytes, may be redefined to limit to support other
00053 // drivers. Limited on disk to <= 4294967296. However, above 2147483647 the
00054 // functions lfs_file_seek, lfs_file_size, and lfs_file_tell will return
00055 // incorrect values due to using signed integers. Stored in superblock and
00056 // must be respected by other littlefs drivers.
00057 #ifndef LFS_FILE_MAX
00058 #define LFS_FILE_MAX 2147483647
00059 #endif
00060
00061 // Maximum size of custom attributes in bytes, may be redefined, but there is
00062 // no real benefit to using a smaller LFS_ATTR_MAX. Limited to <= 1022.
00063 #ifndef LFS_ATTR_MAX
00064 #define LFS_ATTR_MAX 1022
00065 #endif
00066
00067     // Possible error codes, these are negative to allow
00068     // valid positive return values
00069     enum lfs_error
00070     {
00071         LFS_ERR_OK = 0,             // No error
00072         LFS_ERR_IO = -5,            // Error during device operation
00073         LFS_ERR_CORRUPT = -84,      // Corrupted
```

```
00074         LFS_ERR_NOENT = -2,         // No directory entry
00075         LFS_ERR_EXIST = -17,        // Entry already exists
00076         LFS_ERR_NOTDIR = -20,       // Entry is not a dir
00077         LFS_ERR_ISDIR = -21,        // Entry is a dir
00078         LFS_ERR_NOTEMPTY = -39,     // Dir is not empty
00079         LFS_ERR_BADF = -9,          // Bad file number
00080         LFS_ERR_FBIG = -27,         // File too large
00081         LFS_ERR_INVAL = -22,        // Invalid parameter
00082         LFS_ERR_NOSPC = -28,        // No space left on device
00083         LFS_ERR_NOMEM = -12,        // No more memory available
00084         LFS_ERR_NOATTR = -61,       // No data/attr available
00085         LFS_ERR_NAMETOOLONG = -36, // File name too long
00086     };
00087
00088     // File types
00089     enum lfs_type
00090     {
00091         // file types
00092         LFS_TYPE_REG = 0x001,
00093         LFS_TYPE_DIR = 0x002,
00094
00095         // internally used types
00096         LFS_TYPE_SPLICE = 0x400,
00097         LFS_TYPE_NAME = 0x000,
00098         LFS_TYPE_STRUCT = 0x200,
00099         LFS_TYPE_USERATTR = 0x300,
00100         LFS_TYPE_FROM = 0x100,
00101         LFS_TYPE_TAIL = 0x600,
00102         LFS_TYPE_GLOBALS = 0x700,
00103         LFS_TYPE_CRC = 0x500,
00104
00105         // internally used type specializations
00106         LFS_TYPE_CREATE = 0x401,
00107         LFS_TYPE_DELETE = 0x4ff,
00108         LFS_TYPE_SUPERBLOCK = 0x0ff,
00109         LFS_TYPE_DIRSTRUCT = 0x200,
00110         LFS_TYPE_CTZSTRUCT = 0x202,
00111         LFS_TYPE_INLINESTRUCT = 0x201,
00112         LFS_TYPE_SOFTTAIL = 0x600,
00113         LFS_TYPE_HARDTAIL = 0x601,
00114         LFS_TYPE_MOVESTATE = 0x7ff,
00115         LFS_TYPE_CCRC = 0x500,
00116         LFS_TYPE_FCRC = 0x5ff,
00117
00118         // internal chip sources
00119         LFS_FROM_NOOP = 0x000,
00120         LFS_FROM_MOVE = 0x101,
00121         LFS_FROM_USERATTRS = 0x102,
00122     };
00123
00124     // File open flags
00125     enum lfs_open_flags
00126     {
00127         // open flags
00128         LFS_O_RDONLY = 1, // Open a file as read only
00129 #ifndef LFS_READONLY
00130         LFS_O_WRONLY = 2,       // Open a file as write only
00131         LFS_O_RDWR = 3,         // Open a file as read and write
00132         LFS_O_CREAT = 0x0100,   // Create a file if it does not exist
00133         LFS_O_EXCL = 0x0200,    // Fail if a file already exists
00134         LFS_O_TRUNC = 0x0400,   // Truncate the existing file to zero size
00135         LFS_O_APPEND = 0x0800, // Move to end of file on every write
00136 #endif
00137
00138     // internally used flags
00139 #ifndef LFS_READONLY
00140         LFS_F_DIRTY = 0x010000,   // File does not match storage
00141         LFS_F_WRITING = 0x020000, // File has been written since last flush
00142 #endif
00143         LFS_F_READING = 0x040000, // File has been read since last flush
00144 #ifndef LFS_READONLY
00145         LFS_F_ERRED = 0x080000, // An error occurred during write
00146 #endif
00147         LFS_F_INLINE = 0x100000, // Currently inlined in directory entry
00148     };
00149
00150     // File seek flags
00151     enum lfs_whence_flags
00152     {
00153         LFS_SEEK_SET = 0, // Seek relative to an absolute position
00154         LFS_SEEK_CUR = 1, // Seek relative to the current file position
00155         LFS_SEEK_END = 2, // Seek relative to the end of the file
00156     };
00157
00158     // Configuration provided during initialization of the littlefs
00159     struct lfs_config
00160     {
```

```
00161         // Opaque user provided context that can be used to pass
00162         // information to the block device operations
00163         void* context;
00164 
00165         // Read a region in a block. Negative error codes are propagated
00166         // to the user.
00167         int (*read)(const struct lfs_config* c, lfs_block_t block, lfs_off_t off, void* buffer,
      lfs_size_t size);
00168 
00169         // Program a region in a block. The block must have previously
00170         // been erased. Negative error codes are propagated to the user.
00171         // May return LFS_ERR_CORRUPT if the block should be considered bad.
00172         int (*prog)(const struct lfs_config* c, lfs_block_t block, lfs_off_t off, const void* buffer,
      lfs_size_t size);
00173 
00174         // Erase a block. A block must be erased before being programmed.
00175         // The state of an erased block is undefined. Negative error codes
00176         // are propagated to the user.
00177         // May return LFS_ERR_CORRUPT if the block should be considered bad.
00178         int (*erase)(const struct lfs_config* c, lfs_block_t block);
00179 
00180         // Sync the state of the underlying block device. Negative error codes
00181         // are propagated to the user.
00182         int (*sync)(const struct lfs_config* c);
00183 
00184 #ifdef LFS_THREADSAFE
00185         // Lock the underlying block device. Negative error codes
00186         // are propagated to the user.
00187         int (*lock)(const struct lfs_config* c);
00188 
00189         // Unlock the underlying block device. Negative error codes
00190         // are propagated to the user.
00191         int (*unlock)(const struct lfs_config* c);
00192 #endif
00193 
00194         // Minimum size of a block read in bytes. All read operations will be a
00195         // multiple of this value.
00196         lfs_size_t read_size;
00197 
00198         // Minimum size of a block program in bytes. All program operations will be
00199         // a multiple of this value.
00200         lfs_size_t prog_size;
00201 
00202         // Size of an erasable block in bytes. This does not impact ram consumption
00203         // and may be larger than the physical erase size. However, non-inlined
00204         // files take up at minimum one block. Must be a multiple of the read and
00205         // program sizes.
00206         lfs_size_t block_size;
00207 
00208         // Number of erasable blocks on the device.
00209         lfs_size_t block_count;
00210 
00211         // Number of erase cycles before littlefs evicts metadata logs and moves
00212         // the metadata to another block. Suggested values are in the
00213         // range 100-1000, with large values having better performance at the cost
00214         // of less consistent wear distribution.
00215         //
00216         // Set to -1 to disable block-level wear-leveling.
00217         int32_t block_cycles;
00218 
00219         // Size of block caches in bytes. Each cache buffers a portion of a block in
00220         // RAM. The littlefs needs a read cache, a program cache, and one additional
00221         // cache per file. Larger caches can improve performance by storing more
00222         // data and reducing the number of disk accesses. Must be a multiple of the
00223         // read and program sizes, and a factor of the block size.
00224         lfs_size_t cache_size;
00225 
00226         // Size of the lookahead buffer in bytes. A larger lookahead buffer
00227         // increases the number of blocks found during an allocation pass. The
00228         // lookahead buffer is stored as a compact bitmap, so each byte of RAM
00229         // can track 8 blocks. Must be a multiple of 8.
00230         lfs_size_t lookahead_size;
00231 
00232         // Optional statically allocated read buffer. Must be cache_size.
00233         // By default lfs_malloc is used to allocate this buffer.
00234         void* read_buffer;
00235 
00236         // Optional statically allocated program buffer. Must be cache_size.
00237         // By default lfs_malloc is used to allocate this buffer.
00238         void* prog_buffer;
00239 
00240         // Optional statically allocated lookahead buffer. Must be lookahead_size
00241         // and aligned to a 32-bit boundary. By default lfs_malloc is used to
00242         // allocate this buffer.
00243         void* lookahead_buffer;
00244 
00245         // Optional upper limit on length of file names in bytes. No downside for
```

```
00246         // larger names except the size of the info struct which is controlled by
00247         // the LFS_NAME_MAX define. Defaults to LFS_NAME_MAX when zero. Stored in
00248         // superblock and must be respected by other littlefs drivers.
00249         lfs_size_t name_max;
00250 
00251         // Optional upper limit on files in bytes. No downside for larger files
00252         // but must be <= LFS_FILE_MAX. Defaults to LFS_FILE_MAX when zero. Stored
00253         // in superblock and must be respected by other littlefs drivers.
00254         lfs_size_t file_max;
00255 
00256         // Optional upper limit on custom attributes in bytes. No downside for
00257         // larger attributes size but must be <= LFS_ATTR_MAX. Defaults to
00258         // LFS_ATTR_MAX when zero.
00259         lfs_size_t attr_max;
00260 
00261         // Optional upper limit on total space given to metadata pairs in bytes. On
00262         // devices with large blocks (e.g. 128kB) setting this to a low size (2-8kB)
00263         // can help bound the metadata compaction time. Must be <= block_size.
00264         // Defaults to block_size when zero.
00265         lfs_size_t metadata_max;
00266 
00267 #ifdef LFS_MULTIVERSION
00268         // On-disk version to use when writing in the form of 16-bit major version
00269         // + 16-bit minor version. This limiting metadata to what is supported by
00270         // older minor versions. Note that some features will be lost. Defaults to
00271         // to the most recent minor version when zero.
00272         uint32_t disk_version;
00273 #endif
00274     };
00275 
00276     // File info structure
00277     struct lfs_info
00278     {
00279         // Type of the file, either LFS_TYPE_REG or LFS_TYPE_DIR
00280         uint8_t type;
00281 
00282         // Size of the file, only valid for REG files. Limited to 32-bits.
00283         lfs_size_t size;
00284 
00285         // Name of the file stored as a null-terminated string. Limited to
00286         // LFS_NAME_MAX+1, which can be changed by redefining LFS_NAME_MAX to
00287         // reduce RAM. LFS_NAME_MAX is stored in superblock and must be
00288         // respected by other littlefs drivers.
00289         char name[LFS_NAME_MAX + 1];
00290     };
00291 
00292     // Filesystem info structure
00293     struct lfs_fsinfo
00294     {
00295         // On-disk version.
00296         uint32_t disk_version;
00297 
00298         // Upper limit on the length of file names in bytes.
00299         lfs_size_t name_max;
00300 
00301         // Upper limit on the size of files in bytes.
00302         lfs_size_t file_max;
00303 
00304         // Upper limit on the size of custom attributes in bytes.
00305         lfs_size_t attr_max;
00306     };
00307 
00308     // Custom attribute structure, used to describe custom attributes
00309     // committed atomically during file writes.
00310     struct lfs_attr
00311     {
00312         // 8-bit type of attribute, provided by user and used to
00313         // identify the attribute
00314         uint8_t type;
00315 
00316         // Pointer to buffer containing the attribute
00317         void* buffer;
00318 
00319         // Size of attribute in bytes, limited to LFS_ATTR_MAX
00320         lfs_size_t size;
00321     };
00322 
00323     // Optional configuration provided during lfs_file_opencfg
00324     struct lfs_file_config
00325     {
00326         // Optional statically allocated file buffer. Must be cache_size.
00327         // By default lfs_malloc is used to allocate this buffer.
00328         void* buffer;
00329 
00330         // Optional list of custom attributes related to the file. If the file
00331         // is opened with read access, these attributes will be read from disk
00332         // during the open call. If the file is opened with write access, the
```

```
00333         // attributes will be written to disk every file sync or close. This
00334         // write occurs atomically with update to the file's contents.
00335         //
00336         // Custom attributes are uniquely identified by an 8-bit type and limited
00337         // to LFS_ATTR_MAX bytes. When read, if the stored attribute is smaller
00338         // than the buffer, it will be padded with zeros. If the stored attribute
00339         // is larger, then it will be silently truncated. If the attribute is not
00340         // found, it will be created implicitly.
00341         struct lfs_attr* attrs;
00342 
00343         // Number of custom attributes in the list
00344         lfs_size_t attr_count;
00345     };
00346 
00348     typedef struct lfs_cache
00349     {
00350         lfs_block_t block;
00351         lfs_off_t off;
00352         lfs_size_t size;
00353         uint8_t* buffer;
00354     } lfs_cache_t;
00355 
00356     typedef struct lfs_mdir
00357     {
00358         lfs_block_t pair[2];
00359         uint32_t rev;
00360         lfs_off_t off;
00361         uint32_t etag;
00362         uint16_t count;
00363         bool erased;
00364         bool split;
00365         lfs_block_t tail[2];
00366     } lfs_mdir_t;
00367 
00368     // littlefs directory type
00369     typedef struct lfs_dir
00370     {
00371         struct lfs_dir* next;
00372         uint16_t id;
00373         uint8_t type;
00374         lfs_mdir_t m;
00375 
00376         lfs_off_t pos;
00377         lfs_block_t head[2];
00378     } lfs_dir_t;
00379 
00380     // littlefs file type
00381     typedef struct lfs_file
00382     {
00383         struct lfs_file* next;
00384         uint16_t id;
00385         uint8_t type;
00386         lfs_mdir_t m;
00387 
00388         struct lfs_ctz
00389         {
00390             lfs_block_t head;
00391             lfs_size_t size;
00392         } ctz;
00393 
00394         uint32_t flags;
00395         lfs_off_t pos;
00396         lfs_block_t block;
00397         lfs_off_t off;
00398         lfs_cache_t cache;
00399 
00400         const struct lfs_file_config* cfg;
00401     } lfs_file_t;
00402 
00403     typedef struct lfs_superblock
00404     {
00405         uint32_t version;
00406         lfs_size_t block_size;
00407         lfs_size_t block_count;
00408         lfs_size_t name_max;
00409         lfs_size_t file_max;
00410         lfs_size_t attr_max;
00411     } lfs_superblock_t;
00412 
00413     typedef struct lfs_gstate
00414     {
00415         uint32_t tag;
00416         lfs_block_t pair[2];
00417     } lfs_gstate_t;
00418 
00419     // The littlefs filesystem type
00420     typedef struct lfs
```

```
00421     {
00422         lfs_cache_t rcache;
00423         lfs_cache_t pcache;
00424 
00425         lfs_block_t root[2];
00426         struct lfs_mlist
00427         {
00428             struct lfs_mlist* next;
00429             uint16_t id;
00430             uint8_t type;
00431             lfs_mdir_t m;
00432         }* mlist;
00433         uint32_t seed;
00434 
00435         lfs_gstate_t gstate;
00436         lfs_gstate_t gdisk;
00437         lfs_gstate_t gdelta;
00438 
00439         struct lfs_free
00440         {
00441             lfs_block_t off;
00442             lfs_block_t size;
00443             lfs_block_t i;
00444             lfs_block_t ack;
00445             uint32_t* buffer;
00446         } free;
00447 
00448         const struct lfs_config* cfg;
00449         lfs_size_t name_max;
00450         lfs_size_t file_max;
00451         lfs_size_t attr_max;
00452 
00453 #ifdef LFS_MIGRATE
00454         struct lfs1* lfs1;
00455 #endif
00456     } lfs_t;
00457 
00459 
00460 #ifndef LFS_READONLY
00461     // Format a block device with the littlefs
00462     //
00463     // Requires a littlefs object and config struct. This clobbers the littlefs
00464     // object, and does not leave the filesystem mounted. The config struct must
00465     // be zeroed for defaults and backwards compatibility.
00466     //
00467     // Returns a negative error code on failure.
00468     int lfs_format(lfs_t* lfs, const struct lfs_config* config);
00469 #endif
00470 
00471     // Mounts a littlefs
00472     //
00473     // Requires a littlefs object and config struct. Multiple filesystems
00474     // may be mounted simultaneously with multiple littlefs objects. Both
00475     // lfs and config must be allocated while mounted. The config struct must
00476     // be zeroed for defaults and backwards compatibility.
00477     //
00478     // Returns a negative error code on failure.
00479     int lfs_mount(lfs_t* lfs, const struct lfs_config* config);
00480 
00481     // Unmounts a littlefs
00482     //
00483     // Does nothing besides releasing any allocated resources.
00484     // Returns a negative error code on failure.
00485     int lfs_unmount(lfs_t* lfs);
00486 
00488 
00489 #ifndef LFS_READONLY
00490     // Removes a file or directory
00491     //
00492     // If removing a directory, the directory must be empty.
00493     // Returns a negative error code on failure.
00494     int lfs_remove(lfs_t* lfs, const char* path);
00495 #endif
00496 
00497 #ifndef LFS_READONLY
00498     // Rename or move a file or directory
00499     //
00500     // If the destination exists, it must match the source in type.
00501     // If the destination is a directory, the directory must be empty.
00502     //
00503     // Returns a negative error code on failure.
00504     int lfs_rename(lfs_t* lfs, const char* oldpath, const char* newpath);
00505 #endif
00506 
00507     // Find info about a file or directory
00508     //
00509     // Fills out the info structure, based on the specified file or directory.
```

```
00510     // Returns a negative error code on failure.
00511     int lfs_stat(lfs_t* lfs, const char* path, struct lfs_info* info);
00512
00513     // Get a custom attribute
00514     //
00515     // Custom attributes are uniquely identified by an 8-bit type and limited
00516     // to LFS_ATTR_MAX bytes. When read, if the stored attribute is smaller than
00517     // the buffer, it will be padded with zeros. If the stored attribute is larger,
00518     // then it will be silently truncated. If no attribute is found, the error
00519     // LFS_ERR_NOATTR is returned and the buffer is filled with zeros.
00520     //
00521     // Returns the size of the attribute, or a negative error code on failure.
00522     // Note, the returned size is the size of the attribute on disk, irrespective
00523     // of the size of the buffer. This can be used to dynamically allocate a buffer
00524     // or check for existence.
00525     lfs_ssize_t lfs_getattr(lfs_t* lfs, const char* path, uint8_t type, void* buffer, lfs_size_t
      size);
00526
00527 #ifndef LFS_READONLY
00528     // Set custom attributes
00529     //
00530     // Custom attributes are uniquely identified by an 8-bit type and limited
00531     // to LFS_ATTR_MAX bytes. If an attribute is not found, it will be
00532     // implicitly created.
00533     //
00534     // Returns a negative error code on failure.
00535     int lfs_setattr(lfs_t* lfs, const char* path, uint8_t type, const void* buffer, lfs_size_t size);
00536 #endif
00537
00538 #ifndef LFS_READONLY
00539     // Removes a custom attribute
00540     //
00541     // If an attribute is not found, nothing happens.
00542     //
00543     // Returns a negative error code on failure.
00544     int lfs_removeattr(lfs_t* lfs, const char* path, uint8_t type);
00545 #endif
00546
00548
00549 #ifndef LFS_NO_MALLOC
00550     // Open a file
00551     //
00552     // The mode that the file is opened in is determined by the flags, which
00553     // are values from the enum lfs_open_flags that are bitwise-ored together.
00554     //
00555     // Returns a negative error code on failure.
00556     int lfs_file_open(lfs_t* lfs, lfs_file_t* file, const char* path, int flags);
00557
00558 // if LFS_NO_MALLOC is defined, lfs_file_open() will fail with LFS_ERR_NOMEM
00559 // thus use lfs_file_opencfg() with config.buffer set.
00560 #endif
00561
00562     // Open a file with extra configuration
00563     //
00564     // The mode that the file is opened in is determined by the flags, which
00565     // are values from the enum lfs_open_flags that are bitwise-ored together.
00566     //
00567     // The config struct provides additional config options per file as described
00568     // above. The config struct must remain allocated while the file is open, and
00569     // the config struct must be zeroed for defaults and backwards compatibility.
00570     //
00571     // Returns a negative error code on failure.
00572     int lfs_file_opencfg(lfs_t* lfs, lfs_file_t* file, const char* path, int flags, const struct
      lfs_file_config* config);
00573
00574     // Close a file
00575     //
00576     // Any pending writes are written out to storage as though
00577     // sync had been called and releases any allocated resources.
00578     //
00579     // Returns a negative error code on failure.
00580     int lfs_file_close(lfs_t* lfs, lfs_file_t* file);
00581
00582     // Synchronize a file on storage
00583     //
00584     // Any pending writes are written out to storage.
00585     // Returns a negative error code on failure.
00586     int lfs_file_sync(lfs_t* lfs, lfs_file_t* file);
00587
00588     // Read data from file
00589     //
00590     // Takes a buffer and size indicating where to store the read data.
00591     // Returns the number of bytes read, or a negative error code on failure.
00592     lfs_ssize_t lfs_file_read(lfs_t* lfs, lfs_file_t* file, void* buffer, lfs_size_t size);
00593
00594 #ifndef LFS_READONLY
00595     // Write data to file
```

```
00596     //
00597     // Takes a buffer and size indicating the data to write. The file will not
00598     // actually be updated on the storage until either sync or close is called.
00599     //
00600     // Returns the number of bytes written, or a negative error code on failure.
00601     lfs_ssize_t lfs_file_write(lfs_t* lfs, lfs_file_t* file, const void* buffer, lfs_size_t size);
00602 #endif
00603
00604     // Change the position of the file
00605     //
00606     // The change in position is determined by the offset and whence flag.
00607     // Returns the new position of the file, or a negative error code on failure.
00608     lfs_soff_t lfs_file_seek(lfs_t* lfs, lfs_file_t* file, lfs_soff_t off, int whence);
00609
00610 #ifndef LFS_READONLY
00611     // Truncates the size of the file to the specified size
00612     //
00613     // Returns a negative error code on failure.
00614     int lfs_file_truncate(lfs_t* lfs, lfs_file_t* file, lfs_off_t size);
00615 #endif
00616
00617     // Return the position of the file
00618     //
00619     // Equivalent to lfs_file_seek(lfs, file, 0, LFS_SEEK_CUR)
00620     // Returns the position of the file, or a negative error code on failure.
00621     lfs_soff_t lfs_file_tell(lfs_t* lfs, lfs_file_t* file);
00622
00623     // Change the position of the file to the beginning of the file
00624     //
00625     // Equivalent to lfs_file_seek(lfs, file, 0, LFS_SEEK_SET)
00626     // Returns a negative error code on failure.
00627     int lfs_file_rewind(lfs_t* lfs, lfs_file_t* file);
00628
00629     // Return the size of the file
00630     //
00631     // Similar to lfs_file_seek(lfs, file, 0, LFS_SEEK_END)
00632     // Returns the size of the file, or a negative error code on failure.
00633     lfs_soff_t lfs_file_size(lfs_t* lfs, lfs_file_t* file);
00634
00636
00637 #ifndef LFS_READONLY
00638     // Create a directory
00639     //
00640     // Returns a negative error code on failure.
00641     int lfs_mkdir(lfs_t* lfs, const char* path);
00642 #endif
00643
00644     // Open a directory
00645     //
00646     // Once open a directory can be used with read to iterate over files.
00647     // Returns a negative error code on failure.
00648     int lfs_dir_open(lfs_t* lfs, lfs_dir_t* dir, const char* path);
00649
00650     // Close a directory
00651     //
00652     // Releases any allocated resources.
00653     // Returns a negative error code on failure.
00654     int lfs_dir_close(lfs_t* lfs, lfs_dir_t* dir);
00655
00656     // Read an entry in the directory
00657     //
00658     // Fills out the info structure, based on the specified file or directory.
00659     // Returns a positive value on success, 0 at the end of directory,
00660     // or a negative error code on failure.
00661     int lfs_dir_read(lfs_t* lfs, lfs_dir_t* dir, struct lfs_info* info);
00662
00663     // Change the position of the directory
00664     //
00665     // The new off must be a value previous returned from tell and specifies
00666     // an absolute offset in the directory seek.
00667     //
00668     // Returns a negative error code on failure.
00669     int lfs_dir_seek(lfs_t* lfs, lfs_dir_t* dir, lfs_off_t off);
00670
00671     // Return the position of the directory
00672     //
00673     // The returned offset is only meant to be consumed by seek and may not make
00674     // sense, but does indicate the current position in the directory iteration.
00675     //
00676     // Returns the position of the directory, or a negative error code on failure.
00677     lfs_soff_t lfs_dir_tell(lfs_t* lfs, lfs_dir_t* dir);
00678
00679     // Change the position of the directory to the beginning of the directory
00680     //
00681     // Returns a negative error code on failure.
00682     int lfs_dir_rewind(lfs_t* lfs, lfs_dir_t* dir);
00683
```

```
00685
00686     // Find on-disk info about the filesystem
00687     //
00688     // Fills out the fsinfo structure based on the filesystem found on-disk.
00689     // Returns a negative error code on failure.
00690     int lfs_fs_stat(lfs_t* lfs, struct lfs_fsinfo* fsinfo);
00691
00692     // Finds the current size of the filesystem
00693     //
00694     // Note: Result is best effort. If files share COW structures, the returned
00695     // size may be larger than the filesystem actually is.
00696     //
00697     // Returns the number of allocated blocks, or a negative error code on failure.
00698     lfs_ssize_t lfs_fs_size(lfs_t* lfs);
00699
00700     // Traverse through all blocks in use by the filesystem
00701     //
00702     // The provided callback will be called with each block address that is
00703     // currently in use by the filesystem. This can be used to determine which
00704     // blocks are in use or how much of the storage is available.
00705     //
00706     // Returns a negative error code on failure.
00707     int lfs_fs_traverse(lfs_t* lfs, int (*cb)(void*, lfs_block_t), void* data);
00708
00709 #ifndef LFS_READONLY
00710     // Attempt to make the filesystem consistent and ready for writing
00711     //
00712     // Calling this function is not required, consistency will be implicitly
00713     // enforced on the first operation that writes to the filesystem, but this
00714     // function allows the work to be performed earlier and without other
00715     // filesystem changes.
00716     //
00717     // Returns a negative error code on failure.
00718     int lfs_fs_mkconsistent(lfs_t* lfs);
00719 #endif
00720
00721 #ifndef LFS_READONLY
00722 #ifdef LFS_MIGRATE
00723     // Attempts to migrate a previous version of littlefs
00724     //
00725     // Behaves similarly to the lfs_format function. Attempts to mount
00726     // the previous version of littlefs and update the filesystem so it can be
00727     // mounted with the current version of littlefs.
00728     //
00729     // Requires a littlefs object and config struct. This clobbers the littlefs
00730     // object, and does not leave the filesystem mounted. The config struct must
00731     // be zeroed for defaults and backwards compatibility.
00732     //
00733     // Returns a negative error code on failure.
00734     int lfs_migrate(lfs_t* lfs, const struct lfs_config* cfg);
00735 #endif
00736 #endif
00737
00738 #ifdef __cplusplus
00739 } /* extern "C" */
00740 #endif
00741
00742 #endif
```

5.73 lfs_util.h

```
00001 /*
00002  * lfs utility functions
00003  *
00004  * Copyright (c) 2022, The littlefs authors.
00005  * Copyright (c) 2017, Arm Limited. All rights reserved.
00006  * SPDX-License-Identifier: BSD-3-Clause
00007  */
00008 #ifndef LFS_UTIL_H
00009 #define LFS_UTIL_H
00010
00011 // Users can override lfs_util.h with their own configuration by defining
00012 // LFS_CONFIG as a header file to include (-DLFS_CONFIG=lfs_config.h).
00013 //
00014 // If LFS_CONFIG is used, none of the default utils will be emitted and must be
00015 // provided by the config file. To start, I would suggest copying lfs_util.h
00016 // and modifying as needed.
00017 #ifdef LFS_CONFIG
00018 #define LFS_STRINGIZE(x) LFS_STRINGIZE2(x)
00019 #define LFS_STRINGIZE2(x) #x
00020 #include LFS_STRINGIZE(LFS_CONFIG)
00021 #else
00022
```

```
00023 // System includes
00024 #include <stdint.h>
00025 #include <stdbool.h>
00026 #include <string.h>
00027 #include <inttypes.h>
00028
00029 #ifndef LFS_NO_MALLOC
00030 #include <stdlib.h>
00031 #endif
00032 #ifndef LFS_NO_ASSERT
00033 #include <assert.h>
00034 #endif
00035 #if !defined(LFS_NO_DEBUG) || !defined(LFS_NO_WARN) || !defined(LFS_NO_ERROR) ||
      defined(LFS_YES_TRACE)
00036 #include <stdio.h>
00037 #endif
00038
00039 #ifdef __cplusplus
00040 extern "C"
00041 {
00042 #endif
00043
00044 // Macros, may be replaced by system specific wrappers. Arguments to these
00045 // macros must not have side-effects as the macros can be removed for a smaller
00046 // code footprint
00047 // Logging functions
00048 #ifndef LFS_TRACE
00049 #ifdef LFS_YES_TRACE
00050 #define LFS_TRACE_(fmt, ...) printf("%s:%d:trace: " fmt "%s\n", __FILE__, __LINE__, __VA_ARGS__)
00051 #define LFS_TRACE(...) LFS_TRACE_(__VA_ARGS__, "")
00052 #else
00053 #define LFS_TRACE(...)
00054 #endif
00055 #endif
00056
00057 #ifndef LFS_DEBUG
00058 #ifndef LFS_NO_DEBUG
00059 #define LFS_DEBUG_(fmt, ...) printf("%s:%d:debug: " fmt "%s\n", __FILE__, __LINE__, __VA_ARGS__)
00060 #define LFS_DEBUG(...) LFS_DEBUG_(__VA_ARGS__, "")
00061 #else
00062 #define LFS_DEBUG(...)
00063 #endif
00064 #endif
00065
00066 #ifndef LFS_WARN
00067 #ifndef LFS_NO_WARN
00068 #define LFS_WARN_(fmt, ...) printf("%s:%d:warn: " fmt "%s\n", __FILE__, __LINE__, __VA_ARGS__)
00069 #define LFS_WARN(...) LFS_WARN_(__VA_ARGS__, "")
00070 #else
00071 #define LFS_WARN(...)
00072 #endif
00073 #endif
00074
00075 #ifndef LFS_ERROR
00076 #ifndef LFS_NO_ERROR
00077 #define LFS_ERROR_(fmt, ...) printf("%s:%d:error: " fmt "%s\n", __FILE__, __LINE__, __VA_ARGS__)
00078 #define LFS_ERROR(...) LFS_ERROR_(__VA_ARGS__, "")
00079 #else
00080 #define LFS_ERROR(...)
00081 #endif
00082 #endif
00083
00084 // Runtime assertions
00085 #ifndef LFS_ASSERT
00086 #ifndef LFS_NO_ASSERT
00087 #define LFS_ASSERT(test) assert(test)
00088 #else
00089 #define LFS_ASSERT(test)
00090 #endif
00091 #endif
00092
00093     // Builtin functions, these may be replaced by more efficient
00094     // toolchain-specific implementations. LFS_NO_INTRINSICS falls back to a more
00095     // expensive basic C implementation for debugging purposes
00096
00097     // Min/max functions for unsigned 32-bit numbers
00098     static inline uint32_t lfs_max(uint32_t a, uint32_t b)
00099     {
00100         return (a > b) ? a : b;
00101     }
00102
00103     static inline uint32_t lfs_min(uint32_t a, uint32_t b)
00104     {
00105         return (a < b) ? a : b;
00106     }
00107
00108     // Align to nearest multiple of a size
```

```
00109     static inline uint32_t lfs_aligndown(uint32_t a, uint32_t alignment)
00110     {
00111         return a - (a % alignment);
00112     }
00113
00114     static inline uint32_t lfs_alignup(uint32_t a, uint32_t alignment)
00115     {
00116         return lfs_aligndown(a + alignment - 1, alignment);
00117     }
00118
00119     // Find the smallest power of 2 greater than or equal to a
00120     static inline uint32_t lfs_npw2(uint32_t a)
00121     {
00122 #if !defined(LFS_NO_INTRINSICS) && (defined(__GNUC__) || defined(__CC_ARM))
00123         return 32 - __builtin_clz(a - 1);
00124 #else
00125     uint32_t r = 0;
00126     uint32_t s;
00127     a -= 1;
00128     s = (a > 0xffff) << 4;
00129     a >>= s;
00130     r |= s;
00131     s = (a > 0xff) << 3;
00132     a >>= s;
00133     r |= s;
00134     s = (a > 0xf) << 2;
00135     a >>= s;
00136     r |= s;
00137     s = (a > 0x3) << 1;
00138     a >>= s;
00139     r |= s;
00140     return (r | (a >> 1)) + 1;
00141 #endif
00142     }
00143
00144     // Count the number of trailing binary zeros in a
00145     // lfs_ctz(0) may be undefined
00146     static inline uint32_t lfs_ctz(uint32_t a)
00147     {
00148 #if !defined(LFS_NO_INTRINSICS) && defined(__GNUC__)
00149         return __builtin_ctz(a);
00150 #else
00151     return lfs_npw2((a & -a) + 1) - 1;
00152 #endif
00153     }
00154
00155     // Count the number of binary ones in a
00156     static inline uint32_t lfs_popc(uint32_t a)
00157     {
00158 #if !defined(LFS_NO_INTRINSICS) && (defined(__GNUC__) || defined(__CC_ARM))
00159         return __builtin_popcount(a);
00160 #else
00161     a = a - ((a >> 1) & 0x55555555);
00162     a = (a & 0x33333333) + ((a >> 2) & 0x33333333);
00163     return (((a + (a >> 4)) & 0xf0f0f0f) * 0x1010101) >> 24;
00164 #endif
00165     }
00166
00167     // Find the sequence comparison of a and b, this is the distance
00168     // between a and b ignoring overflow
00169     static inline int lfs_scmp(uint32_t a, uint32_t b)
00170     {
00171         return (int)(unsigned)(a - b);
00172     }
00173
00174     // Convert between 32-bit little-endian and native order
00175     static inline uint32_t lfs_fromle32(uint32_t a)
00176     {
00177 #if (defined(BYTE_ORDER) && defined(ORDER_LITTLE_ENDIAN) && BYTE_ORDER == ORDER_LITTLE_ENDIAN) ||
    \
00178     (defined(__BYTE_ORDER) && defined(__ORDER_LITTLE_ENDIAN) && __BYTE_ORDER == __ORDER_LITTLE_ENDIAN)
    ||                                                          \
00179     (defined(__BYTE_ORDER__) && defined(__ORDER_LITTLE_ENDIAN__) && __BYTE_ORDER__ ==
    __ORDER_LITTLE_ENDIAN__)
00180         return a;
00181 #elif !defined(LFS_NO_INTRINSICS) && ((defined(BYTE_ORDER) && defined(ORDER_BIG_ENDIAN) && BYTE_ORDER
    == ORDER_BIG_ENDIAN) ||                                     \
00182                                       (defined(__BYTE_ORDER) && defined(__ORDER_BIG_ENDIAN) &&
    __BYTE_ORDER == __ORDER_BIG_ENDIAN) ||                      \
00183                                       (defined(__BYTE_ORDER__) && defined(__ORDER_BIG_ENDIAN__) &&
    __BYTE_ORDER__ == __ORDER_BIG_ENDIAN__))
00184     return __builtin_bswap32(a);
00185 #else
00186     return (((uint8_t*)&a)[0] << 0) | (((uint8_t*)&a)[1] << 8) | (((uint8_t*)&a)[2] << 16) |
    (((uint8_t*)&a)[3] << 24);
00187 #endif
00188     }
```

```
00189
00190     static inline uint32_t lfs_tole32(uint32_t a)
00191     {
00192         return lfs_fromle32(a);
00193     }
00194
00195     // Convert between 32-bit big-endian and native order
00196     static inline uint32_t lfs_frombe32(uint32_t a)
00197     {
00198 #if !defined(LFS_NO_INTRINSICS) && ((defined(BYTE_ORDER) && defined(ORDER_LITTLE_ENDIAN) && BYTE_ORDER
    == ORDER_LITTLE_ENDIAN) ||                                      \
00199                                     (defined(__BYTE_ORDER) && defined(__ORDER_LITTLE_ENDIAN) &&
    __BYTE_ORDER == __ORDER_LITTLE_ENDIAN) ||                       \
00200                                     (defined(__BYTE_ORDER__) && defined(__ORDER_LITTLE_ENDIAN__) &&
    __BYTE_ORDER__ == __ORDER_LITTLE_ENDIAN__))
00201         return __builtin_bswap32(a);
00202 #elif (defined(BYTE_ORDER) && defined(ORDER_BIG_ENDIAN) && BYTE_ORDER == ORDER_BIG_ENDIAN) ||
    \
00203     (defined(__BYTE_ORDER) && defined(__ORDER_BIG_ENDIAN) && __BYTE_ORDER == __ORDER_BIG_ENDIAN) ||
    \
00204     (defined(__BYTE_ORDER__) && defined(__ORDER_BIG_ENDIAN__) && __BYTE_ORDER__ ==
    __ORDER_BIG_ENDIAN__)
00205     return a;
00206 #else
00207     return (((uint8_t*)&a)[0] « 24) | (((uint8_t*)&a)[1] « 16) | (((uint8_t*)&a)[2] « 8) |
    (((uint8_t*)&a)[3] « 0);
00208 #endif
00209     }
00210
00211     static inline uint32_t lfs_tobe32(uint32_t a)
00212     {
00213         return lfs_frombe32(a);
00214     }
00215
00216     // Calculate CRC-32 with polynomial = 0x04c11db7
00217     uint32_t lfs_crc(uint32_t crc, const void* buffer, size_t size);
00218
00219     // Allocate memory, only used if buffers are not provided to littlefs
00220     // Note, memory must be 64-bit aligned
00221     static inline void* lfs_malloc(size_t size)
00222     {
00223 #ifndef LFS_NO_MALLOC
00224         return malloc(size);
00225 #else
00226     (void)size;
00227     return NULL;
00228 #endif
00229     }
00230
00231     // Deallocate memory, only used if buffers are not provided to littlefs
00232     static inline void lfs_free(void* p)
00233     {
00234 #ifndef LFS_NO_MALLOC
00235         free(p);
00236 #else
00237     (void)p;
00238 #endif
00239     }
00240
00241 #ifdef __cplusplus
00242 } /* extern "C" */
00243 #endif
00244
00245 #endif
00246 #endif
```

5.74 LittleFS_Wrapper.c File Reference

```
#include "AppConfiguration.h"
#include "Console.h"
#include "LittleFS_Wrapper.h"
#include "W25Qxx.h"
```

Macros

- #define **LFS_READ_SIZE** (128)

- #define **LFS_PROG_SIZE** (LFS_READ_SIZE)
- #define **LFS_LOOKAHEAD_SIZE** (LFS_READ_SIZE / 8)
- #define **LFS_BLOCK_SIZE** (65536)
- #define **LFS_BLOCK_COUNT** (128)
- #define **LFS_BLOCK_CYCLES** (-1)
- #define **LFS_CACHE_SIZE** (64 % LFS_PROG_SIZE == 0 ? 64 : LFS_PROG_SIZE)
- #define **LFS_ERASE_VALUE** (0xff)
- #define **LFS_ERASE_CYCLES** (-1)
- #define **LFS_BADBLOCK_BEHAVIOR** (LFS_TESTBD_BADBLOCK_PROGERROR)

Functions

- static **__attribute__** ((aligned(32)))
- static int lfs_device_prog (const struct lfs_config *c, lfs_block_t block, lfs_off_t off, const void *buffer, lfs_size_t size)
- static int lfs_device_erase (const struct lfs_config *c, lfs_block_t block)
- static int lfs_device_sync (const struct lfs_config *c)
- int lfsWrapper_Init (lfs_t *const plfs)

Variables

- static uint8_t **lfs_read_buf** [LFS_READ_SIZE]
- static uint8_t **lfs_prog_buf** [LFS_PROG_SIZE]
- static const struct lfs_config cfg

5.74.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-08-16

Copyright

5.74.2 Function Documentation

5.74.2.1 lfs_device_prog()

```
static int lfs_device_prog (
            const struct lfs_config * c,
            lfs_block_t block,
            lfs_off_t off,
            const void * buffer,
            lfs_size_t size )  [static]
```

read function provided to LitleFS

Parameters

c	
block	
off	
buffer	
size	

Returns

int

5.74.2.2 lfs_device_erase()

```
static int lfs_device_erase (
            const struct lfs_config * c,
            lfs_block_t block )  [static]
```

Erase function provided to LitleFS.

Parameters

c	
block	

Returns

int

5.74.2.3 lfs_device_sync()

```
static int lfs_device_sync (
            const struct lfs_config * c )  [static]
```

Erase function provided to LitleFS. Not supported by W25qxx.

Parameters

c	

Returns

int

5.74.2.4 lfsWrapper_Init()

```
int lfsWrapper_Init (
            lfs_t *const plfs )
```

Wrapper around littleFS initialization.

Parameters

plfs	pointer to lfs structure

Returns

int non zero if error

< reformat if we can't mount the filesystem

< this should only happen on the first bootHere is the caller graph for this function:

5.74.3 Variable Documentation

5.74.3.1 cfg

```
const struct lfs_config cfg  [static]
```

Initial value:

```
= {

    .read = lfs_device_read,
    .prog = lfs_device_prog,
    .erase = lfs_device_erase,
    .sync = lfs_device_sync,


    .read_size = LFS_READ_SIZE,
    .prog_size = LFS_PROG_SIZE,
    .block_size = LFS_BLOCK_SIZE,
    .block_count = LFS_BLOCK_COUNT,
    .cache_size = LFS_CACHE_SIZE,
    .lookahead_size = LFS_LOOKAHEAD_SIZE,
    .block_cycles = LFS_BLOCK_CYCLES,

    .read_buffer = lfs_read_buf,
    .prog_buffer = lfs_prog_buf,
    .lookahead_buffer = lfs_lookahead_buf,
}
```

configuration of the filesystem is provided by this struct

5.75 LittleFS_Wrapper.h File Reference

```
#include <stdbool.h>
#include ¨lfs.h¨
```

Functions

- int lfsWrapper_Init (lfs_t *const plfs)
- bool **lfs_Test** (void)

5.75.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-08-16

Copyright

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5.75.2 Function Documentation

5.75.2.1 lfsWrapper_Init()

```
int lfsWrapper_Init (
            lfs_t *const plfs )
```

Wrapper around littleFS initialization.

Parameters

plfs	pointer to lfs structure

Returns

int non zero if error

< reformat if we can't mount the filesystem

< this should only happen on the first bootHere is the caller graph for this function:

5.76 LittleFS_Wrapper.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef _LITTLE_FS_WRAPPER_H
00015 #define _LITTLE_FS_WRAPPER_H
00016
00018
00019 #include <stdbool.h>
00020 #include "lfs.h"
00021
00023
00024 int lfsWrapper_Init(lfs_t* const plfs);
00025 bool lfs_Test(void);
00026
00028
00029 #endif /* _LITTLE_FS_WRAPPER_H */
```

5.77 W25Qxx.c File Reference

```
#include <stddef.h>
#include <stdbool.h>
#include <stdlib.h>
#include <stdio.h>
#include <string.h>
#include ¨AppConfiguration.h¨
#include ¨DebugPrint.h¨
#include ¨W25qxx.h¨
```

Macros

- #define **W25QXX_DUMMY_BYTE** (0xA5)
- #define **W25qxx_Delay**(delay) HAL_Delay(delay)

Functions

- void **FLASH_SS_Clear** ()
- void **FLASH_SS_Set** ()
- uint8_t **W25qxx_Spi** (uint8_t Data)
- uint32_t **W25qxx_ReadID** (void)
- void **W25qxx_ReadUniqID** (void)
- void **W25qxx_WriteEnable** (void)
- void **W25qxx_WriteDisable** (void)
- uint8_t **W25qxx_ReadStatusRegister** (uint8_t SelectStatusRegister_1_2_3)
- void **W25qxx_WriteStatusRegister** (uint8_t SelectStatusRegister_1_2_3, uint8_t Data)
- void **W25qxx_WaitForWriteEnd** (void)
- eW25qxxStatus **W25qxx_Init** (void)
- eW25qxxStatus **W25qxx_EraseChip** (void)
- eW25qxxStatus **W25qxx_EraseSector** (uint32_t SectorAddr)
- eW25qxxStatus **W25qxx_EraseBlock** (uint32_t BlockAddr)
- uint32_t **W25qxx_PageToSector** (uint32_t PageAddress)
- uint32_t **W25qxx_PageToBlock** (uint32_t PageAddress)
- uint32_t **W25qxx_SectorToBlock** (uint32_t SectorAddress)
- uint32_t **W25qxx_SectorToPage** (uint32_t SectorAddress)

- uint32_t **W25qxx_BlockToPage** (uint32_t BlockAddress)
- bool **W25qxx_IsEmptyPage** (uint32_t Page_Address, uint32_t OffsetInByte, uint32_t NumByteToCheck_↩ up_to_PageSize)
- bool **W25qxx_IsEmptySector** (uint32_t Sector_Address, uint32_t OffsetInByte, uint32_t NumByteTo↩ Check_up_to_SectorSize)
- bool **W25qxx_IsEmptyBlock** (uint32_t Block_Address, uint32_t OffsetInByte, uint32_t NumByteToCheck↩ _up_to_BlockSize)
- eW25qxxStatus **W25qxx_WriteByte** (uint8_t pBuffer, uint32_t WriteAddr_inBytes)
- eW25qxxStatus **W25qxx_WritePage** (uint8_t ∗pBuffer, uint32_t Page_Address, uint32_t OffsetInByte, uint32_t NumByteToWrite_up_to_PageSize)
- eW25qxxStatus **W25qxx_WriteSector** (uint8_t ∗pBuffer, uint32_t Sector_Address, uint32_t OffsetInByte, uint32_t NumByteToWrite_up_to_SectorSize)
- eW25qxxStatus **W25qxx_WriteBlock** (uint8_t ∗pBuffer, uint32_t Block_Address, uint32_t OffsetInByte, uint32_t NumByteToWrite_up_to_BlockSize)
- eW25qxxStatus **W25qxx_ReadByte** (uint8_t ∗pBuffer, uint32_t Bytes_Address)
- eW25qxxStatus **W25qxx_ReadBytes** (uint8_t ∗pBuffer, uint32_t ReadAddr, uint32_t NumByteToRead)
- eW25qxxStatus **W25qxx_ReadPage** (uint8_t ∗pBuffer, uint32_t Page_Address, uint32_t OffsetInByte, uint32_t NumByteToRead_up_to_PageSize)
- eW25qxxStatus **W25qxx_ReadSector** (uint8_t ∗pBuffer, uint32_t Sector_Address, uint32_t OffsetInByte, uint32_t NumByteToRead_up_to_SectorSize)
- eW25qxxStatus **W25qxx_ReadBlock** (uint8_t ∗pBuffer, uint32_t Block_Address, uint32_t OffsetInByte, uint32_t NumByteToRead_up_to_BlockSize)

Variables

- static w25qxx_t **gW25qxxDev**

5.77.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-08-05

Copyright

5.78 W25Qxx.h File Reference

```
#include <stdint.h>
#include <stdbool.h>
#include ¨spi.h¨
#include ¨gpio.h¨
```

Data Structures

- struct df_dataparam
- struct w25qxx_t

Macros

- #define **W25QXXH_SPI_HANDLE** (&hspi2)
- #define **W25QXXH_SPI_TIMEOUT_MS** (100)
- #define **W25QXXH_SPI_CS_PORT** (SPI2_NSS_GPIO_Port)
- #define **W25XXH_SPI_CS_PIN** (SPI2_NSS_Pin)
- #define **_W25QXX_DEBUG** (0)
- #define **DF_MAX_NO_PAGE** 65536
- #define **DF_PAGE_SIZE** 256
- #define **DF_SECTOR_SIZE** 16
- #define **DF_SBLOCK_SIZE** 128
- #define **DF_BBLOCK_SIZE** 256
- #define **DF_MAX_SECTOR** 4096
- #define **DF_MAX_SBLOCK** 512
- #define **DF_MAX_BBLOCK** 256

Typedefs

- typedef int32_t **eW25qxxStatus**

Enumerations

- enum **W25QXX_ID_t** {
  W25Q10 = 1 , **W25Q20** , **W25Q40** , **W25Q80** ,
  W25Q16 , **W25Q32** , **W25Q64** , **W25Q128** ,
  W25Q256 , **W25Q512** }

Functions

- eW25qxxStatus **W25qxx_Init** (void)
- eW25qxxStatus **W25qxx_EraseChip** (void)
- eW25qxxStatus **W25qxx_EraseSector** (uint32_t SectorAddr)
- eW25qxxStatus **W25qxx_EraseBlock** (uint32_t BlockAddr)
- uint32_t **W25qxx_PageToSector** (uint32_t PageAddress)
- uint32_t **W25qxx_PageToBlock** (uint32_t PageAddress)
- uint32_t **W25qxx_SectorToBlock** (uint32_t SectorAddress)
- uint32_t **W25qxx_SectorToPage** (uint32_t SectorAddress)
- uint32_t **W25qxx_BlockToPage** (uint32_t BlockAddress)
- bool **W25qxx_IsEmptyPage** (uint32_t Page_Address, uint32_t OffsetInByte, uint32_t NumByteToCheck_↩ up_to_PageSize)
- bool **W25qxx_IsEmptySector** (uint32_t Sector_Address, uint32_t OffsetInByte, uint32_t NumByteTo↩ Check_up_to_SectorSize)
- bool **W25qxx_IsEmptyBlock** (uint32_t Block_Address, uint32_t OffsetInByte, uint32_t NumByteToCheck↩ _up_to_BlockSize)
- eW25qxxStatus **W25qxx_WriteByte** (uint8_t pBuffer, uint32_t WriteAddr_inBytes)
- eW25qxxStatus **W25qxx_WritePage** (uint8_t ∗pBuffer, uint32_t Page_Address, uint32_t OffsetInByte, uint32_t NumByteToWrite_up_to_PageSize)

- eW25qxxStatus **W25qxx_WriteSector** (uint8_t *pBuffer, uint32_t Sector_Address, uint32_t OffsetInByte, uint32_t NumByteToWrite_up_to_SectorSize)
- eW25qxxStatus **W25qxx_WriteBlock** (uint8_t *pBuffer, uint32_t Block_Address, uint32_t OffsetInByte, uint32_t NumByteToWrite_up_to_BlockSize)
- eW25qxxStatus **W25qxx_ReadByte** (uint8_t *pBuffer, uint32_t Bytes_Address)
- eW25qxxStatus **W25qxx_ReadBytes** (uint8_t *pBuffer, uint32_t ReadAddr, uint32_t NumByteToRead)
- eW25qxxStatus **W25qxx_ReadPage** (uint8_t *pBuffer, uint32_t Page_Address, uint32_t OffsetInByte, uint32_t NumByteToRead_up_to_PageSize)
- eW25qxxStatus **W25qxx_ReadSector** (uint8_t *pBuffer, uint32_t Sector_Address, uint32_t OffsetInByte, uint32_t NumByteToRead_up_to_SectorSize)
- eW25qxxStatus **W25qxx_ReadBlock** (uint8_t *pBuffer, uint32_t Block_Address, uint32_t OffsetInByte, uint32_t NumByteToRead_up_to_BlockSize)
- bool **W25qxx_Test** ()

5.78.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2023-08-05

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5.79 W25Qxx.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef _W25Qxx_H /* Guard against multiple inclusion */
00015 #define _W25Qxx_H
00016
00018
00019 #include <stdint.h>
00020 #include <stdbool.h>
00021 #include "spi.h"
00022 #include "gpio.h"
00023
00025
00026 #define W25QXXH_SPI_HANDLE (&hspi2)
00027 #define W25QXXH_SPI_TIMEOUT_MS (100)
00028
00029 #define W25QXXH_SPI_CS_PORT (SPI2_NSS_GPIO_Port)
00030 #define W25XXH_SPI_CS_PIN (SPI2_NSS_Pin)
00031
00033
00034 #define _W25QXX_DEBUG (0)
00035
00036 #define DF_MAX_NO_PAGE 65536
00037 #define DF_PAGE_SIZE 256
00038 #define DF_SECTOR_SIZE 16  // 16 pages in one sector = 4KB
00039 #define DF_SBLOCK_SIZE 128 // 32KB
```

```
00040 #define DF_BBLOCK_SIZE 256 // 64KB
00041 #define DF_MAX_SECTOR 4096
00042 #define DF_MAX_SBLOCK 512
00043 #define DF_MAX_BBLOCK 256
00044
00046
00047 typedef enum
00048 {
00049     W25Q10 = 1,
00050     W25Q20,
00051     W25Q40,
00052     W25Q80,
00053     W25Q16,
00054     W25Q32,
00055     W25Q64,
00056     W25Q128,
00057     W25Q256,
00058     W25Q512,
00059 } W25QXX_ID_t;
00060
00062
00063 typedef struct
00064 {
00065     uint8_t txbuff[DF_PAGE_SIZE];
00066     uint8_t rxbuff[DF_PAGE_SIZE];
00067     uint8_t txbuff_len;
00068     uint8_t rxbuff_len;
00069     uint8_t rxbuff_readpos;
00070     uint8_t df_mode;
00071
00072 } df_dataparam;
00073
00074 typedef struct
00075 {
00076     W25QXX_ID_t ID;
00077     uint8_t UniqID[8];
00078     uint16_t PageSize;
00079     uint32_t PageCount;
00080     uint32_t SectorSize;
00081     uint32_t SectorCount;
00082     uint32_t BlockSize;
00083     uint32_t BlockCount;
00084     uint32_t CapacityInKiloByte;
00085     uint8_t StatusRegister1;
00086     uint8_t StatusRegister2;
00087     uint8_t StatusRegister3;
00088     uint8_t Lock;
00089 } w25qxx_t;
00090
00091 typedef int32_t eW25qxxStatus;
00092
00094
00095 // ###########################################################################
00096 //  in Page,Sector and block read/write functions, can put 0 to read maximum bytes
00097 // ###########################################################################
00098 eW25qxxStatus W25qxx_Init(void);
00099
00100 eW25qxxStatus W25qxx_EraseChip(void);
00101 eW25qxxStatus W25qxx_EraseSector(uint32_t SectorAddr);
00102 eW25qxxStatus W25qxx_EraseBlock(uint32_t BlockAddr);
00103
00104 uint32_t W25qxx_PageToSector(uint32_t PageAddress);
00105 uint32_t W25qxx_PageToBlock(uint32_t PageAddress);
00106 uint32_t W25qxx_SectorToBlock(uint32_t SectorAddress);
00107 uint32_t W25qxx_SectorToPage(uint32_t SectorAddress);
00108 uint32_t W25qxx_BlockToPage(uint32_t BlockAddress);
00109
00110 bool W25qxx_IsEmptyPage(uint32_t Page_Address, uint32_t OffsetInByte, uint32_t
      NumByteToCheck_up_to_PageSize);
00111 bool W25qxx_IsEmptySector(uint32_t Sector_Address, uint32_t OffsetInByte, uint32_t
      NumByteToCheck_up_to_SectorSize);
00112 bool W25qxx_IsEmptyBlock(uint32_t Block_Address, uint32_t OffsetInByte, uint32_t
      NumByteToCheck_up_to_BlockSize);
00113
00114 eW25qxxStatus W25qxx_WriteByte(uint8_t pBuffer, uint32_t WriteAddr_inBytes);
00115 eW25qxxStatus W25qxx_WritePage(uint8_t* pBuffer, uint32_t Page_Address, uint32_t OffsetInByte,
      uint32_t NumByteToWrite_up_to_PageSize);
00116 eW25qxxStatus W25qxx_WriteSector(uint8_t* pBuffer, uint32_t Sector_Address, uint32_t OffsetInByte,
      uint32_t NumByteToWrite_up_to_SectorSize);
00117 eW25qxxStatus W25qxx_WriteBlock(uint8_t* pBuffer, uint32_t Block_Address, uint32_t OffsetInByte,
      uint32_t NumByteToWrite_up_to_BlockSize);
00118
00119 eW25qxxStatus W25qxx_ReadByte(uint8_t* pBuffer, uint32_t Bytes_Address);
00120 eW25qxxStatus W25qxx_ReadBytes(uint8_t* pBuffer, uint32_t ReadAddr, uint32_t NumByteToRead);
00121 eW25qxxStatus W25qxx_ReadPage(uint8_t* pBuffer, uint32_t Page_Address, uint32_t OffsetInByte, uint32_t
      NumByteToRead_up_to_PageSize);
00122 eW25qxxStatus W25qxx_ReadSector(uint8_t* pBuffer, uint32_t Sector_Address, uint32_t OffsetInByte,
```

```
      uint32_t NumByteToRead_up_to_SectorSize);
00123 eW25qxxStatus W25qxx_ReadBlock(uint8_t* pBuffer, uint32_t Block_Address, uint32_t OffsetInByte,
      uint32_t NumByteToRead_up_to_BlockSize);
00124
00125 bool W25qxx_Test();
00126
00128
00129 #endif /* _W25Qxx_H */
```

5.80 AppSD_API.c File Reference

```
#include <assert.h>
#include <string.h>
#include ¨AppConfiguration.h¨
#include ¨AppSD_API.h¨
```

Functions

- static sSDFS_t * SDFs_GetInstance ()
- static eStorageFSStatus_t SDFs_Init (sSDFS_t *const pMe)
- __attribute__ ((unused))
- static eStorageFSStatus_t SDFs_OpenFileRaw (sSDFS_t *const pMe, eStorageFileNamesEnums_t file↩ Enum, eStorageFSOperationModes_t modeEnum)
- static eStorageFSStatus_t SDFs_CloseFileRaw (sSDFS_t *const pMe, eStorageFileNamesEnums_t file↩ Enum)
- static eStorageFSStatus_t SDFs_ReadFileRaw (sSDFS_t *const pMe, eStorageFileNamesEnums_t file↩ Enum, char *const pOutReadBuf, uint32_t bytesToRead, uint32_t *const pOutBytesRead)
- static eStorageFSStatus_t SDFs_ComputeFileCRC (sSDFS_t *const pMe, eStorageFileNamesEnums_t fileEnum, uint8_t *const pInOutRamBuf, uint32_t RamBufSize, uint32_t *const pOutCRC)
- static eStorageFSStatus_t SDFs_GetFileSizeRaw (const sSDFS_t *const pMe, eStorageFileNamesEnums_t fileEnum, uint32_t *const pOutFileSizeInBytes)
- static eStorageFSStatus_t SDFs_GetFilePresent (sSDFS_t *const pMe, eStorageFileNamesEnums_t file↩ Enum)
- eStorageFSStatus_t SDFs_API_Init ()
- eStorageFSStatus_t SDFs_API_GetGoldenFileStatus ()
- eStorageFSStatus_t SDFs_API_OpenGoldenImageFile ()
- eStorageFSStatus_t SDFs_API_GetGoldenImageFileSize (uint32_t *pOutFileSizeInBytes)
- eStorageFSStatus_t SDFs_API_ReadGoldenImageFile (char *const pOutReadBuf, uint32_t bytesToRead, uint32_t *const pOutBytesRead)
- eStorageFSStatus_t SDFs_API_CloseGoldenImageFile ()
- eStorageFSStatus_t SDFs_API_ComputeGoldenImageFileCRC (uint8_t *const pInOutRamBuf, uint32_↩ t RamBufSize, uint32_t *const pOutCRC)

Variables

- static const uint8_t gcOpModeToFatFSModeConverterTable [eFS_MODE_MAX]
- static const char *const gcFilesNamesTable [eFS_MAX]
- static sSDFS_t **gSDCard** = {.IsMounted = false}

5.80.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-06-07

Copyright

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5.80.2 Function Documentation

5.80.2.1 SDFs_GetInstance()

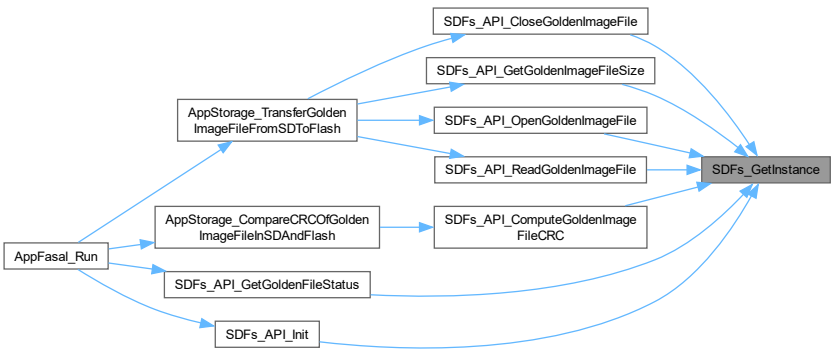
```
static sSDFS_t * SDFs_GetInstance ( )  [static]
```

Get storage media instance.

Returns

sSDFS_t*

Here is the caller graph for this function:



5.80.2.2 SDFs_Init()

```
static eStorageFSStatus_t SDFs_Init (
            sSDFS_t *const pMe )  [static]
```

Initialize FatFS file-system.

Parameters

pMe	FatFS wrapper instance

Returns

eStorageFSStatus_t

Here is the caller graph for this function:

5.80.2.3 __attribute__()

```
__attribute__ (
              (unused)  )
```

Initialize FatFS file-system.

Delete File.

Write to a file in FatFS.

Parameters

pMe	FatFS wrapper instance

Returns

eStorageFSStatus_t

Parameters

pMe	FatFS wrapper instance
fileEnum	File to be written into
IsAppend	Pass true to write to file in append mode, false to write at the beginning
pInWriteBuf	data to be written is passed here
bufSize	size of buffer passed for writing

Returns

eStorageFSStatus_t

Parameters

pMe	FatFS wrapper instance
fileEnum	File to be deleted

Returns

eStorageFSStatus_t

5.80.2.4 SDFs_OpenFileRaw()

```
static eStorageFSStatus_t SDFs_OpenFileRaw (
            sSDFS_t *const pMe,
            eStorageFileNamesEnums_t fileEnum,
            eStorageFSOperationModes_t modeEnum )  [static]
```

Open a file in FatFS.

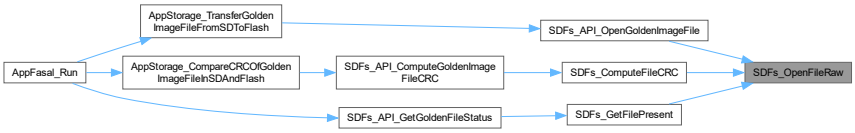
Parameters

pMe	FatFS wrapper instance
fileEnum	File to be opened
modeEnum	Mode to open file In

Returns

eStorageFSStatus_t

Here is the caller graph for this function:



5.80.2.5 SDFs_CloseFileRaw()

```
static eStorageFSStatus_t SDFs_CloseFileRaw (
            sSDFS_t *const pMe,
            eStorageFileNamesEnums_t fileEnum )  [static]
```

Close a file in FatFS.

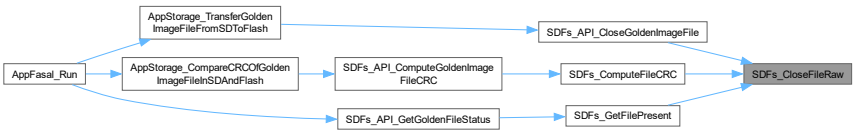
Parameters

pMe	FatFS wrapper instance
fileEnum	file to be closed

Returns

eStorageFSStatus_t

Here is the caller graph for this function:



5.80.2.6 SDFs_ReadFileRaw()

```
static eStorageFSStatus_t SDFs_ReadFileRaw (
            sSDFS_t *const pMe,
            eStorageFileNamesEnums_t fileEnum,
            char *const pOutReadBuf,
            uint32_t bytesToRead,
            uint32_t *const pOutBytesRead )  [static]
```

read from a file in FatFS

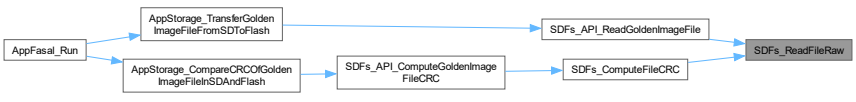
Parameters

pMe	FatFS wrapper instance
fileEnum	File to be read from
pOutReadBuf	read data will be saved here
bytesToRead	number of bytes to read
pOutBytesRead	number of bytes read is saved here

Returns

eStorageFSStatus_t

Here is the caller graph for this function:



5.80.2.7 SDFs_ComputeFileCRC()

```
static eStorageFSStatus_t SDFs_ComputeFileCRC (
            sSDFS_t *const pMe,
            eStorageFileNamesEnums_t fileEnum,
            uint8_t *const pInOutRamBuf,
            uint32_t RamBufSize,
            uint32_t *const pOutCRC )  [static]
```

compute CRC of requested file

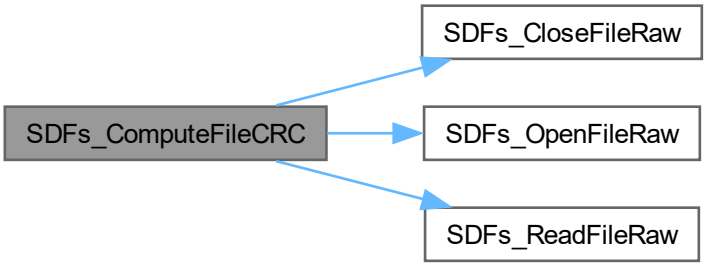
Parameters

pMe	FatFS wrapper instance
fileEnum	File whose CRC needs to be computed
pInOutRamBuf	Ram buffer that will be used as temporary storage while computing CRC
RamBufSize	ram buffer size passed
pOutCRC	computed CRC will be saved here

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.80.2.8 SDFs_GetFileSizeRaw()

```
static eStorageFSStatus_t SDFs_GetFileSizeRaw (
            const sSDFS_t *const pMe,
            eStorageFileNamesEnums_t fileEnum,
            uint32_t *const pOutFileSizeInBytes )  [static]
```

Get size of file in FatFS.

Parameters

pMe	FatFS wrapper instance
fileEnum	file whose size needs to be determined
pOutFileSizeInBytes	size of the file computed is saved here

Returns

eStorageFSStatus_t

Here is the caller graph for this function:

5.80.2.9 SDFs_GetFilePresent()

```
static eStorageFSStatus_t SDFs_GetFilePresent (
            sSDFS_t *const pMe,
            eStorageFileNamesEnums_t fileEnum )  [static]
```

Determine if a file is present in FatFS.

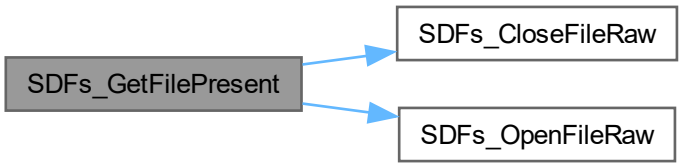
Parameters

pMe	FatFS wrapper instance
fileEnum	file to be checked

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.80.2.10 SDFs_API_Init()

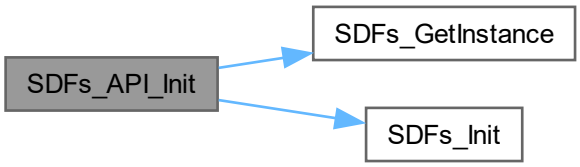
```
eStorageFSStatus_t SDFs_API_Init ( )
```

Initialize FatFS wrapper.

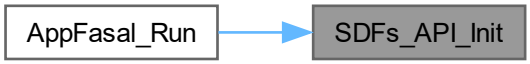
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.80.2.11 SDFs_API_GetGoldenFileStatus()

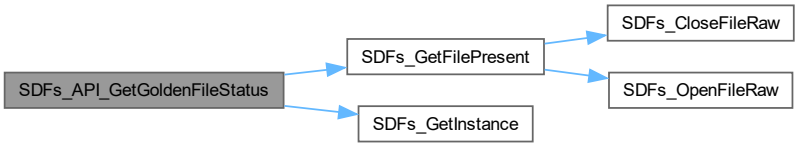
```
eStorageFSStatus_t SDFs_API_GetGoldenFileStatus ( )
```

Check if file exists.

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.80.2.12 SDFs_API_OpenGoldenImageFile()

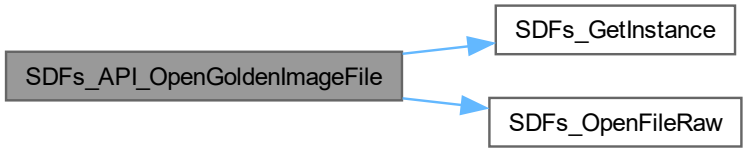
```
eStorageFSStatus_t SDFs_API_OpenGoldenImageFile ( )
```

Open Golden Image file.

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.80.2.13 SDFs_API_GetGoldenImageFileSize()

```
eStorageFSStatus_t SDFs_API_GetGoldenImageFileSize (
            uint32_t * pOutFileSizeInBytes )
```

Get size of golden image.

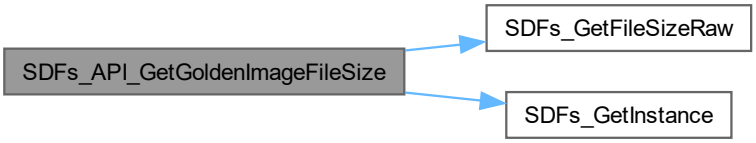
Parameters

pOutFileSizeInBytes	size of the golden image is saved here

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.80.2.14 SDFs_API_ReadGoldenImageFile()

```
eStorageFSStatus_t SDFs_API_ReadGoldenImageFile (
            char *const pOutReadBuf,
            uint32_t bytesToRead,
            uint32_t *const pOutBytesRead )
```

Read golden image file.

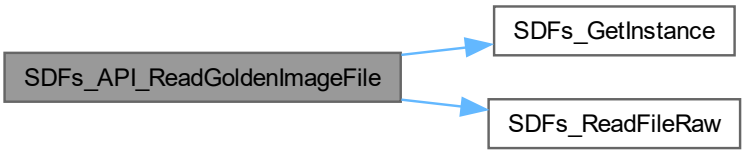
Parameters

pOutReadBuf	contents of read file are saved here
bytesToRead	bytes to read from file
pOutBytesRead	bytes read from file

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.80.2.15 SDFs_API_CloseGoldenImageFile()

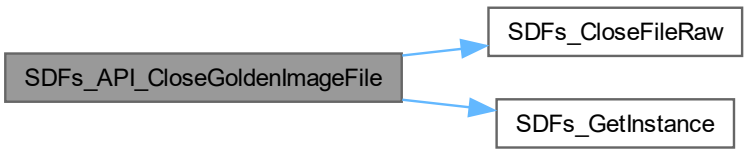
```
eStorageFSStatus_t SDFs_API_CloseGoldenImageFile ( )
```

Close golden Image file.

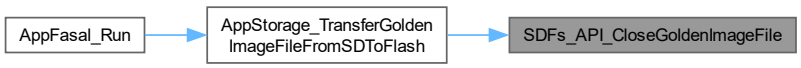
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.80.2.16 SDFs_API_ComputeGoldenImageFileCRC()

```
eStorageFSStatus_t SDFs_API_ComputeGoldenImageFileCRC (
            uint8_t *const pInOutRamBuf,
            uint32_t RamBufSize,
            uint32_t *const pOutCRC )
```

compute CRC of golden Image file

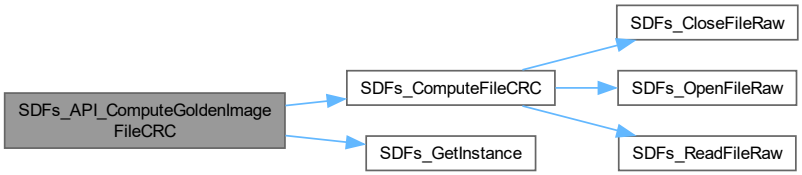
Parameters

pInOutRamBuf	Ram buffer that will be used as temporary storage while computing CRC
RamBufSize	ram buffer size passed
pOutCRC	computed CRC will be saved here

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.80.3 Variable Documentation

5.80.3.1 gcOpModeToFatFSModeConverterTable

```
const uint8_t gcOpModeToFatFSModeConverterTable[eFS_MODE_MAX]  [static]
```

Initial value:

```
=
{
        [eFS_READONLY]          = (FA_READ),
        [eFS_WRITEONLY]         = (FA_WRITE),
        [eFS_READ_WRITE]        = (FA_READ  | FA_WRITE),
        [eFS_READ_CREATE]       = (FA_READ  | FA_OPEN_ALWAYS),
        [eFS_WRITE_CREATE]      = (FA_WRITE | FA_OPEN_ALWAYS),
        [eFS_WRITE_APPEND]      = (FA_WRITE | FA_OPEN_ALWAYS),
        [eFS_READ_WRITE_CREATE] = (FA_READ  | FA_WRITE | FA_OPEN_ALWAYS),
}
```

Utility table to convert user file-system modes to FatFS file system modes.

5.80.3.2 gcFilesNamesTable

```
const char* const gcFilesNamesTable[eFS_MAX]  [static]
```

Initial value:

```
=
{
        [eFS_GOLDEN_IMAGE]  = "fallback.txt",
        [eFS_FILE_2]        = "file2.txt"
}
```

Map Name enums to File name strings.

5.81 AppSD_API.h File Reference

```
#include <stdbool.h>
#include ¨AppStorageDataStructures.h¨
#include ¨crc.h¨
#include ¨ff.h¨
```

Data Structures

- struct sSDFS_t

Macros

- #define **SDFS_CRC_INSTANCE** (&hcrc)

Functions

- eStorageFSStatus_t SDFs_API_Init ()
- eStorageFSStatus_t SDFs_API_GetGoldenFileStatus ()
- eStorageFSStatus_t SDFs_API_OpenGoldenImageFile ()
- eStorageFSStatus_t SDFs_API_GetGoldenImageFileSize (uint32_t *pOutFileSizeInBytes)
- eStorageFSStatus_t SDFs_API_ReadGoldenImageFile (char *const pOutReadBuf, uint32_t bytesToRead, uint32_t *const pOutBytesRead)
- eStorageFSStatus_t SDFs_API_CloseGoldenImageFile ()
- eStorageFSStatus_t SDFs_API_ComputeGoldenImageFileCRC (uint8_t *const pInOutRamBuf, uint32_t RamBufSize, uint32_t *const pOutCRC)

5.81.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-06-07

Copyright

5.81.2 Function Documentation

5.81.2.1 SDFs_API_Init()

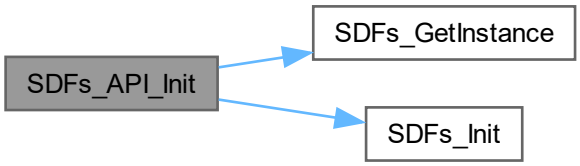
```
eStorageFSStatus_t SDFs_API_Init ( )
```

Initialize FatFS wrapper.

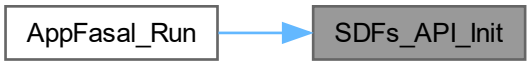
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.81.2.2 SDFs_API_GetGoldenFileStatus()

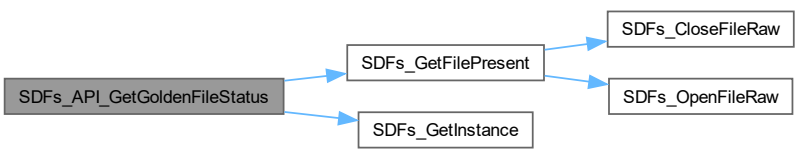
```
eStorageFSStatus_t SDFs_API_GetGoldenFileStatus ( )
```

Check if file exists.

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.81.2.3 SDFs_API_OpenGoldenImageFile()

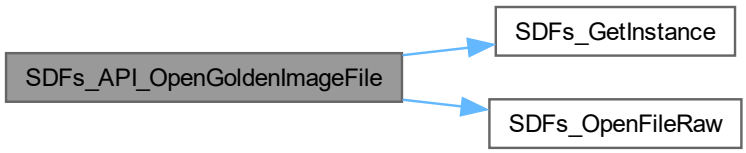
```
eStorageFSStatus_t SDFs_API_OpenGoldenImageFile ( )
```

Open Golden Image file.

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.81.2.4 SDFs_API_GetGoldenImageFileSize()

```
eStorageFSStatus_t SDFs_API_GetGoldenImageFileSize (
            uint32_t * pOutFileSizeInBytes )
```

Get size of golden image.

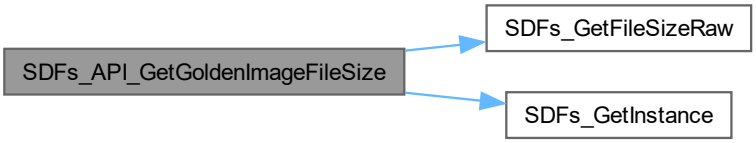
Parameters

pOutFileSizeInBytes	size of the golden image is saved here

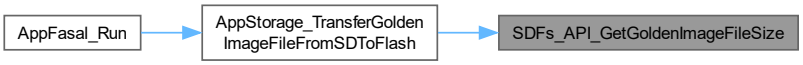
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.81.2.5 SDFs_API_ReadGoldenImageFile()

```
eStorageFSStatus_t SDFs_API_ReadGoldenImageFile (
            char *const pOutReadBuf,
            uint32_t bytesToRead,
            uint32_t *const pOutBytesRead )
```

Read golden image file.

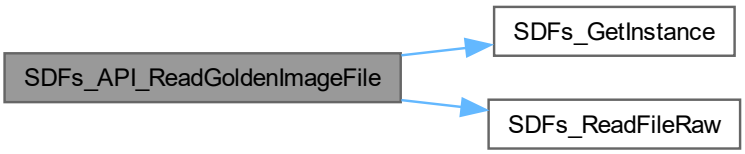
Parameters

pOutReadBuf	contents of read file are saved here
bytesToRead	bytes to read from file
pOutBytesRead	bytes read from file

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.81.2.6 SDFs_API_CloseGoldenImageFile()

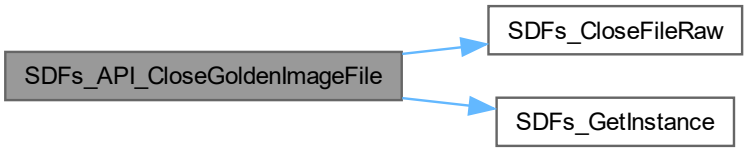
```
eStorageFSStatus_t SDFs_API_CloseGoldenImageFile ( )
```

Close golden Image file.

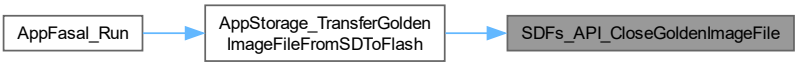
Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:



5.81.2.7 SDFs_API_ComputeGoldenImageFileCRC()

```
eStorageFSStatus_t SDFs_API_ComputeGoldenImageFileCRC (
            uint8_t *const pInOutRamBuf,
            uint32_t RamBufSize,
            uint32_t *const pOutCRC )
```

compute CRC of golden Image file

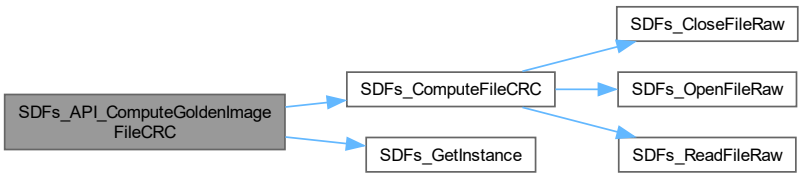
Parameters

pInOutRamBuf	Ram buffer that will be used as temporary storage while computing CRC
RamBufSize	ram buffer size passed
pOutCRC	computed CRC will be saved here

Returns

eStorageFSStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.82 AppSD_API.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPSTORAGE_APPSDFS_APPSD_API_H_
00015 #define APPSTORAGE_APPSDFS_APPSD_API_H_
00016
00018
00019 #include <stdbool.h>
00020 #include "AppStorageDataStructures.h"
00021 #include "crc.h"
00022 #include "ff.h"
```

```
00023
00025
00026 #define SDFS_CRC_INSTANCE (&hcrc)
00027
00029
00034 typedef struct
00035 {
00036     bool IsMounted;
00037     FATFS fs;
00038     FIL fileHandles[eFS_MAX];
00039 } sSDFS_t;
00040
00042
00043 eStorageFSStatus_t SDFs_API_Init();
00044 eStorageFSStatus_t SDFs_API_GetGoldenFileStatus();
00045 eStorageFSStatus_t SDFs_API_OpenGoldenImageFile();
00046 eStorageFSStatus_t SDFs_API_GetGoldenImageFileSize(uint32_t* pOutFileSizeInBytes);
00047 eStorageFSStatus_t SDFs_API_ReadGoldenImageFile(char* const pOutReadBuf, uint32_t bytesToRead,
      uint32_t* const pOutBytesRead);
00048 eStorageFSStatus_t SDFs_API_CloseGoldenImageFile();
00049 eStorageFSStatus_t SDFs_API_ComputeGoldenImageFileCRC(uint8_t* const pInOutRamBuf, uint32_t
      RamBufSize, uint32_t* const pOutCRC);
00050
00052
00053 #endif /* APPSTORAGE_APPSDFS_APPSD_API_H_ */
```

5.83 AppStorage.c File Reference

```
#include <assert.h>
#include <stdbool.h>
#include "AppFlash_API.h"
#include "AppSD_API.h"
#include "AppStorage.h"
#include "AppStorageDataStructures.h"
#include "Console.h"
#include "SPI.h"
#include "W25Qxx.h"
```

Functions

- static __attribute__ ((aligned(4)))
- void AppStorage_SetPower (bool IsEnable)
- eTransferMode_t AppStorage_GetCurrentTransferMode ()
- eStorageFSStatus_t AppStorage_TransferGoldenImageFileFromSDToFlash ()
- eStorageFSStatus_t AppStorage_CompareCRCOfGoldenImageFileInSDAndFlash (bool *const pOutIs↩ CRCMatching)

Variables

- static const eTransferMode_t gcTransferModeToGPIOStatusMappingTable [eTX_MODE_MAX]

5.83.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-06-07

Copyright

5.83.2 Function Documentation

5.83.2.1 AppStorage_SetPower()

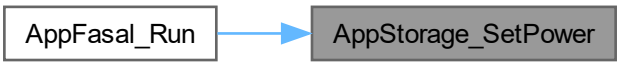
```
void AppStorage_SetPower (
            bool IsEnable )
```

Set Sensor power for powering external flash board.

Note

SPI also needs to be controlled here since power on SPI pins is also powering the external board

Here is the caller graph for this function:



5.83.2.2 AppStorage_GetCurrentTransferMode()

```
eTransferMode_t AppStorage_GetCurrentTransferMode ( )
```

Get current transfer mode setting.

Returns

eTransferMode_t

Here is the caller graph for this function:

5.83.2.3 AppStorage_TransferGoldenImageFileFromSDToFlash()

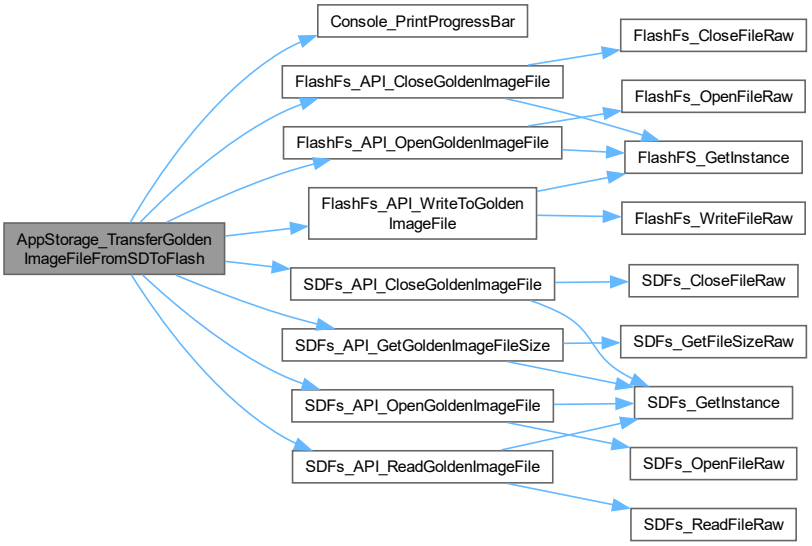
```
eStorageFSStatus_t AppStorage_TransferGoldenImageFileFromSDToFlash ( )
```

Transfer Golden Image from SD card to flash.

Returns

eAppStorageStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.83.2.4 AppStorage_CompareCRCOfGoldenImageFileInSDAndFlash()

```
eStorageFSStatus_t AppStorage_CompareCRCOfGoldenImageFileInSDAndFlash (
            bool *const pOutIsCRCMatching )
```

Compute and compare CRC of golden Image file stored in CRC and Flash.

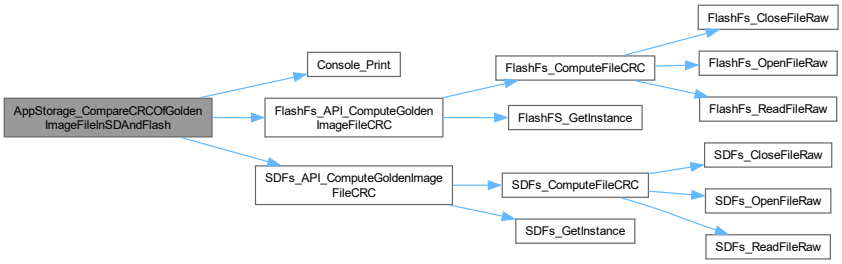
Parameters

pOutIsCRCMatching	Set to true if CRC matches, False otherwise.

Returns

eAppStorageStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.83.3 Variable Documentation

5.83.3.1 gcTransferModeToGPIOStatusMappingTable

```
const eTransferMode_t gcTransferModeToGPIOStatusMappingTable[eTX_MODE_MAX]  [static]
```

Initial value:

```
= {
    [eTX_MODE_SDCARD_TO_FLASH] = GPIO_PIN_RESET, [eTX_MODE_XMODEM_TO_FLASH] = GPIO_PIN_SET}
```

utility mapping of Transfer mode eTransferMode_t to GPIO state

5.84 AppStorage.h File Reference

```
#include <stdbool.h>
#include ¨AppStorageDataStructures.h¨
```

Macros

- #define **FLASH_POWER_EN_PORT** (SENSOR_POWER_ENABLE_GPIO_Port)
- #define **FLASH_POWER_EN_PIN** (SENSOR_POWER_ENABLE_Pin)
- #define **TRANSFER_MODE_PORT** (TRANSFER_MODE_GPIO_Port)
- #define **TRANSFE_MODE_PIN** (TRANSFER_MODE_Pin)

Enumerations

- enum eTransferMode_t { **eTX_MODE_SDCARD_TO_FLASH** , **eTX_MODE_XMODEM_TO_FLASH** , **eTX↩ _MODE_MAX** }

Functions

- void AppStorage_SetPower (bool IsEnable)
- eStorageFSStatus_t AppStorage_TransferGoldenImageFileFromSDToFlash ()
- eStorageFSStatus_t AppStorage_CompareCRCOfGoldenImageFileInSDAndFlash (bool *const pOutIs↩ CRCMatching)
- eTransferMode_t AppStorage_GetCurrentTransferMode ()

5.84.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-06-07

Copyright

5.84.2 Enumeration Type Documentation

5.84.2.1 eTransferMode_t

```
enum eTransferMode_t
```

Enumerations for storage mediums supported.

5.84.3 Function Documentation

5.84.3.1 AppStorage_SetPower()

```
void AppStorage_SetPower (
            bool IsEnable )
```

Set Sensor power for powering external flash board.

Note

SPI also needs to be controlled here since power on SPI pins is also powering the external board

Here is the caller graph for this function:

5.84.3.2 AppStorage_TransferGoldenImageFileFromSDToFlash()

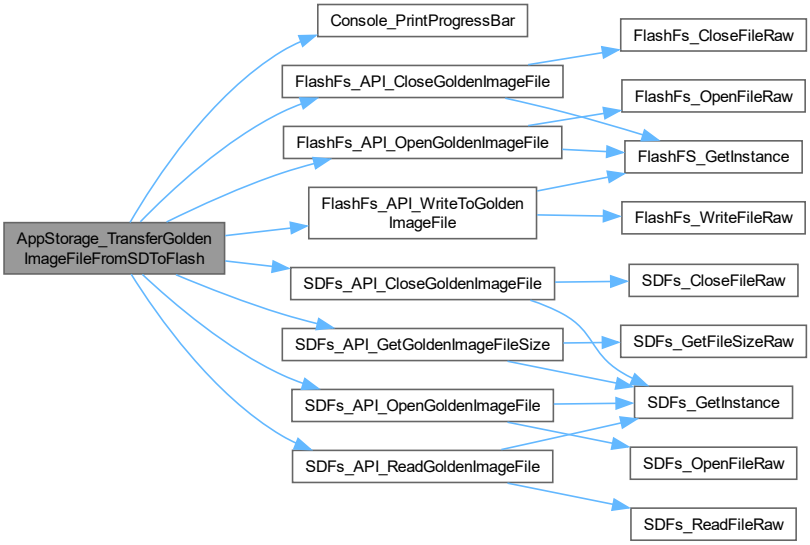
```
eStorageFSStatus_t AppStorage_TransferGoldenImageFileFromSDToFlash ( )
```

Transfer Golden Image from SD card to flash.

Returns

eAppStorageStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.84.3.3 AppStorage_CompareCRCOfGoldenImageFileInSDAndFlash()

```
eStorageFSStatus_t AppStorage_CompareCRCOfGoldenImageFileInSDAndFlash (
            bool *const pOutIsCRCMatching )
```

Compute and compare CRC of golden Image file stored in CRC and Flash.

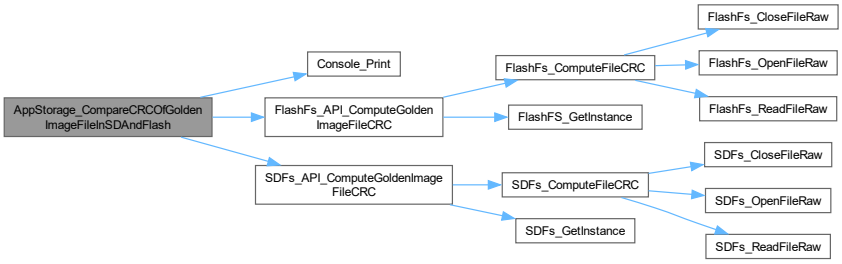
Parameters

pOutIsCRCMatching	Set to true if CRC matches, False otherwise.

Returns

eAppStorageStatus_t

Here is the call graph for this function:



Here is the caller graph for this function:

5.84.3.4 AppStorage_GetCurrentTransferMode()

```
eTransferMode_t AppStorage_GetCurrentTransferMode ( )
```

Get current transfer mode setting.

Returns

eTransferMode_t

Here is the caller graph for this function:

5.85 AppStorage.h

Go to the documentation of this file.

```
00001
00012
00013 #ifndef APPSTORAGE_APPSTORAGE_H_
00014 #define APPSTORAGE_APPSTORAGE_H_
00015
00017
00018 #include <stdbool.h>
00019 #include "AppStorageDataStructures.h"
00020
00022
00023 #define FLASH_POWER_EN_PORT (SENSOR_POWER_ENABLE_GPIO_Port)
00024 #define FLASH_POWER_EN_PIN (SENSOR_POWER_ENABLE_Pin)
00025
00026 #define TRANSFER_MODE_PORT (TRANSFER_MODE_GPIO_Port)
00027 #define TRANSFE_MODE_PIN (TRANSFER_MODE_Pin)
00028
00030
00035 typedef enum
00036 {
00037     eTX_MODE_SDCARD_TO_FLASH,
00038     eTX_MODE_XMODEM_TO_FLASH,
00039     eTX_MODE_MAX
00040 } eTransferMode_t;
00041
00043
00044 void AppStorage_SetPower(bool IsEnable);
00045 eStorageFSStatus_t AppStorage_TransferGoldenImageFileFromSDToFlash();
00046 eStorageFSStatus_t AppStorage_CompareCRCOfGoldenImageFileInSDAndFlash(bool* const pOutIsCRCMatching);
00047 eTransferMode_t AppStorage_GetCurrentTransferMode();
00048
00050
00051 #endif /* APPSTORAGE_APPSTORAGE_H_ */
```

5.86 AppStorageDataStructures.h File Reference

Enumerations

- enum eStorageFSStatus_t { **eFS_SUCCESS** , **eFS_ERROR** , **eFS_NO_FILE** , **eFS_MAX_STATUS** }
- enum eStorageFSOperationModes_t { **eFS_READONLY** , **eFS_WRITEONLY** , **eFS_READ_WRITE** , **eFS_READ_CREATE** , **eFS_WRITE_CREATE** , **eFS_WRITE_APPEND** , **eFS_READ_WRITE_CREATE** , **eFS_MODE_MAX** }
- enum eStorageFileNamesEnums_t { **eFS_GOLDEN_IMAGE** , **eFS_FILE_2** , **eFS_MAX** }

5.86.1 Detailed Description

Author

Vishal Keshava Murthy

Version

0.1

Date

2024-06-07

Copyright

5.86.2 Enumeration Type Documentation

5.86.2.1 eStorageFSStatus_t

`enum eStorageFSStatus_t`

Storage medium statuses defined here.

5.86.2.2 eStorageFSOperationModes_t

`enum eStorageFSOperationModes_t`

User enumeration RW modes for file system.

5.86.2.3 eStorageFileNamesEnums_t

`enum eStorageFileNamesEnums_t`

Enums for all the files to be saved in Storage medium.

5.87 AppStorageDataStructures.h

Go to the documentation of this file.

```
00001
00013
00014 #ifndef APPSTORAGE_APPSTORAGEDATASTRUCTURES_H_
00015 #define APPSTORAGE_APPSTORAGEDATASTRUCTURES_H_
00016
00018
00023 typedef enum
00024 {
00025     eFS_SUCCESS,
00026     eFS_ERROR,
00027     eFS_NO_FILE,
00028     eFS_MAX_STATUS,
00029 } eStorageFSStatus_t;
00030
00035 typedef enum
00036 {
00037     eFS_READONLY,
00038     eFS_WRITEONLY,
00039     eFS_READ_WRITE,
00040     eFS_READ_CREATE,
00041     eFS_WRITE_CREATE,
00042     eFS_WRITE_APPEND,
00043     eFS_READ_WRITE_CREATE,
00044     eFS_MODE_MAX
00045 } eStorageFSOperationModes_t;
00046
00051 typedef enum
00052 {
00053     eFS_GOLDEN_IMAGE,
00054     eFS_FILE_2,
00055     eFS_MAX
00056 } eStorageFileNamesEnums_t;
00057
00059
00060 #endif /* APPSTORAGE_APPSTORAGEDATASTRUCTURES_H_ */
```

Index